

## Project Data Page:





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## **DESIGN CALCULATION**

*In Accordance with ASME Section VIII Division 1*

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Liquid Ammonia Drainage Tank (200-F)Rev 1-7%.pvdb

Date of Analysis : Sep 16, 2026 2:46pm

**PV Elite 28 U1, March 2026**



### Vessel Design Summary:

#### ASME Code, Section VIII Division 1, 2025

Diameter Spec : 3000.000 mm. ID  
 Vessel Design Length, Tangent to Tangent 9000.00 mm.  
 Specified Datum Line Distance 0.00 mm.  
 Internal Design Temperature 80 °C  
 Internal Design Pressure 4.500 bars  
 External Design Temperature 0 °C  
 Maximum Allowable Working Pressure 5 bars  
 Shop Test Pressure 9.733 bars  
 Field Test Pressure 6.951 bars  
 Required Minimum Design Metal Temperature -40.0 °C  
 Warmest Computed Minimum Design Metal Temperature -48.0 °C  
 Wind Design Code ASCE/SEI 7-16  
 Earthquake Design Code ASCE/SEI 7-16

### Materials of Construction:

| Component Type | Material  | Class | Thickness     | UNS #  | Material Treatment Options |
|----------------|-----------|-------|---------------|--------|----------------------------|
| Shell          | SA-516 70 | ...   | ...           | K02700 |                            |
| Head           | SA-516 70 | ...   | ...           | K02700 |                            |
| Nozzle         | SA-105    | ...   | ...           | K03504 | +IT                        |
| Nozzle         | SA-106 B  | ...   | ...           | K03006 |                            |
| Nozzle Flg     | SA-105    | ...   | ...           | K03504 |                            |
| Hrz Bolting    | SA-307 A  | ...   | 6 <= t <= 100 |        |                            |
| Saddles        | SA-516 70 | ...   | ...           | K02700 |                            |

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized  
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

### Element Pressures and MAWP (bars & mm.):

| Element Description or Type | Design Pressure + Stat. head | Ext. Press. | Element M.A.W.P | Total Corrosion Allowance | Str. Flg. Gov. | In Creep Range |
|-----------------------------|------------------------------|-------------|-----------------|---------------------------|----------------|----------------|
| BOTTOM HEAD                 | 4.600                        | 0.00        | 5.000           | 3.0000                    | No             | No             |
| SHELL-1                     | 4.600                        | 0.00        | 5.000           | 3.0000                    | N/A            | No             |
| SHELL-2                     | 4.600                        | 0.00        | 5.000           | 3.0000                    | N/A            | No             |
| SHELL-3                     | 4.600                        | 0.00        | 5.000           | 3.0000                    | N/A            | No             |
| TOP HEAD                    | 4.600                        | 0.00        | 5.000           | 3.0000                    | No             | No             |

Liquid Level: 1500.00 mm. Dens.: 0.001 kg./cm3 Sp. Gr.: 0.682

### Element Types and Properties:

| Element Type | "To" Elev mm. | Element Length mm. | Nominal Thickness mm. | Finished Thickness mm. | Reqd Thk Internal mm. | Reqd Thk External mm. | Long Eff | Circ Eff |
|--------------|---------------|--------------------|-----------------------|------------------------|-----------------------|-----------------------|----------|----------|
| Ellipse      | 50.0          | 50.0               | 12.0                  | 10.0                   | 8.9                   | ...                   | 0.85     | 0.85     |
| Cylinder     | 3050.0        | 3000.0             | 10.0                  | 10.0                   | 8.9                   | ...                   | 0.85     | 0.85     |
| Cylinder     | 6050.0        | 3000.0             | 10.0                  | 10.0                   | 8.9                   | ...                   | 0.85     | 0.85     |
| Cylinder     | 8950.0        | 2900.0             | 10.0                  | 10.0                   | 8.9                   | ...                   | 0.85     | 0.85     |
| Ellipse      | 9000.0        | 50.0               | 12.0                  | 10.0                   | 8.9                   | ...                   | 0.85     | 0.85     |

### Local Stress Analysis Results:

| Description | Analysis Type | Max Stress Ratio | High Stress Location | Pass Fail |
|-------------|---------------|------------------|----------------------|-----------|
| Lift Lug    | WRC-107/537   | 0.860            | n/a                  | Passed    |
| Lift Lug    | WRC-107/537   | 0.808            | n/a                  | Passed    |
| N1-6 Inch   | WRC-297       | 1.078            | Shell                | Failed    |
| N2-10 Inch  | WRC-297       | 1.061            | Shell                | Failed    |
| N4-4 Inch   | WRC-297       | 0.746            | Shell                | Passed    |
| N3-10 Inch  | WRC-297       | 1.061            | Shell                | Failed    |

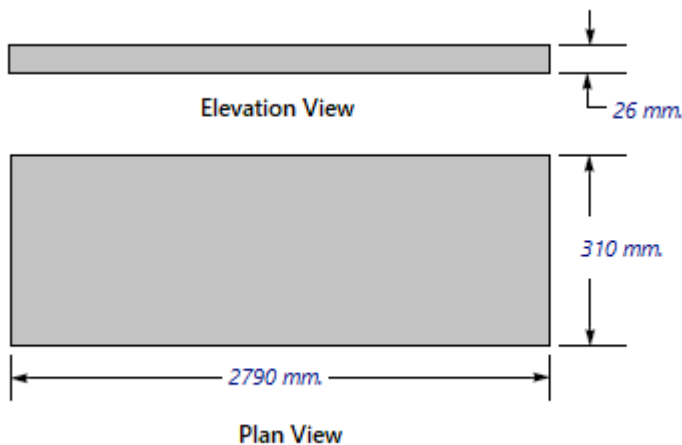
### Foundation/Support Summary

#### Saddle Parameters:

|                                 |              |
|---------------------------------|--------------|
| Saddle Width                    | 280.000 mm.  |
| Saddle Bearing Angle            | 120.000 deg. |
| Centerline Dimension            | 2133.000 mm. |
| Wear Pad Width                  | 475.000 mm.  |
| Wear Pad Thickness              | 10.000 mm.   |
| Wear Pad Bearing Angle          | 132.000 deg. |
| Distance from Saddle to Tangent | 1800.000 mm. |

|                                         |              |
|-----------------------------------------|--------------|
| Baseplate Length                        | 2790.000 mm. |
| Baseplate Thickness                     | 26.000 mm.   |
| Baseplate Width                         | 310.000 mm.  |
| Number of Ribs (including outside ribs) | 6            |
| Rib Thickness                           | 20.000 mm.   |
| Web Thickness                           | 20.000 mm.   |
| Height of Center Web                    | 607.000 mm.  |
| Number of Bolts in Baseplate            | 10           |

#### Baseplate Sketch



#### Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

|                                        |             |
|----------------------------------------|-------------|
| Maximum Vertical Saddle Load           | 26405.72 Kg |
| Maximum Transverse Saddle Shear Load   | 1258.38 Kg  |
| Maximum Longitudinal Saddle Shear Load | 2516.75 Kg  |

#### Summary of Maximum Saddle Loads, Pressure Test Case:

|                                        |             |
|----------------------------------------|-------------|
| Maximum Vertical Saddle Load           | 47343.66 Kg |
| Maximum Transverse Saddle Shear Load   | 570.47 Kg   |
| Maximum Longitudinal Saddle Shear Load | 622.44 Kg   |

#### Weights:

|                                              |             |
|----------------------------------------------|-------------|
| Fabricated - Bare w/o removable internals    | 12854.8 kg. |
| Shop Test - Fabricated + water ( full )      | 83510.2 kg. |
| Shipping - Fab. + rem. intls.+ shipping app. | 12854.8 kg. |
| Erected - Fab. + rem. intls.+ insul. (etc)   | 20908.7 kg. |



|            |                                     |             |
|------------|-------------------------------------|-------------|
| Empty      | - Fab. + intls. + details + wghts.  | 20908.7 kg. |
| Operating  | - Empty + operating liquid (no ca)  | 45001.9 kg. |
| Field Test | - Empty weight + test medium (full) | 91564.0 kg. |

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### Nozzle Calculation Summary (bars & mm.):

| Description | MAWP | Ext | MAPNC | UG-45 | [tr] | Weld Path | Areas or Stresses |
|-------------|------|-----|-------|-------|------|-----------|-------------------|
| K6-2 Inch   | 5    | ... | ...   | OK    | 7.80 | OK        | Passed            |
| N1-6 Inch   | 5    | ... | ...   | OK    | 8.95 | OK        | Passed            |
| K4-2 Inch   | 5    | ... | ...   | OK    | 7.80 | OK        | Passed            |
| K5-2 Inch   | 5    | ... | ...   | OK    | 7.80 | OK        | Passed            |
| M1-24 Inch  | 5    | ... | ...   | ...   |      | OK        | Passed            |
| D1-2 Inch   | 5    | ... | ...   | OK    | 7.80 | OK        | Passed            |
| N2-10 Inch  | 5    | ... | ...   | OK    | 8.95 | OK        | Passed            |
| K3-2 Inch   | 5    | ... | ...   | OK    | 7.80 | OK        | Passed            |
| N4-4 Inch   | 5    | ... | ...   | OK    | 8.02 | OK        | Passed            |
| K1-6 Inch   | 5    | ... | ...   | OK    | 8.95 | OK        | Passed            |
| N3-10 Inch  | 5    | ... | ...   | OK    | 8.95 | OK        | Passed            |
| V1-2 Inch   | 5    | ... | ...   | OK    | 7.80 | OK        | Passed            |

### Nozzle MAWP Summary:

Minimum MAWP Nozzles : 5 Nozzle : V1-2 Inch  
 Minimum MAWP Shells/Flanges : 5 Element : SHELL-3  
 Minimum MAPnc Shells/Flanges : 8 Element : SHELL-3

Computed Vessel M.A.W.P. : 5 bars

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

### Check the spatial relationship between the nozzles:

| From Node | Nozzle Description | X Coordinate mm. | Layout Angle deg | Dia. Limit mm. |
|-----------|--------------------|------------------|------------------|----------------|
| 20        | K6-2 Inch          | 300.000          | 0.000            | 113.600        |
| 20        | N1-6 Inch          | 900.000          | 0.000            | 268.800        |
| 20        | K4-2 Inch          | 1400.000         | 0.000            | 113.600        |
| 20        | K5-2 Inch          | 1800.000         | 0.000            | 113.600        |
| 20        | M1-24 Inch         | 2400.000         | 0.000            | 1231.200       |
| 20        | D1-2 Inch          | 2500.000         | 180.000          | 113.600        |
| 30        | N2-10 Inch         | 3500.000         | 0.000            | 472.000        |
| 30        | K3-2 Inch          | 4200.000         | 0.000            | 113.600        |
| 30        | N4-4 Inch          | 4550.000         | 180.000          | 175.200        |
| 40        | K1-6 Inch          | 6700.000         | 0.000            | 268.800        |
| 40        | N3-10 Inch         | 7550.001         | 0.000            | 472.000        |
| 40        | V1-2 Inch          | 8450.000         | 0.000            | 113.600        |

### The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.  
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

*Diameter Limit clash between : M1-24 Inch and K5-2 Inch.  
 The Centerline Dist. is 600.00 while the sum of the limit Radii is 672.40 mm..*

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### Nozzle Schedule:

| Description | Nominal or Actual Size | Schd or FVC Type | Flg Type | Nozzle O/Dia in | Wall Thk mm | Reinforcing Pad Diameter | Pad Thk mm | Cut Length mm | Flg Class |
|-------------|------------------------|------------------|----------|-----------------|-------------|--------------------------|------------|---------------|-----------|
| K6-2 Inch   | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 340.59        | 300       |
| K4-2 Inch   | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 340.59        | 300       |
| K5-2 Inch   | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 340.59        | 300       |
| D1-2 Inch   | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 340.59        | 300       |
| K3-2 Inch   | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 340.59        | 300       |
| V1-2 Inch   | 2.000 in               | Actual           | LWN      | 3.320           | 16.764      | ...                      | ...        | 340.59        | 300       |
| N4-4 Inch   | 4.000 in               | Actual           | WNF      | 4.000           | 10.000      | ...                      | ...        | 344.86        | 150       |
| N1-6 Inch   | 6.000 in               | Actual           | WNF      | 6.000           | 12.000      | ...                      | ...        | 345.94        | 150       |
| K1-6 Inch   | 6.000 in               | Actual           | WNF      | 6.000           | 12.000      | ...                      | ...        | 341.94        | 150       |
| N2-10 Inch  | 10.000 in              | Actual           | WNF      | 10.000          | 12.000      | ...                      | ...        | 351.39        | 150       |
| N3-10 Inch  | 10.000 in              | Actual           | WNF      | 10.000          | 12.000      | ...                      | ...        | 351.39        | 150       |
| M1-24 Inch  | 24.000 in              | Actual           | WNF      | 24.945          | 12.000      | ...                      | ...        | 492.89        | 150       |

### General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

### Nozzle Material and Weld Fillet Leg Size Details (mm.):

| Description | Material | Shl Grve Weld | Noz Shl/Pad Weld | Pad OD Weld | Pad Grve Weld | Inside Weld |
|-------------|----------|---------------|------------------|-------------|---------------|-------------|
| K6-2 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| K4-2 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| K5-2 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| D1-2 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| K3-2 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| V1-2 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| N4-4 Inch   | SA-106 B | 10.000        | 10.000           | ...         | ...           | ...         |
| N1-6 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| K1-6 Inch   | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| N2-10 Inch  | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| N3-10 Inch  | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |
| M1-24 Inch  | SA-105   | 10.000        | 10.000           | ...         | ...           | ...         |

Note: The Outside projections below do not include the flange thickness.

### Nozzle Miscellaneous Data:

| Description | Elev/Distance From Datum mm. | Layout Angle deg | Proj Outside mm. | Proj Inside mm. | Installed in Component |
|-------------|------------------------------|------------------|------------------|-----------------|------------------------|
| K6-2 Inch   | 300.000                      | 0.0              | 330.00           | 0.00            | SHELL-1                |
| K4-2 Inch   | 1400.000                     | 0.0              | 330.00           | 0.00            | SHELL-1                |
| K5-2 Inch   | 1800.000                     | 0.0              | 330.00           | 0.00            | SHELL-1                |
| D1-2 Inch   | 2500.000                     | 180.0            | 330.00           | 0.00            | SHELL-1                |
| K3-2 Inch   | 4200.000                     | 0.0              | 330.00           | 0.00            | SHELL-2                |
| V1-2 Inch   | 8450.000                     | 0.0              | 330.00           | 0.00            | SHELL-3                |
| N4-4 Inch   | 4550.000                     | 180.0            | 330.00           | 0.00            | SHELL-2                |
| N1-6 Inch   | 900.000                      | 0.0              | 330.00           | 0.00            | SHELL-1                |
| K1-6 Inch   | 6700.000                     | 0.0              | 330.00           | 0.00            | SHELL-3                |
| N2-10 Inch  | 3500.000                     | 0.0              | 330.00           | 0.00            | SHELL-2                |
| N3-10 Inch  | 7550.000                     | 0.0              | 330.00           | 0.00            | SHELL-3                |
| M1-24 Inch  | 2400.000                     | 0.0              | 430.00           | 0.00            | SHELL-1                |





# **Bill of Materials:**

| QTY | DESCRIPTION                                                    | MATERIAL  |
|-----|----------------------------------------------------------------|-----------|
| 2   | ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 3000.0mm. ID X 50.0mm. | SA-516 70 |
| 2   | INSULATION: 800mm X 80mm THK                                   |           |
| 2   | SADDLE: 280.0mm X 120 DEG                                      | SA-516 70 |
| 1   | PLATFORM: 1000mm X 3050mm WIDE                                 |           |
| 1   | LADDER: 5450.0mm                                               | ...       |
| 2   | INSULATION: 3000mm X 80mm THK                                  |           |
| 4   | LIFTING LUGS:                                                  | SA-516 70 |
| 4   | LIFTING LUG PADS: 450.0mm X 700.0mm X 10.0mm THK               |           |
| 1   | INSULATION: 2900mm X 80mm THK                                  |           |
| 1   | CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)                        | SA-105    |
| 2   | CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)                         | SA-105    |
| 1   | CLASS 150 GR 1.1, 6.0" BLIND FLANGE(S)                         | SA-105    |
| 2   | CYLINDRICAL SEGMENT 10.0mm. THK X 3743.7mm. X 3000.0mm.        | SA-516 70 |
| 1   | CYLINDRICAL SEGMENT 10.0mm. THK X 1937.3mm. X 3000.0mm.        | SA-516 70 |
| 2   | CYLINDRICAL SEGMENT 10.0mm. THK X 2042.0mm. X 3000.0mm.        | SA-516 70 |
| 2   | CYLINDRICAL SEGMENT 10.0mm. THK X 2015.8mm. X 3000.0mm.        | SA-516 70 |
| 1   | CYLINDRICAL SEGMENT 10.0mm. THK X 1309.0mm. X 3000.0mm.        | SA-516 70 |
| 1   | CYLINDRICAL SEGMENT 10.0mm. THK X 3036.8mm. X 2900.0mm.        | SA-516 70 |
| 2   | CYLINDRICAL SEGMENT 10.0mm. THK X 3063.0mm. X 2900.0mm.        | SA-516 70 |
| 1   | CYLINDRICAL SEGMENT 10.0mm. THK X 6387.8mm. X 2900.0mm.        | SA-516 70 |
| 1   | NAMEPLATE                                                      | ...       |



### Minimum Design Metal Temperature Results Summary (°C):

| Description                                       | Notes | Curve | Basic MDMT | Reduced MDMT | UG-20(f) MDMT | Thickness ratio | Gov Thk mm. | E*   | PWHT reqd |
|---------------------------------------------------|-------|-------|------------|--------------|---------------|-----------------|-------------|------|-----------|
| BOTTOM HEAD                                       | [10]  | D     | -48        | -48          | -29           | 0.846           | 10.000      | 0.85 | Yes       |
| BOTTOM HEAD                                       | [7]   | D     | -48        | -48          | -29           | 0.661           | 12.000      | 0.85 | Yes       |
| SHELL-1                                           | [8]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 0.85 | Yes       |
| SHELL-2                                           | [8]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 0.85 | Yes       |
| SHELL-3                                           | [8]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 0.85 | Yes       |
| TOP HEAD                                          | [10]  | D     | -48        | -48          | -29           | 0.846           | 10.000      | 0.85 | Yes       |
| TOP HEAD                                          | [7]   | D     | -48        | -48          | -29           | 0.661           | 12.000      | 0.85 | Yes       |
| Left Saddle                                       | [24]  | D     | -48        | -48          | -29           | 0.850           | 10.000      | 0.85 | Yes       |
| New Saddle                                        | [24]  | D     | -48        | -48          | -29           | 0.850           | 10.000      | 0.85 | Yes       |
| K6-2 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| N1-6 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.541           | 14.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| K4-2 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| K5-2 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| M1-24 Inch                                        | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| D1-2 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| N2-10 Inch                                        | [1]   | D     | -46        | -48          | -29           | 0.458           | 16.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| K3-2 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| N4-4 Inch                                         | [1]   | C     | -36        | -48          | -29           | 0.541           | 14.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| K1-6 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| N3-10 Inch                                        | [1]   | D     | -46        | -48          | -29           | 0.458           | 16.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| V1-2 Inch                                         | [1]   | D     | -48        | -48          | -29           | 0.850           | 10.000      | 1.00 | Yes       |
| Nozzle Flg                                        | [5]   | A     | -18        | -104         |               |                 |             |      |           |
| Nozl. Bolting[22]                                 |       |       | -48        |              |               | 1.000           | 64.000      | 1.00 | Yes       |
| Warmest MDMT:                                     |       |       | -18        | -48          |               |                 |             |      |           |
| Required Minimum Design Metal Temperature         |       |       |            |              |               | -40.0           | °C          |      |           |
| Warmest Computed Minimum Design Metal Temperature |       |       |            |              |               | -48.0           | °C          |      |           |

### Notes:

- [ ! ] - This was an impact tested material.
- [ 1 ] - Governing Nozzle Weld.
- [ 4 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [ 5 ] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [ 6 ] - MDMT Calculations at the Shell/Head Joint.
- [ 7 ] - MDMT Calculations for the Straight Flange.
- [ 8 ] - Cylinder/Cone/Flange Junction MDMT.
- [ 9 ] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



UG-84(b)(2) was not considered.  
UCS-66(g) was not considered.  
UCS-66(i) was not considered.

**Notes:**

Impact test temps were not entered in and not considered in the analysis.  
UCS-66(i) applies to impact tested materials not by specification and  
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).  
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

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Class From To: Check of Additional Element Data

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There were no geometry errors or warnings.

**PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.**

### Information Regarding Nozzle Loads

*Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.*



*One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.*



*The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.*



*The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.*

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#### Units used in this Analysis (barsUniCode):

| Name           | System Unit | Constant   | User Unit |
|----------------|-------------|------------|-----------|
| Length         | Feet        | 304.8000   | mm.       |
| Force          | Pounds      | 0.4536     | Kgf       |
| Mass           | Pounds      | 0.4536     | kg.       |
| Area           | sq. inches  | 6.4516     | cm2       |
| Moment         | ft. lbs.    | 0.1383     | Kg-m.     |
| Stress         | lbs./sq.in. | 0.0069     | N./mm2    |
| Temperature    | Degrees F   | 0.5556     | °C        |
| Pressure       | psig        | 0.0689     | bars      |
| Elast. Modulus | lbs./sq.in. | 6.8946     | KPa.      |
| Pipe Density   | lbs./cu.in. | 0.0277     | kg./cm3   |
| Ins. Density   | lbs./cu.ft. | 0.1600E-04 | kg./cm3   |
| Fluid Density  | lbs./cu.ft. | 0.1600E-04 | kg./cm3   |
| Wind Speed     | miles/hr    | 1.6093     | Km/hr     |
| Tray Weight    | lbs./sq.ft. | 0.0005     | Kgs/cm2   |
| Inertia        | in.4        | 41.6231    | cm4       |
| G Load         | G's         | 1.0000     | g's       |
| Wind Load      | lbs./sq.ft. | 4.8825     | Kgs/m2    |
| Elevation      | Feet        | 304.8000   | mm.       |
| Volume         | in.3        | 16.3900    | cm3       |
| Diameter       | inches      | 25.4000    | mm.       |
| Thickness      | inches      | 25.4000    | mm.       |

#### PV Elite Vessel Analysis Program: Input Data

|                                          |        |      |
|------------------------------------------|--------|------|
| Design Internal Pressure (for Hydrotest) | 4.4997 | bars |
| Design Internal Temperature              | 80.0   | °C   |
| Projection of Nozzle from Vessel Top     | 500    | mm.  |
| Projection of Nozzle from Vessel Bottom  | 500    | mm.  |
| Minimum Design Metal Temperature         | -40.0  | °C   |
| Type of Construction                     | Welded |      |
| Special Service                          | Lethal |      |
| Degree of Radiography                    | RT-1   |      |
| Use Higher Longitudinal Stresses (Flag)  | Y      |      |
| Select t for Internal Pressure (Flag)    | N      |      |
| Select t for External Pressure (Flag)    | N      |      |
| Select t for Axial Stress (Flag)         | N      |      |
| Select Location for Stiff. Rings (Flag)  | N      |      |
| Consider Vortex Shedding                 | N      |      |

#### Shop Pressure Test:

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(c)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | No         |

#### Field Pressure Test:

|                                      |            |
|--------------------------------------|------------|
| Type of Pressure Test                | UG-99(b)   |
| Pressure Test Position               | Horizontal |
| Test Performed in Corroded Condition | Yes        |

|              |                |
|--------------|----------------|
| Load Case 1  | NP+EW+WI+FW+BW |
| Load Case 2  | NP+EW+EE+FS+BS |
| Load Case 3  | NP+OW+WI+FW+BW |
| Load Case 4  | NP+OW+EQ+FS+BS |
| Load Case 5  | NP+HW+HI       |
| Load Case 6  | NP+HW+HE       |
| Load Case 7  | IP+OW+WI+FW+BW |
| Load Case 8  | IP+OW+EQ+FS+BS |
| Load Case 9  | EP+OW+WI+FW+BW |
| Load Case 10 | EP+OW+EQ+FS+BS |
| Load Case 11 | HP+HW+HI       |
| Load Case 12 | HP+HW+HE       |
| Load Case 13 | IP+WE+EW       |
| Load Case 14 | IP+WF+CW       |
| Load Case 15 | IP+VO+OW       |



|              |             |
|--------------|-------------|
| Load Case 16 | IP+VE+EW    |
| Load Case 17 | NP+VO+OW    |
| Load Case 18 | FS+BS+IP+OW |
| Load Case 19 | FS+BS+EP+OW |
| Load Case 20 | BL+IP+OW    |

|                                         |                   |
|-----------------------------------------|-------------------|
| Wind Design Code                        | ASCE/SEI 7-16     |
| Wind Load Reduction Scale Factor        | 1.100             |
| Basic Wind Speed                        | [V] 97.2 Km/hr    |
| Surface Roughness Category              | C: Open Terrain   |
| Type of Surface                         | Moderately Smooth |
| Base Elevation                          | 0 mm.             |
| Percent Wind for Hydrotest              | 50.0              |
| Using User defined Wind Press. Vs Elev. | N                 |
| Height of Hill or Escarpment H or Hh    | 0 mm.             |
| Distance Upwind of Crest Lh             | 0 mm.             |
| Distance from Crest to the Vessel x     | 0 mm.             |
| Type of Terrain ( Hill, Escarpment )    | Flat              |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100            |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000            |
| Damping Factor (Beta) for Wind (Filled) | 0.0000            |
| Site Elevation above Sea Level          | 0 mm.             |

|                                     |               |
|-------------------------------------|---------------|
| Seismic Design Code                 | ASCE/SEI 7-16 |
| Seismic Load Reduction Scale Factor | 1.000         |
| Importance Factor                   | 1.250         |
| Table Value Fa                      | 1.600         |
| Table Value Fv                      | 2.400         |
| Max. Mapped Res. Acceleration [Ss]  | 0.147         |
| Max. Eff. Ground Acceleration [S]   | 0.065         |
| Force Modification Factor R         | 3.000         |
| Site Class                          | D             |
| Component Elevation Ratio z/h       | 0.000         |
| Amplification Factor Ap             | 1.000         |
| Force Factor                        | 0.000         |
| Consider Vertical Acceleration      | Yes           |
| Minimum Acceleration Multiplier     | 0.000         |
| User Value of Sds (used if > 0 )    | 0.000         |
| User Value of Sd1 (used if > 0 )    | 0.000         |
| Moment Reduction Factor Tau         | 1.000         |

|                                          |   |
|------------------------------------------|---|
| Overall M.A.W.P. + Static Head           | Y |
| Consider MAP New and Cold in Noz. Design | N |
| Consider External Loads for Nozzle Des.  | Y |
| Use ASME VIII-1 Appendix 1-9             | N |

|                             |                                |
|-----------------------------|--------------------------------|
| Perform Blast Load Analysis | No                             |
| Material Database Year      | Current w/Addenda or Code Year |

#### Configuration Directives:

|                                                          |     |
|----------------------------------------------------------|-----|
| g0 is taken as min( g0, attached shell thk )             | Yes |
| Do not use Nozzle MDMT Interpretation VIII-1 01-37       | No  |
| Use Table G instead of exact equation for "A"            | Yes |
| Shell Head Joints are Tapered                            | Yes |
| Compute "K" in corroded condition                        | Yes |
| Use Code Case 2286                                       | No  |
| Use the MAWP to compute the MDMT                         | Yes |
| For thickness ratios <= 0.35, MDMT will be -155F (-104C) | Yes |
| For PWHT & P1 Materials the MDMT can be < -55F (-48C)    | No  |
| Using Metric Material Databases, ASME II D               | Yes |
| Calculate B31.3 type stress for Nozzles with Loads       | Yes |
| Reduce the MDMT due to lower membrane stress             | Yes |
| Consider Longitudinal Stress in MDMT Calculations        | Yes |

#### Complete Listing of Vessel Elements and Details:

|                   |     |
|-------------------|-----|
| Element#          | 1/5 |
| Element From Node | 10  |



|                                      |                 |
|--------------------------------------|-----------------|
| Element To Node                      | 20              |
| Element Type                         | Elliptical      |
| Description                          | BOTTOM HEAD     |
| Distance "FROM" to "TO"              | 50 mm.          |
| Inside Diameter                      | 3000 mm.        |
| Element Thickness                    | 10 mm.          |
| Internal Corrosion Allowance         | 3 mm.           |
| Nominal Thickness                    | 12 mm.          |
| External Corrosion Allowance         | 0 mm.           |
| Design Internal Pressure             | 4.4997 bars     |
| Design Temperature Internal Pressure | 80 °C           |
| Design External Pressure             | 0 bars          |
| Design Temperature External Pressure | 0 °C            |
| Effective Diameter Multiplier        | 1.2             |
| Material Name                        | SA-516 70       |
| Allowable Stress, Ambient            | 138 N./mm2      |
| Allowable Stress, Operating          | 138 N./mm2      |
| Allowable Stress, Hydrotest          | 234 N./mm2      |
| Material Density                     | 0.00775 kg./cm3 |
| P Number Thickness                   | 32.004 mm.      |
| Yield Stress, Operating              | 243 N./mm2      |
| UCS-66 Chart Curve Designation       | D               |
| External Pressure Chart Name         | CS-2            |
| UNS Number                           | K02700          |
| Product Form                         | Plate           |
| Efficiency, Longitudinal Seam        | 0.85            |
| Efficiency, Circumferential Seam     | 0.85            |
| Elliptical Head Factor               | 2.0             |
| Weld is pre-Heated                   | No              |

|                                      |                   |
|--------------------------------------|-------------------|
| Element From Node                    | 10                |
| Detail Type                          | Liquid            |
| Detail ID                            | Liquid 10         |
| Dist. from "FROM" Node / Offset dist | 0 mm.             |
| Height/Length of Liquid              | 1500 mm.          |
| Liquid Density                       | 0.0006809 kg./cm3 |

|                                      |                 |
|--------------------------------------|-----------------|
| Element From Node                    | 10              |
| Detail Type                          | Insulation      |
| Detail ID                            | Insulation      |
| Dist. from "FROM" Node / Offset dist | -750 mm.        |
| Height/Length of Insulation          | 800 mm.         |
| Thickness of Insulation              | 80 mm.          |
| Density                              | 0.00032 kg./cm3 |

---

|                                      |             |
|--------------------------------------|-------------|
| Element#                             | 2/5         |
| Element From Node                    | 20          |
| Element To Node                      | 30          |
| Element Type                         | Cylinder    |
| Description                          | SHELL-1     |
| Distance "FROM" to "TO"              | 3000 mm.    |
| Inside Diameter                      | 3000 mm.    |
| Element Thickness                    | 10 mm.      |
| Internal Corrosion Allowance         | 3 mm.       |
| Nominal Thickness                    | 10 mm.      |
| External Corrosion Allowance         | 0 mm.       |
| Design Internal Pressure             | 4.4997 bars |
| Design Temperature Internal Pressure | 80 °C       |
| Design External Pressure             | 0 bars      |
| Design Temperature External Pressure | 0 °C        |
| Effective Diameter Multiplier        | 1.2         |
| Material Name                        | SA-516 70   |
| Efficiency, Longitudinal Seam        | 0.85        |
| Efficiency, Circumferential Seam     | 0.85        |
| Weld is pre-Heated                   | No          |

|                   |        |
|-------------------|--------|
| Element From Node | 20     |
| Detail Type       | Saddle |



|                                      |               |         |
|--------------------------------------|---------------|---------|
| Detail ID                            | Left Saddle   |         |
| Dist. from "FROM" Node / Offset dist | 1750          | mm.     |
| Width of Saddle                      | 280           | mm.     |
| Height of Saddle at Bottom           | 2133          | mm.     |
| Saddle Contact Angle                 | 120.0         |         |
| Height of Composite Ring Stiffener   | 0             | mm.     |
| Width of Wear Plate                  | 475           | mm.     |
| Thickness of Wear Plate              | 10            | mm.     |
| Contact Angle, Wear Plate (degrees)  | 132.0         |         |
| Friction coefficient                 | 0.0           |         |
| Moment Factor                        | 3.0           |         |
| Dimension E at base (optional)       | 0             | mm.     |
| Circumferential Eff. over Saddle     | 1.0           |         |
| Circumferential Eff. at Midspan      | 1.0           |         |
| Tangent to Tangent dist. (optional)  | 0             | mm.     |
| Element From Node                    | 20            |         |
| Detail Type                          | Platform      |         |
| Detail ID                            | PLAT:[1 OF 1] |         |
| Dist. from "FROM" Node / Offset dist | 0             | mm.     |
| Platform Start Angle (degrees)       | 0.0           |         |
| Platform End Angle (degrees)         | 0.0           |         |
| Platform Wind Area                   | 90000         | cm2     |
| Platform Weight                      | 4779.9        | Kgf     |
| Platform Railing Weight              | 0             | Kgf/mm. |
| Platform Grating Weight              | 0.017         | Kgs/cm2 |
| Platform Width                       | 3050          | mm.     |
| Platform Height                      | 1000          | mm.     |
| Platform Clearance or End Offset     | 85            | mm.     |
| Platform Force Coefficient           | 1.0           |         |
| Ladder Layout Angle                  | 0.0           |         |
| Ladder Start Elevation               | 0             | mm.     |
| Ladder End Elevation                 | 5450          | mm.     |
| Unit Weight of Ladder                | 0.037         | Kgf/mm. |
| Platform Length (top head platform)  | 9000          | mm.     |
| Element From Node                    | 20            |         |
| Detail Type                          | Liquid        |         |
| Detail ID                            | Liquid 20     |         |
| Dist. from "FROM" Node / Offset dist | 0             | mm.     |
| Height/Length of Liquid              | 1500          | mm.     |
| Liquid Density                       | 0.0006809     | kg./cm3 |
| Element From Node                    | 20            |         |
| Detail Type                          | Insulation    |         |
| Detail ID                            | Ins: 20       |         |
| Dist. from "FROM" Node / Offset dist | 0             | mm.     |
| Height/Length of Insulation          | 3000          | mm.     |
| Thickness of Insulation              | 80            | mm.     |
| Density                              | 0.00032       | kg./cm3 |
| Element From Node                    | 20            |         |
| Detail Type                          | Nozzle        |         |
| Detail ID                            | K6-2 Inch     |         |
| Dist. from "FROM" Node / Offset dist | 250           | mm.     |
| Nozzle Diameter                      | 2             | in.     |
| Nozzle Schedule                      | None          |         |
| Nozzle Class                         | 300           |         |
| Layout Angle                         | 0.0           |         |
| Blind Flange (Y/N)                   | N             |         |
| Weight of Nozzle ( Used if > 0 )     | 13.916        | Kgf     |
| Grade of Attached Flange             | GR 1.1        |         |
| Nozzle Matl                          | SA-105 +IT    |         |
| Element From Node                    | 20            |         |
| Detail Type                          | Nozzle        |         |
| Detail ID                            | N1-6 Inch     |         |
| Dist. from "FROM" Node / Offset dist | 850           | mm.     |
| Nozzle Diameter                      | 6             | in.     |
| Nozzle Schedule                      | 80            |         |



|                                  |        |     |
|----------------------------------|--------|-----|
| Nozzle Class                     | 150    |     |
| Layout Angle                     | 0.0    |     |
| Blind Flange (Y/N)               | N      |     |
| Weight of Nozzle ( Used if > 0 ) | 30.316 | Kgf |
| Grade of Attached Flange         | GR 1.1 |     |
| Nozzle Matl                      | SA-105 | +IT |

|                                      |        |      |
|--------------------------------------|--------|------|
| Element From Node                    | 20     |      |
| Detail Type                          | Nozzle |      |
| Detail ID                            | K4-2   | Inch |
| Dist. from "FROM" Node / Offset dist | 1350   | mm.  |
| Nozzle Diameter                      | 2      | in.  |
| Nozzle Schedule                      | None   |      |
| Nozzle Class                         | 300    |      |
| Layout Angle                         | 0.0    |      |
| Blind Flange (Y/N)                   | N      |      |
| Weight of Nozzle ( Used if > 0 )     | 13.916 | Kgf  |
| Grade of Attached Flange             | GR 1.1 |      |
| Nozzle Matl                          | SA-105 | +IT  |

|                                      |        |      |
|--------------------------------------|--------|------|
| Element From Node                    | 20     |      |
| Detail Type                          | Nozzle |      |
| Detail ID                            | K5-2   | Inch |
| Dist. from "FROM" Node / Offset dist | 1750   | mm.  |
| Nozzle Diameter                      | 2      | in.  |
| Nozzle Schedule                      | None   |      |
| Nozzle Class                         | 300    |      |
| Layout Angle                         | 0.0    |      |
| Blind Flange (Y/N)                   | N      |      |
| Weight of Nozzle ( Used if > 0 )     | 13.916 | Kgf  |
| Grade of Attached Flange             | GR 1.1 |      |
| Nozzle Matl                          | SA-105 | +IT  |

|                                      |        |      |
|--------------------------------------|--------|------|
| Element From Node                    | 20     |      |
| Detail Type                          | Nozzle |      |
| Detail ID                            | M1-24  | Inch |
| Dist. from "FROM" Node / Offset dist | 2350   | mm.  |
| Nozzle Diameter                      | 24     | in.  |
| Nozzle Schedule                      | XS     |      |
| Nozzle Class                         | 150    |      |
| Layout Angle                         | 0.0    |      |
| Blind Flange (Y/N)                   | Y      |      |
| Weight of Nozzle ( Used if > 0 )     | 536.48 | Kgf  |
| Grade of Attached Flange             | GR 1.1 |      |
| Nozzle Matl                          | SA-105 | +IT  |

|                                      |        |      |
|--------------------------------------|--------|------|
| Element From Node                    | 20     |      |
| Detail Type                          | Nozzle |      |
| Detail ID                            | D1-2   | Inch |
| Dist. from "FROM" Node / Offset dist | 2450   | mm.  |
| Nozzle Diameter                      | 2      | in.  |
| Nozzle Schedule                      | None   |      |
| Nozzle Class                         | 300    |      |
| Layout Angle                         | 180.0  |      |
| Blind Flange (Y/N)                   | Y      |      |
| Weight of Nozzle ( Used if > 0 )     | 17.625 | Kgf  |
| Grade of Attached Flange             | GR 1.1 |      |
| Nozzle Matl                          | SA-105 | +IT  |

|                                      |          |     |
|--------------------------------------|----------|-----|
| Element From Node                    | 20       |     |
| Detail Type                          | Weight   |     |
| Detail ID                            | MH Davit |     |
| Dist. from "FROM" Node / Offset dist | 2350     | mm. |
| Miscellaneous Weight                 | 25       | Kgf |
| Offset from Element Centerline       | 1860     | mm. |

|                                      |                 |     |
|--------------------------------------|-----------------|-----|
| Element From Node                    | 20              |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | Insert Plate-N1 |     |
| Dist. from "FROM" Node / Offset dist | 850             | mm. |



Miscellaneous Weight 55 Kgf  
 Offset from Element Centerline 1500 mm.

Element# 3/5  
 Element From Node 30  
 Element To Node 40  
 Element Type Cylinder  
 Description SHELL-2  
 Distance "FROM" to "TO" 3000 mm.  
 Inside Diameter 3000 mm.  
 Element Thickness 10 mm.  
 Internal Corrosion Allowance 3 mm.  
 Nominal Thickness 10 mm.  
 External Corrosion Allowance 0 mm.  
 Design Internal Pressure 4.4997 bars  
 Design Temperature Internal Pressure 80 °C  
 Design External Pressure 0 bars  
 Design Temperature External Pressure 0 °C  
 Effective Diameter Multiplier 1.2  
 Material Name SA-516 70  
 Efficiency, Longitudinal Seam 0.85  
 Efficiency, Circumferential Seam 0.85  
 Weld is pre-Heated No

Element From Node 30  
 Detail Type Liquid  
 Detail ID Liquid: 30  
 Dist. from "FROM" Node / Offset dist 0 mm.  
 Height/Length of Liquid 1500 mm.  
 Liquid Density 0.0006809 kg./cm3

Element From Node 30  
 Detail Type Insulation  
 Detail ID Ins: 30  
 Dist. from "FROM" Node / Offset dist 0 mm.  
 Height/Length of Insulation 3000 mm.  
 Thickness of Insulation 80 mm.  
 Density 0.00032 kg./cm3

Element From Node 30  
 Detail Type Nozzle  
 Detail ID N2-10 Inch  
 Dist. from "FROM" Node / Offset dist 450 mm.  
 Nozzle Diameter 10 in.  
 Nozzle Schedule XS  
 Nozzle Class 150  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 75.321 Kgf  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-105 +IT

Element From Node 30  
 Detail Type Nozzle  
 Detail ID K3-2 Inch  
 Dist. from "FROM" Node / Offset dist 1150 mm.  
 Nozzle Diameter 2 in.  
 Nozzle Schedule None  
 Nozzle Class 300  
 Layout Angle 0.0  
 Blind Flange (Y/N) N  
 Weight of Nozzle ( Used if > 0 ) 13.916 Kgf  
 Grade of Attached Flange GR 1.1  
 Nozzle Matl SA-105 +IT

Element From Node 30  
 Detail Type Nozzle  
 Detail ID N4-4 Inch



|                                      |          |     |
|--------------------------------------|----------|-----|
| Dist. from "FROM" Node / Offset dist | 1500     | mm. |
| Nozzle Diameter                      | 4        | in. |
| Nozzle Schedule                      | 120      |     |
| Nozzle Class                         | 150      |     |
| Layout Angle                         | 180.0    |     |
| Blind Flange (Y/N)                   | N        |     |
| Weight of Nozzle ( Used if > 0 )     | 21.273   | Kgf |
| Grade of Attached Flange             | GR 1.1   |     |
| Nozzle Matl                          | SA-106 B |     |

|                                      |             |     |
|--------------------------------------|-------------|-----|
| Element From Node                    | 30          |     |
| Detail Type                          | Weight      |     |
| Detail ID                            | Clip Weight |     |
| Dist. from "FROM" Node / Offset dist | 1500        | mm. |
| Miscellaneous Weight                 | 190         | Kgf |
| Offset from Element Centerline       | 0           | mm. |

|                                      |                 |     |
|--------------------------------------|-----------------|-----|
| Element From Node                    | 30              |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | Insert Plate-N2 |     |
| Dist. from "FROM" Node / Offset dist | 450             | mm. |
| Miscellaneous Weight                 | 95.002          | Kgf |
| Offset from Element Centerline       | 1500            | mm. |

|                                      |                 |     |
|--------------------------------------|-----------------|-----|
| Element From Node                    | 30              |     |
| Detail Type                          | Weight          |     |
| Detail ID                            | Insert Plate-N4 |     |
| Dist. from "FROM" Node / Offset dist | 1500            | mm. |
| Miscellaneous Weight                 | 20              | Kgf |
| Offset from Element Centerline       | 1500            | mm. |

|                                      |           |      |
|--------------------------------------|-----------|------|
| Element#                             | 4/5       |      |
| Element From Node                    | 40        |      |
| Element To Node                      | 50        |      |
| Element Type                         | Cylinder  |      |
| Description                          | SHELL-3   |      |
| Distance "FROM" to "TO"              | 2900      | mm.  |
| Inside Diameter                      | 3000      | mm.  |
| Element Thickness                    | 10        | mm.  |
| Internal Corrosion Allowance         | 3         | mm.  |
| Nominal Thickness                    | 10        | mm.  |
| External Corrosion Allowance         | 0         | mm.  |
| Design Internal Pressure             | 4.4997    | bars |
| Design Temperature Internal Pressure | 80        | °C   |
| Design External Pressure             | 0         | bars |
| Design Temperature External Pressure | 0         | °C   |
| Effective Diameter Multiplier        | 1.2       |      |
| Material Name                        | SA-516 70 |      |
| Efficiency, Longitudinal Seam        | 0.85      |      |
| Efficiency, Circumferential Seam     | 0.85      |      |
| Weld is pre-Heated                   | No        |      |

|                                      |            |     |
|--------------------------------------|------------|-----|
| Element From Node                    | 40         |     |
| Detail Type                          | Saddle     |     |
| Detail ID                            | New Saddle |     |
| Dist. from "FROM" Node / Offset dist | 1150       | mm. |
| Width of Saddle                      | 280        | mm. |
| Height of Saddle at Bottom           | 2133       | mm. |
| Saddle Contact Angle                 | 120.0      |     |
| Height of Composite Ring Stiffener   | 0          | mm. |
| Width of Wear Plate                  | 475        | mm. |
| Thickness of Wear Plate              | 10         | mm. |
| Contact Angle, Wear Plate (degrees)  | 132.0      |     |
| Friction coefficient                 | 0.40000001 |     |
| Moment Factor                        | 3.0        |     |
| Dimension E at base (optional)       | 0          | mm. |
| Circumferential Eff. over Saddle     | 1.0        |     |
| Circumferential Eff. at Midspan      | 1.0        |     |



|                                      |                 |         |
|--------------------------------------|-----------------|---------|
| Tangent to Tangent dist. (optional)  | 0               | mm.     |
| Element From Node                    | 40              |         |
| Detail Type                          | Liquid          |         |
| Detail ID                            | Liquid 40       |         |
| Dist. from "FROM" Node / Offset dist | 0               | mm.     |
| Height/Length of Liquid              | 1500            | mm.     |
| Liquid Density                       | 0.0006809       | kg./cm3 |
| Element From Node                    | 40              |         |
| Detail Type                          | Insulation      |         |
| Detail ID                            | Ins: 40         |         |
| Dist. from "FROM" Node / Offset dist | 0               | mm.     |
| Height/Length of Insulation          | 2900            | mm.     |
| Thickness of Insulation              | 80              | mm.     |
| Density                              | 0.00032         | kg./cm3 |
| Element From Node                    | 40              |         |
| Detail Type                          | Nozzle          |         |
| Detail ID                            | K1-6 Inch       |         |
| Dist. from "FROM" Node / Offset dist | 650             | mm.     |
| Nozzle Diameter                      | 6               | in.     |
| Nozzle Schedule                      | 80              |         |
| Nozzle Class                         | 150             |         |
| Layout Angle                         | 0.0             |         |
| Blind Flange (Y/N)                   | Y               |         |
| Weight of Nozzle ( Used if > 0 )     | 42.957          | Kgf     |
| Grade of Attached Flange             | GR 1.1          |         |
| Nozzle Matl                          | SA-105 +IT      |         |
| Element From Node                    | 40              |         |
| Detail Type                          | Nozzle          |         |
| Detail ID                            | N3-10 Inch      |         |
| Dist. from "FROM" Node / Offset dist | 1500            | mm.     |
| Nozzle Diameter                      | 10              | in.     |
| Nozzle Schedule                      | XS              |         |
| Nozzle Class                         | 150             |         |
| Layout Angle                         | 0.0             |         |
| Blind Flange (Y/N)                   | N               |         |
| Weight of Nozzle ( Used if > 0 )     | 75.321          | Kgf     |
| Grade of Attached Flange             | GR 1.1          |         |
| Nozzle Matl                          | SA-105 +IT      |         |
| Element From Node                    | 40              |         |
| Detail Type                          | Nozzle          |         |
| Detail ID                            | V1-2 Inch       |         |
| Dist. from "FROM" Node / Offset dist | 2400            | mm.     |
| Nozzle Diameter                      | 2               | in.     |
| Nozzle Schedule                      | None            |         |
| Nozzle Class                         | 300             |         |
| Layout Angle                         | 0.0             |         |
| Blind Flange (Y/N)                   | Y               |         |
| Weight of Nozzle ( Used if > 0 )     | 17.625          | Kgf     |
| Grade of Attached Flange             | GR 1.1          |         |
| Nozzle Matl                          | SA-105 +IT      |         |
| Element From Node                    | 40              |         |
| Detail Type                          | Weight          |         |
| Detail ID                            | GWR Level Trans |         |
| Dist. from "FROM" Node / Offset dist | 650             | mm.     |
| Miscellaneous Weight                 | 7               | Kgf     |
| Offset from Element Centerline       | 1500            | mm.     |
| Element From Node                    | 40              |         |
| Detail Type                          | Weight          |         |
| Detail ID                            | Insert Plate-N3 |         |
| Dist. from "FROM" Node / Offset dist | 1500            | mm.     |
| Miscellaneous Weight                 | 95.002          | Kgf     |
| Offset from Element Centerline       | 1500            | mm.     |



|                                      |            |         |
|--------------------------------------|------------|---------|
| Element#                             | 5/5        |         |
| Element From Node                    | 50         |         |
| Element To Node                      | 60         |         |
| Element Type                         | Elliptical |         |
| Description                          | TOP HEAD   |         |
| Distance "FROM" to "TO"              | 50         | mm.     |
| Inside Diameter                      | 3000       | mm.     |
| Element Thickness                    | 10         | mm.     |
| Internal Corrosion Allowance         | 3          | mm.     |
| Nominal Thickness                    | 12         | mm.     |
| External Corrosion Allowance         | 0          | mm.     |
| Design Internal Pressure             | 4.4997     | bars    |
| Design Temperature Internal Pressure | 80         | °C      |
| Design External Pressure             | 0          | bars    |
| Design Temperature External Pressure | 0          | °C      |
| Effective Diameter Multiplier        | 1.2        |         |
| Material Name                        | SA-516 70  |         |
| Efficiency, Longitudinal Seam        | 0.85       |         |
| Efficiency, Circumferential Seam     | 0.85       |         |
| Elliptical Head Factor               | 2.0        |         |
| Weld is pre-Heated                   | No         |         |
|                                      |            |         |
| Element From Node                    | 50         |         |
| Detail Type                          | Liquid     |         |
| Detail ID                            | Liquid 50  |         |
| Dist. from "FROM" Node / Offset dist | 0          | mm.     |
| Height/Length of Liquid              | 1500       | mm.     |
| Liquid Density                       | 0.0006809  | kg./cm3 |
|                                      |            |         |
| Element From Node                    | 50         |         |
| Detail Type                          | Insulation |         |
| Detail ID                            | Ins: 50    |         |
| Dist. from "FROM" Node / Offset dist | 0          | mm.     |
| Height/Length of Insulation          | 800        | mm.     |
| Thickness of Insulation              | 80         | mm.     |
| Density                              | 0.00032    | kg./cm3 |



**XY Coordinate Calculations:**

| From        | To | X (Horiz.)<br>mm. | Y (Vert.)<br>mm. | DX (Horiz.)<br>mm. | DY (Vert.)<br>mm. |
|-------------|----|-------------------|------------------|--------------------|-------------------|
| BOTTOM HEAD |    | 50                | ...              | 50                 | ...               |
| SHELL-1     |    | 3050              | ...              | 3000               | ...               |
| SHELL-2     |    | 6050              | ...              | 3000               | ...               |
| SHELL-3     |    | 8950              | ...              | 2900               | ...               |
| TOP HEAD    |    | 9000              | ...              | 50                 | ...               |

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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### Internal Pressure Results Summary:

#### Element Thickness, Pressure, Diameter and Allowable Stress :

| From        | To | Int. Press<br>+ Liq. Hd<br>bars | Nominal<br>Thickness<br>mm. | Total Corr<br>Allowance<br>mm. | Element<br>Diameter<br>mm. | Allowable<br>Stress(SE)<br>N./mm <sup>2</sup> | Element# |
|-------------|----|---------------------------------|-----------------------------|--------------------------------|----------------------------|-----------------------------------------------|----------|
| BOTTOM HEAD |    | 4.6002                          | 12                          | 3                              | 3000                       | 117.3                                         | 1        |
| SHELL-1     |    | 4.6002                          | 10                          | 3                              | 3000                       | 117.3                                         | 2        |
| SHELL-2     |    | 4.6002                          | 10                          | 3                              | 3000                       | 117.3                                         | 3        |
| SHELL-3     |    | 4.6002                          | 10                          | 3                              | 3000                       | 117.3                                         | 4        |
| TOP HEAD    |    | 4.6002                          | 12                          | 3                              | 3000                       | 117.3                                         | 5        |

#### Element Required Thickness and MAWP :

| From        | To | Design<br>Pressure<br>bars | M.A.W.P.<br>Corroded<br>bars | M.A.P.<br>New & Cold<br>bars | Minimum<br>Thickness<br>mm. | Required<br>Thickness<br>mm. | Element# |
|-------------|----|----------------------------|------------------------------|------------------------------|-----------------------------|------------------------------|----------|
| BOTTOM HEAD |    | 4.49974                    | 5.37426                      | 7.81434                      | 10                          | 8.8814                       | 1        |
| SHELL-1     |    | 4.49974                    | 5.34706                      | 7.78839                      | 10                          | 8.90864                      | 2        |
| SHELL-2     |    | 4.49974                    | 5.34706                      | 7.78839                      | 10                          | 8.90864                      | 3        |
| SHELL-3     |    | 4.49974                    | 5.34706                      | 7.78839                      | 10                          | 8.90864                      | 4        |
| TOP HEAD    |    | 4.49974                    | 5.37426                      | 7.81434                      | 10                          | 8.8814                       | 5        |

Minimum 5 8

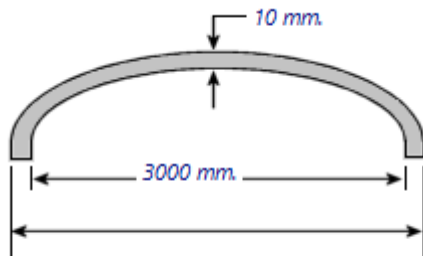
MAWP: 5 bars, limited by: SHELL-3.

Elements Suitable for Design Internal Pressure.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: BOTTOM HEAD

| Elliptical Head Design Information: |    |               |                         |    |               |
|-------------------------------------|----|---------------|-------------------------|----|---------------|
| Design Pressure                     | P  | 4.600 bars    | Int. Design Temperature |    | 80 °C         |
| Material                            |    | SA-516 70     | Ext. Design Temperature |    | 0 °C          |
| Length                              |    | 50.000 mm.    | External Pressure Chart |    | CS-2          |
|                                     |    |               | UNS Number              |    | K02700        |
| Int. Corr. All.                     | c  | 3.0000 mm.    | Allowable Stress (ope)  | S  | 138.0 N./mm^2 |
| Ext. Corr. All.                     | ce | 0.0000 mm.    | Allowable Stress (amb)  | Sa | 138.0 N./mm^2 |
| Inside Diameter                     | Di | 3000.000 mm.  | Actual Stress           |    | 116.0 N./mm^2 |
| Spec. Min. Thk t,e, or sa           |    | 10.0000 mm.   | Spec. Nominal Thk.      |    | 12.000 mm.    |
| Surface Area                        |    | 103876.0 cm^2 | SG of Contents          |    | 0.682         |
| Element Volume                      |    | 3888417.8 cm3 | Weight of Contents      |    | 1325.1 Kg     |
| Empty Weight                        |    | 1053.8 Kg     | Operating Weight        |    | 2378.9 Kg     |
|                                     |    |               | Comp. All. Ope. Stress  |    | 69.5 N./mm^2  |
|                                     |    |               | Aspect Ratio            |    | 2.00          |

| Radiography Information: |      |                  |             |
|--------------------------|------|------------------|-------------|
| User Defined             |      | User Defined     |             |
| Circ. Joint Efficiency   | 85 % | Long. Joint Eff. | E or v 85 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm.

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.6 \cdot 3006.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 4.6)$$

$$= 5.8814 + 3.0000 = 8.8814 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.100 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 7.0) / (0.997 \cdot 3006.0 + 0.2 \cdot 7.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)$$

$$= 8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.6 \cdot (0.997 \cdot 3006.0 + 0.2 \cdot 7.0)) / (2 \cdot 0.85 \cdot 7.0)$$

$$= 115.956 \text{ N./mm}^2$$

Straight Flange Required Thickness:  
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$  per UG-27 (c)(1)  
 $= (4.6 \cdot 1503.0) / (138.0 \cdot 0.85 - 0.6 \cdot 4.6) + 3.0$   
 $= 8.909 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:  
 Less Operating Hydrostatic Head Pressure of 0.100 bars  
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$  per UG-27 (c)(1)  
 $= (138.0 \cdot 0.85 \cdot 9.0) / (1503.0 + 0.6 \cdot 9.0)$   
 $= 7 - 0 = 7 \text{ bars}$

Factor K, corroded condition [Kcor]:  
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$   
 $= (2 + (3006.0 / (2 \cdot 753.0))^2) / 6$   
 $= 0.997347$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  1.744 %

#### MDMT Calculations in the Knuckle Portion:

Govrn. thk,  $t_g = 10.0$ ,  $t_r = 6.965$ ,  $c = 3.0 \text{ mm.}$ ,  $E^* = 0.85$   
 Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.846$ , Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

#### MDMT Calculations in the Head Straight Flange:

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 7.0$ ,  $c = 3.0 \text{ mm.}$ ,  $E^* = 0.85$   
 Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.661$ , Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Note:  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

| MDMT Design Information: |            |                        |            |
|--------------------------|------------|------------------------|------------|
| MDMT Curve               | D          | Press. at MDMT         | 5.448 bars |
| UCS-66(b) red.           | No         | Min. Des. Mtl. Temp.   | -40 °C     |
| Governing Thk.           | 12.000 mm. | Basic MDMT             | -48 °C     |
| Stress/Thk. Ratio        | 0.661      | Temperature Difference | 0 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C     |
|                          |            | MDMT per UG-20(f)      | -29 °C     |
|                          |            | Joint Efficiency $E^*$ | 85 %       |

Note: Temperature values displayed above are for the location with the highest basic MDMT

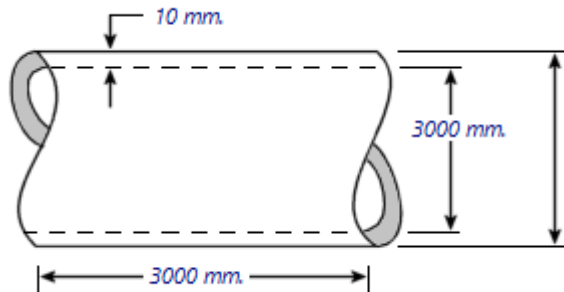
#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm.

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: SHELL-1

| Cylindrical Shell Design Information:             |    |                |                         |    |               |
|---------------------------------------------------|----|----------------|-------------------------|----|---------------|
| Design Pressure                                   | P  | 4.600 bars     | Int. Design Temperature |    | 80 °C         |
| Material                                          |    | SA-516 70      | Ext. Design Temperature |    | 0 °C          |
| Length                                            |    | 3000.000 mm.   | External Pressure Chart |    | CS-2          |
|                                                   |    |                | UNS Number              |    | K02700        |
| Int. Corr. All.                                   | c  | 3.0000 mm.     | Allowable Stress (ope)  | S  | 138.0 N./mm^2 |
| Ext. Corr. All.                                   | ce | 0.0000 mm.     | Allowable Stress (amb)  | Sa | 138.0 N./mm^2 |
| Inside Diameter                                   | Di | 3000.000 mm.   | Actual Stress           |    | 116.5 N./mm^2 |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 10.0000 mm.    | Spec. Nominal Thk.      |    | 10.000 mm.    |
| Surface Area                                      |    | 284628.3 cm^2  | SG of Contents          |    | 0.682         |
| Element Volume                                    |    | 21209550.0 cm3 | Weight of Contents      |    | 7228.0 Kg     |
| Empty Weight                                      |    | 2198.6 Kg      | Operating Weight        |    | 9426.5 Kg     |
|                                                   |    |                | Comp. All. Ope. Stress  |    | 69.5 N./mm^2  |

| Radiography Information: |      |                  |             |
|--------------------------|------|------------------|-------------|
| User Defined             |      | User Defined     |             |
| Circ. Joint Efficiency   | 85 % | Long. Joint Eff. | E or v 85 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.6 \cdot 1503.0) / (138.0 \cdot 0.85 - 0.6 \cdot 4.6)$$

$$= 5.9086 + 3.0000 = 8.9086 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.100 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 7.0) / (1503.0 + 0.6 \cdot 7.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4.6 \cdot (1503.0 + 0.6 \cdot 7.0)) / (0.85 \cdot 7.0)$$

$$= 116.535 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.332 %

#### Minimum Design Metal Temperature Results:

Govrn. thk, t<sub>g</sub> = 10.0, t<sub>r</sub> = 7.0, c = 3.0 mm., E\* = 0.85



Thickness Ratio =  $tr * E^* / (tg - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D    -48 °C

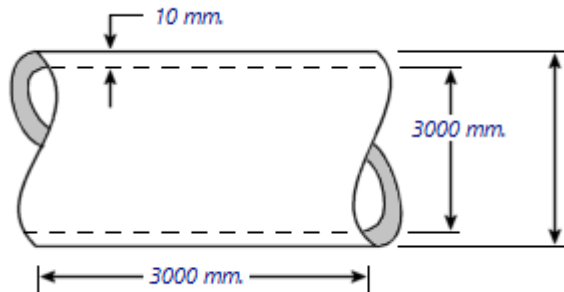
**Note:**  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

| MDMT Design Information: |            |                        |            |
|--------------------------|------------|------------------------|------------|
| MDMT Curve               | D          | Press. at MDMT         | 5.448 bars |
| UCS-66(b) red.           | No         | Min. Des. Mtl. Temp.   | -40 °C     |
| Governing Thk.           | 10.000 mm. | Basic MDMT             | -48 °C     |
| Stress/Thk. Ratio        | 0.850      | Temperature Difference | 0 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C     |
|                          |            | MDMT per UG-20(f)      | -29 °C     |
|                          |            | Joint Efficiency E*    | 85 %       |

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



### Internal Pressure Results for: SHELL-2

| Cylindrical Shell Design Information:             |    |                |                         |    |               |
|---------------------------------------------------|----|----------------|-------------------------|----|---------------|
| Design Pressure                                   | P  | 4.600 bars     | Int. Design Temperature |    | 80 °C         |
| Material                                          |    | SA-516 70      | Ext. Design Temperature |    | 0 °C          |
| Length                                            |    | 3000.000 mm.   | External Pressure Chart |    | CS-2          |
|                                                   |    |                | UNS Number              |    | K02700        |
| Int. Corr. All.                                   | c  | 3.0000 mm.     | Allowable Stress (ope)  | S  | 138.0 N./mm^2 |
| Ext. Corr. All.                                   | ce | 0.0000 mm.     | Allowable Stress (amb)  | Sa | 138.0 N./mm^2 |
| Inside Diameter                                   | Di | 3000.000 mm.   | Actual Stress           |    | 116.5 N./mm^2 |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 10.0000 mm.    | Spec. Nominal Thk.      |    | 10.000 mm.    |
| Surface Area                                      |    | 284628.3 cm^2  | SG of Contents          |    | 0.682         |
| Element Volume                                    |    | 21209550.0 cm3 | Weight of Contents      |    | 7228.0 Kg     |
| Empty Weight                                      |    | 2198.6 Kg      | Operating Weight        |    | 9426.5 Kg     |
|                                                   |    |                | Comp. All. Ope. Stress  |    | 69.5 N./mm^2  |

| Radiography Information: |      |                  |             |
|--------------------------|------|------------------|-------------|
| User Defined             |      | User Defined     |             |
| Circ. Joint Efficiency   | 85 % | Long. Joint Eff. | E or v 85 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.6 \cdot 1503.0) / (138.0 \cdot 0.85 - 0.6 \cdot 4.6)$$

$$= 5.9086 + 3.0000 = 8.9086 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.100 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 7.0) / (1503.0 + 0.6 \cdot 7.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4.6 \cdot (1503.0 + 0.6 \cdot 7.0)) / (0.85 \cdot 7.0)$$

$$= 116.535 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.332 %

### Minimum Design Metal Temperature Results:

Govern. thk, t<sub>g</sub> = 10.0, t<sub>r</sub> = 7.0, c = 3.0 mm., E\* = 0.85



Thickness Ratio =  $tr * E^* / (tg - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D    -48 °C

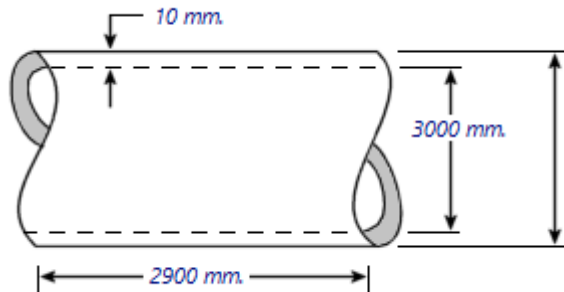
**Note:**  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

| MDMT Design Information: |            |                        |            |
|--------------------------|------------|------------------------|------------|
| MDMT Curve               | D          | Press. at MDMT         | 5.448 bars |
| UCS-66(b) red.           | No         | Min. Des. Mtl. Temp.   | -40 °C     |
| Governing Thk.           | 10.000 mm. | Basic MDMT             | -48 °C     |
| Stress/Thk. Ratio        | 0.850      | Temperature Difference | 0 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C     |
|                          |            | MDMT per UG-20(f)      | -29 °C     |
|                          |            | Joint Efficiency E*    | 85 %       |

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



### Internal Pressure Results for: SHELL-3

| Cylindrical Shell Design Information:             |    |                |                         |    |               |
|---------------------------------------------------|----|----------------|-------------------------|----|---------------|
| Design Pressure                                   | P  | 4.600 bars     | Int. Design Temperature |    | 80 °C         |
| Material                                          |    | SA-516 70      | Ext. Design Temperature |    | 0 °C          |
| Length                                            |    | 2900.000 mm.   | External Pressure Chart |    | CS-2          |
|                                                   |    |                | UNS Number              |    | K02700        |
| Int. Corr. All.                                   | c  | 3.0000 mm.     | Allowable Stress (ope)  | S  | 138.0 N./mm^2 |
| Ext. Corr. All.                                   | ce | 0.0000 mm.     | Allowable Stress (amb)  | Sa | 138.0 N./mm^2 |
| Inside Diameter                                   | Di | 3000.000 mm.   | Actual Stress           |    | 116.5 N./mm^2 |
| Spec. Min. Thk t <sub>e</sub> , or s <sub>a</sub> |    | 10.0000 mm.    | Spec. Nominal Thk.      |    | 10.000 mm.    |
| Surface Area                                      |    | 275140.7 cm^2  | SG of Contents          |    | 0.682         |
| Element Volume                                    |    | 20502566.0 cm3 | Weight of Contents      |    | 6987.0 Kg     |
| Empty Weight                                      |    | 2125.3 Kg      | Operating Weight        |    | 9112.3 Kg     |
|                                                   |    |                | Comp. All. Ope. Stress  |    | 69.5 N./mm^2  |

| Radiography Information: |      |                  |             |
|--------------------------|------|------------------|-------------|
| User Defined             |      | User Defined     |             |
| Circ. Joint Efficiency   | 85 % | Long. Joint Eff. | E or v 85 % |

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (4.6 \cdot 1503.0) / (138.0 \cdot 0.85 - 0.6 \cdot 4.6)$$

$$= 5.9086 + 3.0000 = 8.9086 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.100 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 7.0) / (1503.0 + 0.6 \cdot 7.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 10.0) / (1500.0 + 0.6 \cdot 10.0)$$

$$= 8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (4.6 \cdot (1503.0 + 0.6 \cdot 7.0)) / (0.85 \cdot 7.0)$$

$$= 116.535 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 \* t<sub>nom</sub> / R<sub>f</sub> \* (1 - R<sub>f</sub> / R<sub>o</sub>)) 0.332 %

### Minimum Design Metal Temperature Results:

Govern. thk, t<sub>g</sub> = 10.0, t<sub>r</sub> = 7.0, c = 3.0 mm., E\* = 0.85



Thickness Ratio =  $tr * E^* / (tg - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

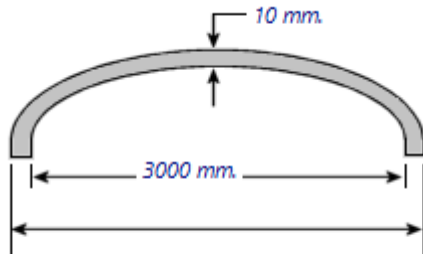
Note:  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

| MDMT Design Information: |            |                        |            |
|--------------------------|------------|------------------------|------------|
| MDMT Curve               | D          | Press. at MDMT         | 5.448 bars |
| UCS-66(b) red.           | No         | Min. Des. Mtl. Temp.   | -40 °C     |
| Governing Thk.           | 10.000 mm. | Basic MDMT             | -48 °C     |
| Stress/Thk. Ratio        | 0.850      | Temperature Difference | 0 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C     |
|                          |            | MDMT per UG-20(f)      | -29 °C     |
|                          |            | Joint Efficiency E*    | 85 %       |

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### Internal Pressure Calculation Results:

#### ASME Code, Section VIII Division 1, 2025



#### Internal Pressure Results for: TOP HEAD

| Elliptical Head Design Information: |    |               |                         |    |               |
|-------------------------------------|----|---------------|-------------------------|----|---------------|
| Design Pressure                     | P  | 4.600 bars    | Int. Design Temperature |    | 80 °C         |
| Material                            |    | SA-516 70     | Ext. Design Temperature |    | 0 °C          |
| Length                              |    | 50.000 mm.    | External Pressure Chart |    | CS-2          |
|                                     |    |               | UNS Number              |    | K02700        |
| Int. Corr. All.                     | c  | 3.0000 mm.    | Allowable Stress (ope)  | S  | 138.0 N./mm^2 |
| Ext. Corr. All.                     | ce | 0.0000 mm.    | Allowable Stress (amb)  | Sa | 138.0 N./mm^2 |
| Inside Diameter                     | Di | 3000.000 mm.  | Actual Stress           |    | 116.0 N./mm^2 |
| Spec. Min. Thk t,e, or sa           |    | 10.0000 mm.   | Spec. Nominal Thk.      |    | 12.000 mm.    |
| Surface Area                        |    | 103876.0 cm^2 | SG of Contents          |    | 0.682         |
| Element Volume                      |    | 3888417.8 cm3 | Weight of Contents      |    | 1325.1 Kg     |
| Empty Weight                        |    | 1053.8 Kg     | Operating Weight        |    | 2378.9 Kg     |
|                                     |    |               | Comp. All. Ope. Stress  |    | 69.5 N./mm^2  |
|                                     |    |               | Aspect Ratio            |    | 2.00          |

| Radiography Information: |      |                  |             |
|--------------------------|------|------------------|-------------|
| User Defined             |      | User Defined     |             |
| Circ. Joint Efficiency   | 85 % | Long. Joint Eff. | E or v 85 % |

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (4.6 \cdot 3006.0 \cdot 0.997) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 4.6)$$

$$= 5.8814 + 3.0000 = 8.8814 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.100 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 7.0) / (0.997 \cdot 3006.0 + 0.2 \cdot 7.0)$$

$$= 5 - 0 = 5 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 10.0) / (1.0 \cdot 3000.0 + 0.2 \cdot 10.0)$$

$$= 8 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (4.6 \cdot (0.997 \cdot 3006.0 + 0.2 \cdot 7.0)) / (2 \cdot 0.85 \cdot 7.0)$$

$$= 115.956 \text{ N./mm}^2$$



Straight Flange Required Thickness:  
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$  per UG-27 (c)(1)  
 $= (4.6 \cdot 1503.0) / (138.0 \cdot 0.85 - 0.6 \cdot 4.6) + 3.0$   
 $= 8.909 \text{ mm.}$

Straight Flange Maximum Allowable Working Pressure:  
 Less Operating Hydrostatic Head Pressure of 0.100 bars  
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$  per UG-27 (c)(1)  
 $= (138.0 \cdot 0.85 \cdot 9.0) / (1503.0 + 0.6 \cdot 9.0)$   
 $= 7 - 0 = 7 \text{ bars}$

Factor K, corroded condition [Kcor]:  
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$   
 $= (2 + (3006.0 / (2 \cdot 753.0))^2) / 6$   
 $= 0.997347$

Percent Elong. per UCS-79, VIII-1-01-57  $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$  1.744 %

#### MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 10.0, tr = 6.965, c = 3.0 mm., E\* = 0.85  
 Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.846$ , Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

#### MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12.0, tr = 7.0, c = 3.0 mm., E\* = 0.85  
 Thickness Ratio =  $tr \cdot E* / (tg - c) = 0.661$ , Temp. Reduction = 19 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Note:  
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

| MDMT Design Information: |            |                        |            |
|--------------------------|------------|------------------------|------------|
| MDMT Curve               | D          | Press. at MDMT         | 5.448 bars |
| UCS-66(b) red.           | No         | Min. Des. Mtl. Temp.   | -40 °C     |
| Governing Thk.           | 12.000 mm. | Basic MDMT             | -48 °C     |
| Stress/Thk. Ratio        | 0.661      | Temperature Difference | 0 °C       |
| UCS-66(c) red.           | No         | Computed Min. Temp.    | -48 °C     |
|                          |            | MDMT per UG-20(f)      | -29 °C     |
|                          |            | Joint Efficiency E*    | 85 %       |

Note: Temperature values displayed above are for the location with the highest basic MDMT

#### Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 37.500 mm.  
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 18.750 mm.

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#### Hydrostatic Test Pressure Results:

|                 |                               |            |
|-----------------|-------------------------------|------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 6.951 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 5.850 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 9.733 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 5.882 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 6.435 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 6.435 bars |
| App 27-4        | = M.A.W.P.                    | 5.347 bars |

#### Horizontal Test performed per: UG-99c

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

#### Stresses on Elements due to Test Pressure (N./mm<sup>2</sup> & bars):

| From To     | Stress | Allowable | Ratio | Pressure |
|-------------|--------|-----------|-------|----------|
| BOTTOM HEAD | 177.1  | 234.0     | 0.757 | 10.03    |
| SHELL-1     | 177.7  | 234.0     | 0.759 | 10.03    |
| SHELL-2     | 177.7  | 234.0     | 0.759 | 10.03    |
| SHELL-3     | 177.7  | 234.0     | 0.759 | 10.03    |
| TOP HEAD    | 177.1  | 234.0     | 0.757 | 10.03    |

#### Stress ratios for Nozzle and Pad Materials (N./mm<sup>2</sup>):

| Description | Pad/Nozzle | Ambient | Operating | Ratio |
|-------------|------------|---------|-----------|-------|
| K6-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N1-6 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| K4-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| K5-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| M1-24 Inch  | Nozzle     | 138.00  | 138.00    | 1.000 |
| D1-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N2-10 Inch  | Nozzle     | 138.00  | 138.00    | 1.000 |
| K3-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N4-4 Inch   | Nozzle     | 118.00  | 118.00    | 1.000 |
| K1-6 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N3-10 Inch  | Nozzle     | 137.98  | 137.98    | 1.000 |
| V1-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Stress ratios for Pressurized Vessel Elements (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| BOTTOM HEAD | 138.00  | 138.00    | 1.000 |
| SHELL-1     | 138.00  | 138.00    | 1.000 |
| SHELL-2     | 138.00  | 138.00    | 1.000 |
| SHELL-3     | 138.00  | 138.00    | 1.000 |
| TOP HEAD    | 138.00  | 138.00    | 1.000 |

Minimum 1.000

#### Hoop Stress in Nozzle Wall during Pressure Test (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| K6-2 Inch   | 2.12    | 225.00    | 0.009 |
| N1-6 Inch   | 5.97    | 225.00    | 0.027 |
| K4-2 Inch   | 2.12    | 225.00    | 0.009 |
| K5-2 Inch   | 2.12    | 225.00    | 0.009 |
| M1-24 Inch  | 26.07   | 225.00    | 0.116 |
| D1-2 Inch   | 2.12    | 225.00    | 0.009 |
| N2-10 Inch  | 10.21   | 225.00    | 0.045 |



|            |       |        |       |
|------------|-------|--------|-------|
| K3-2 Inch  | 2.12  | 225.00 | 0.009 |
| N4-4 Inch  | 4.69  | 216.00 | 0.022 |
| K1-6 Inch  | 5.97  | 225.00 | 0.027 |
| N3-10 Inch | 10.21 | 224.97 | 0.045 |
| V1-2 Inch  | 2.12  | 225.00 | 0.009 |

Elements Suitable for Test Pressure.

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### Hydrostatic Test Pressure Results:

|                 |                               |            |
|-----------------|-------------------------------|------------|
| UG99(b)         | = 1.30 * M.A.W.P. * Sa/S      | 6.951 bars |
| UG99b(b)(35)    | = 1.30 * Design Pres * Sa/S   | 5.850 bars |
| UG99(c)         | = 1.30 * M.A.P. - Head(Hyd)   | 9.733 bars |
| UG100           | = 1.10 * M.A.W.P. * Sa/S      | 5.882 bars |
| PED             | = max(1.43*DP, 1.25*DP*ratio) | 6.435 bars |
| PED&UG99(b)(35) | = max(1.43*DP, 1.3*DP*ratio)  | 6.435 bars |
| App 27-4        | = M.A.W.P.                    | 5.347 bars |

UG-99(b), Test Pressure Calculation:  
 = Test Factor • MAWP • Stress Ratio  
 = 1.3 • 5.347 • 1.0  
 = 6.951 bars

### Horizontal Test performed per: UG-99b

*Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.*

### Stresses on Elements due to Test Pressure (N./mm<sup>2</sup> & bars):

| From To     | Stress | Allowable | Ratio | Pressure |
|-------------|--------|-----------|-------|----------|
| BOTTOM HEAD | 183.9  | 234.0     | 0.786 | 7.29     |
| SHELL-1     | 184.8  | 234.0     | 0.790 | 7.29     |
| SHELL-2     | 184.8  | 234.0     | 0.790 | 7.29     |
| SHELL-3     | 184.8  | 234.0     | 0.790 | 7.29     |
| TOP HEAD    | 183.9  | 234.0     | 0.786 | 7.29     |

### Stress ratios for Nozzle and Pad Materials (N./mm<sup>2</sup>):

| Description | Pad/Nozzle | Ambient | Operating | Ratio |
|-------------|------------|---------|-----------|-------|
| K6-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N1-6 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| K4-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| K5-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| M1-24 Inch  | Nozzle     | 138.00  | 138.00    | 1.000 |
| D1-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N2-10 Inch  | Nozzle     | 138.00  | 138.00    | 1.000 |
| K3-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N4-4 Inch   | Nozzle     | 118.00  | 118.00    | 1.000 |
| K1-6 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |
| N3-10 Inch  | Nozzle     | 137.98  | 137.98    | 1.000 |
| V1-2 Inch   | Nozzle     | 138.00  | 138.00    | 1.000 |

Minimum 1.000

### Stress ratios for Pressurized Vessel Elements (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| BOTTOM HEAD | 138.00  | 138.00    | 1.000 |
| SHELL-1     | 138.00  | 138.00    | 1.000 |
| SHELL-2     | 138.00  | 138.00    | 1.000 |
| SHELL-3     | 138.00  | 138.00    | 1.000 |
| TOP HEAD    | 138.00  | 138.00    | 1.000 |

Minimum 1.000

### Hoop Stress in Nozzle Wall during Pressure Test (N./mm<sup>2</sup>):

| Description | Ambient | Operating | Ratio |
|-------------|---------|-----------|-------|
| K6-2 Inch   | 1.54    | 225.00    | 0.007 |
| N1-6 Inch   | 4.34    | 225.00    | 0.019 |



|            |       |        |       |
|------------|-------|--------|-------|
| K4-2 Inch  | 1.54  | 225.00 | 0.007 |
| K5-2 Inch  | 1.54  | 225.00 | 0.007 |
| M1-24 Inch | 18.97 | 225.00 | 0.084 |
| D1-2 Inch  | 1.54  | 225.00 | 0.007 |
| N2-10 Inch | 7.43  | 225.00 | 0.033 |
| K3-2 Inch  | 1.54  | 225.00 | 0.007 |
| N4-4 Inch  | 3.41  | 216.00 | 0.016 |
| K1-6 Inch  | 4.34  | 225.00 | 0.019 |
| N3-10 Inch | 7.43  | 224.97 | 0.033 |
| V1-2 Inch  | 1.54  | 225.00 | 0.007 |

Elements Suitable for Test Pressure.

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## External Pressure Calculation Results Summary:

### ASME Code, Section VIII Division 1, 2025

#### External Pressure Calculations:

| From | To | Section Length<br>mm. | Outside Diameter<br>mm. | Corroded Thickness<br>mm. | Factor A    | Factor B<br>N./mm <sup>2</sup> |
|------|----|-----------------------|-------------------------|---------------------------|-------------|--------------------------------|
| 10   | 20 | No Calc               | 3020                    | 7                         | 0.00032193  | 32.1855                        |
| 20   | 30 | 9500                  | 3020                    | 7                         | 0.45524E-04 | 4.55135                        |
| 30   | 40 | 9500                  | 3020                    | 7                         | 0.45524E-04 | 4.55135                        |
| 40   | 50 | 9500                  | 3020                    | 7                         | 0.45524E-04 | 4.55135                        |
| 50   | 60 | No Calc               | 3020                    | 7                         | 0.00032193  | 32.1855                        |

#### External Pressure Calculations:

| From | To | External Actual T.<br>mm. | External Required T.<br>mm. | External Design Pressure<br>bars | External M.A.W.P.<br>bars |
|------|----|---------------------------|-----------------------------|----------------------------------|---------------------------|
| 10   | 20 | 10                        | No Calc                     | ...                              | 0.82887                   |
| 20   | 30 | 10                        | No Calc                     | ...                              | 0.14065                   |
| 30   | 40 | 10                        | No Calc                     | ...                              | 0.14065                   |
| 40   | 50 | 10                        | No Calc                     | ...                              | 0.14065                   |
| 50   | 60 | 10                        | No Calc                     | ...                              | 0.82887                   |

Minimum

0

#### External Pressure Calculations:

| From | To | Actual Length<br>Bet. Stiffeners<br>mm. | Allowable Length<br>Bet. Stiffeners<br>mm. | Ring Inertia<br>Required<br>cm <sup>4</sup> | Ring Inertia<br>Available<br>cm <sup>4</sup> |
|------|----|-----------------------------------------|--------------------------------------------|---------------------------------------------|----------------------------------------------|
| 10   | 20 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |
| 20   | 30 | 9500                                    | No Calc                                    | No Calc                                     | No Calc                                      |
| 30   | 40 | 9500                                    | No Calc                                    | No Calc                                     | No Calc                                      |
| 40   | 50 | 9500                                    | No Calc                                    | No Calc                                     | No Calc                                      |
| 50   | 60 | No Calc                                 | No Calc                                    | No Calc                                     | No Calc                                      |

Elements Suitable for External Pressure.

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### Element and Detail Weights:

| From  | To | Element<br>Metal Wgt.<br>kg. | Element<br>ID Volume<br>cm3 | Corroded<br>Metal Wgt.<br>kg. | Corroded<br>ID Volume<br>cm3 | Extra due<br>Misc %<br>kg. |
|-------|----|------------------------------|-----------------------------|-------------------------------|------------------------------|----------------------------|
| 10    | 20 | 1053.81                      | 3888417                     | 790.358                       | 3911084                      | 73.7667                    |
| 20    | 30 | 2198.59                      | 21209550                    | 1540.56                       | 21294472                     | 153.901                    |
| 30    | 40 | 2198.59                      | 21209550                    | 1540.56                       | 21294472                     | 153.901                    |
| 40    | 50 | 2125.31                      | 20502566                    | 1489.2                        | 20584658                     | 148.771                    |
| 50    | 60 | 1053.81                      | 3888417                     | 790.358                       | 3911084                      | 73.7667                    |
| Total |    | 8630                         | 70698504.00                 | 6151                          | 70995776.00                  | 604                        |

### Weight of Details:

| From | Type | Weight of<br>Detail<br>kg. | X Offset,<br>Dtl. Cent.<br>mm. | Y Offset,<br>Dtl. Cent.<br>mm. | Z Offset<br>Dtl. Cent.<br>mm. | Description     |
|------|------|----------------------------|--------------------------------|--------------------------------|-------------------------------|-----------------|
| 10   | Liqd | 1325.12                    | -250                           | 750                            | ...                           | Liquid 10       |
| 10   | Ins1 | 358.921                    | -350                           | ...                            | ...                           | Insulation      |
| 20   | Sad1 | 1099.98                    | 1750                           | 1806.5                         | ...                           | Left Saddle     |
| 20   | Plat | 5114.48                    | 4500                           | 1586.2                         | ...                           | PLAT:[1 OF 1]   |
| 20   | Liqd | 7227.95                    | 1500                           | 750                            | ...                           | Liquid 20       |
| 20   | Ins1 | 748.827                    | 1500                           | ...                            | ...                           | Ins: 20         |
| 20   | Noz1 | 14.8905                    | 250                            | 1525.4                         | ...                           | K6-2 Inch       |
| 20   | Noz1 | 32.4385                    | 850                            | 1576.2                         | ...                           | N1-6 Inch       |
| 20   | Noz1 | 14.8905                    | 1350                           | 1525.4                         | ...                           | K4-2 Inch       |
| 20   | Noz1 | 14.8905                    | 1750                           | 1525.4                         | ...                           | K5-2 Inch       |
| 20   | Noz1 | 574.03                     | 2350                           | 1804.8                         | ...                           | M1-24 Inch      |
| 20   | Noz1 | 18.8589                    | 2450                           | -1525.4                        | ...                           | D1-2 Inch       |
| 20   | Wght | 25                         | 2350                           | 1860                           | ...                           | MH Davit        |
| 20   | Wght | 55                         | 850                            | 1500                           | ...                           | Insert Plate-N1 |
| 30   | Liqd | 7227.95                    | 1500                           | 750                            | ...                           | Liquid: 30      |
| 30   | Ins1 | 748.827                    | 1500                           | ...                            | ...                           | Ins: 30         |
| 30   | Noz1 | 80.5939                    | 450                            | 1627                           | ...                           | N2-10 Inch      |
| 30   | Noz1 | 14.8905                    | 1150                           | 1525.4                         | ...                           | K3-2 Inch       |
| 30   | Noz1 | 22.7622                    | 1500                           | -1550.8                        | ...                           | N4-4 Inch       |
| 30   | Wght | 190                        | 1500                           | ...                            | ...                           | Clip Weight     |
| 30   | Wght | 95.0017                    | 450                            | 1500                           | ...                           | Insert Plate-N2 |
| 30   | Wght | 20.0004                    | 1500                           | -1500                          | ...                           | Insert Plate-N4 |
| 40   | Sad1 | 1099.98                    | 1150                           | 1806.5                         | ...                           | New Saddle      |
| 40   | Liqd | 6987.02                    | 1450                           | 750                            | ...                           | Liquid 40       |
| 40   | Ins1 | 723.866                    | 1450                           | ...                            | ...                           | Ins: 40         |
| 40   | Noz1 | 45.9645                    | 650                            | 1576.2                         | ...                           | K1-6 Inch       |
| 40   | Noz1 | 80.5939                    | 1500                           | 1627                           | ...                           | N3-10 Inch      |
| 40   | Noz1 | 18.8589                    | 2400                           | 1525.4                         | ...                           | V1-2 Inch       |
| 40   | Wght | 7                          | 650                            | 1500                           | ...                           | GWR Level Trans |
| 40   | Wght | 95.0017                    | 1500                           | 1500                           | ...                           | Insert Plate-N3 |
| 50   | Liqd | 1325.12                    | 300                            | 750                            | ...                           | Liquid 50       |
| 50   | Ins1 | 358.921                    | 400                            | ...                            | ...                           | Ins: 50         |

### Total Weight of Each Detail Type:

|            |         |
|------------|---------|
| Saddles    | 2200.0  |
| Platforms  | 5114.5  |
| Liquid     | 24093.2 |
| Insulation | 2939.4  |
| Nozzles    | 933.7   |
| Weights    | 487.0   |

Sum of the Detail Weights 35767.6 kg.

### Weight Summation Results: (kg.)

| Fabricated | Shop Test | Shipping | Erected | Empty | Operating |
|------------|-----------|----------|---------|-------|-----------|
|------------|-----------|----------|---------|-------|-----------|



|               |         |         |         |         |         |         |
|---------------|---------|---------|---------|---------|---------|---------|
| Main Elements | 9234.2  | 9234.2  | 9234.2  | 9234.2  | 9234.2  | 9234.2  |
| Saddles       | 2200.0  | 2200.0  | 2200.0  | 2200.0  | 2200.0  | 2200.0  |
| Nozzles       | 933.7   | 933.7   | 933.7   | 933.7   | 933.7   | 933.7   |
| Wld Weights   | 487.0   | 487.0   | 487.0   | 487.0   | 487.0   | 487.0   |
| Platforms     | ...     | ...     | ...     | 5114.5  | 5114.5  | 5114.5  |
| Insulation    | ...     | ...     | ...     | 2939.4  | 2939.4  | 2939.4  |
| Ope. Liquid   | ...     | ...     | ...     | ...     | ...     | 24093.2 |
| Test Liquid   | ...     | 70655.4 | ...     | ...     | ...     | ...     |
| Totals        | 12854.8 | 83510.2 | 12854.8 | 20908.7 | 20908.7 | 45001.9 |

#### Field Installation Options:

Platforms installed before lifting.

Insulation installed before lifting.

#### Miscellaneous Weight Percent: 7.0 %

*Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.*

#### Weight Summary:

|                 |                                                     |             |
|-----------------|-----------------------------------------------------|-------------|
| Fabricated Wt.  | - Bare weight without removable internals           | 12854.8 kg. |
| Shop Test Wt.   | - Fabricated weight + water ( full )                | 83510.2 kg. |
| Shipping Wt.    | - Fab. weight + removable intls.+ shipping app.     | 12854.8 kg. |
| Erected Wt.     | - Fab. wt + or - loose items (trays,platforms etc.) | 20908.7 kg. |
| Ope. Wt. no Liq | - Fab. weight + internals. + details + weights      | 20908.7 kg. |
| Operating Wt.   | - Empty weight + operating liq. uncorroded          | 45001.9 kg. |
| Oper. Wt. + CA  | - Corr wt. + operating liquid                       | 42349.3 kg. |
| Field Test Wt.  | - Empty weight + test medium (full)                 | 91564.0 kg. |

#### Surface Areas of Elements:

| From | To | Outside Surface Area<br>cm <sup>2</sup> | Inside Surface Area<br>cm <sup>2</sup> |
|------|----|-----------------------------------------|----------------------------------------|
| 10   | 20 | 103876                                  | 102271                                 |
| 20   | 30 | 284628                                  | 282743                                 |
| 30   | 40 | 284628                                  | 282743                                 |
| 40   | 50 | 275141                                  | 273319                                 |
| 50   | 60 | 103876                                  | 102271                                 |

|       |           |                           |
|-------|-----------|---------------------------|
| Total | 1052149.4 | 1043347.5 cm <sup>2</sup> |
| Total | 1132.5    | 1123.0 ft <sup>2</sup>    |

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### Nozzle Flange MAWP Results: (bars & °C)

| Nozzle Description | Flange Rating |                             | Des. Temp | Class | Grade/Group | Equiv. Press | Max Pressure |     |     |
|--------------------|---------------|-----------------------------|-----------|-------|-------------|--------------|--------------|-----|-----|
|                    | Ope.          | Ambient                     |           |       |             |              | UG-44        | 50% | DNV |
| K6-2 Inch          | 47.94         | 51.00                       | 80        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| N1-6 Inch          | 18.26         | 19.60                       | 80        | 150   | GR 1.1      | ...          | ...          | ... | ... |
| K4-2 Inch          | 47.94         | 51.00                       | 80        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| K5-2 Inch          | 47.94         | 51.00                       | 80        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| M1-24 Inch         | 18.26         | 19.60                       | 80        | 150   | GR 1.1      | ...          | ...          | ... | ... |
| D1-2 Inch          | 47.94         | 51.00                       | 80        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| N2-10 Inch         | 18.26         | 19.60                       | 80        | 150   | GR 1.1      | ...          | ...          | ... | ... |
| K3-2 Inch          | 47.94         | 51.00                       | 80        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| N4-4 Inch          | 18.26         | 19.60                       | 80        | 150   | GR 1.1      | ...          | ...          | ... | ... |
| K1-6 Inch          | 18.26         | 19.60                       | 80        | 150   | GR 1.1      | ...          | ...          | ... | ... |
| N3-10 Inch         | 18.26         | 19.60                       | 80        | 150   | GR 1.1      | ...          | ...          | ... | ... |
| V1-2 Inch          | 47.94         | 51.00                       | 80        | 300   | GR 1.1      | ...          | ...          | ... | ... |
| Min Rating         | 18.260        | 19.600, [for Core Elements] |           |       |             |              | ...          | ... | ... |

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

#### Warning:

*There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.*

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## Wind Load Calculation Results:

### Input Values:

|                                         |                   |
|-----------------------------------------|-------------------|
| Wind Design Code                        | ASCE/SEI 7-16     |
| Wind Load Reduction Scale Factor        | 1.100             |
| Basic Wind Speed [V]                    | 97.2 Km/hr        |
| Surface Roughness Category              | C: Open Terrain   |
| Type of Surface                         | Moderately Smooth |
| Base Elevation                          | 0 mm.             |
| Percent Wind for Hydrotest              | 50.0              |
| Using User defined Wind Press. Vs Elev. | N                 |
| Height of Hill or Escarpment H or Hh    | 0 mm.             |
| Distance Upwind of Crest Lh             | 0 mm.             |
| Distance from Crest to the Vessel x     | 0 mm.             |
| Type of Terrain ( Hill, Escarpment )    | Flat              |
| Damping Factor (Beta) for Wind (Ope)    | 0.0100            |
| Damping Factor (Beta) for Wind (Empty)  | 0.0000            |
| Damping Factor (Beta) for Wind (Filled) | 0.0000            |
| Site Elevation above Sea Level          | 0 mm.             |

### Wind Analysis Results

*The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.*

Static Gust-Effect Factor, Operating Case [G]:  
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$   
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.228 \cdot 0.971) / (1 + 1.7 \cdot 3.4 \cdot 0.228)))$   
 $= \min(0.85, 0.91)$   
 $= 0.850$

|                                         |           |
|-----------------------------------------|-----------|
| Natural Frequency of Vessel (Operating) | 33.000 Hz |
| Natural Frequency of Vessel (Empty)     | 33.000 Hz |
| Natural Frequency of Vessel (Test)      | 33.000 Hz |

|                                    |       |
|------------------------------------|-------|
| User Entered Importance Factor is  | 1.000 |
| Force Coefficient [Cf]             | 0.536 |
| Structure Height to Diameter ratio | 3.152 |

*This is classified as a rigid structure. Static analysis performed.*

### Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of  $[\alpha]$  and  $[Zg]$ :  
 Exposure Category: C from Table 26.11-1  
 $\alpha = 9.5$ :  $Zg = 274320.0$  mm.

Effective Height [z]:  
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$   
 $= 2133.0 + 0.0 = 2133.0$  mm.  
 $= 6.998$  ft. Imperial Units

Velocity Pressure coefficient evaluated at height z [Kz]:  
 Because  $z$  (6.998 ft.) < 15 ft.  
 $= 2.01(15 / Zg)^{2/\alpha}$   
 $= 2.01(15/900.0)^{2/9.5}$   
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor [Kd]:  
 $= 1$ , per Table 26.6-1

As there is No Hill Present: [Kzt]:  
 $K1 = 0$ ,  $K2 = 0$ ,  $K3 = 0$



Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf } [78.12] \text{ Kgs/m}^2$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.85 \cdot 0.536 \cdot 24.636$$

$$= 179.5 \text{ lbs. } [81.4] \text{ Kg}$$

| Element     | Hgt (z)<br>mm. | K1    | K2    | K3    | Kz    | Kzt   | qz<br>Kgs/m2 |
|-------------|----------------|-------|-------|-------|-------|-------|--------------|
| BOTTOM HEAD | 2133.0         | 0.000 | 0.000 | 0.000 | 0.849 | 1.000 | 78.120       |
| SHELL-1     | 2133.0         | 0.000 | 0.000 | 0.000 | 0.849 | 1.000 | 78.120       |
| SHELL-2     | 2133.0         | 0.000 | 0.000 | 0.000 | 0.849 | 1.000 | 78.120       |
| SHELL-3     | 2133.0         | 0.000 | 0.000 | 0.000 | 0.849 | 1.000 | 78.120       |
| TOP HEAD    | 2133.0         | 0.000 | 0.000 | 0.000 | 0.849 | 1.000 | 78.120       |

#### **Platform Load Calculations**

| ID            | Wind Area<br>cm2 | Elevation<br>mm. | Pressure<br>Kgs/m2 | Force<br>Kgf | Cf   |
|---------------|------------------|------------------|--------------------|--------------|------|
| PLAT:[1 OF 1] | 90000.00         | 2133.00          | 78.12              | 703.08       | 1.00 |

#### **Wind Loads on Masses/Equipment/Piping**

| ID            | Wind Area<br>cm2 | Elevation<br>mm. | Pressure<br>Kgs/m2 | Force<br>Kgf |
|---------------|------------------|------------------|--------------------|--------------|
| MH Davit      | 0.00             | 3993.00          | 78.12              | 0.00         |
| Insert Plate- | 0.00             | 3633.00          | 78.12              | 0.00         |
| Clip Weight   | 0.00             | 2133.00          | 78.12              | 0.00         |
| Insert Plate- | 0.00             | 3633.00          | 78.12              | 0.00         |
| Insert Plate- | 0.00             | 0.00             | 78.12              | 0.00         |
| GWR Level Tra | 0.00             | 3633.00          | 78.12              | 0.00         |
| Insert Plate- | 0.00             | 3633.00          | 78.12              | 0.00         |

#### **Wind Load Calculation:**

| From | To | Wind<br>Height<br>mm. | Wind<br>Diameter<br>mm. | Wind<br>Area<br>cm^2 | Wind<br>Pressure<br>Kgs/m^2 | Element<br>Wind Load<br>Kgf |
|------|----|-----------------------|-------------------------|----------------------|-----------------------------|-----------------------------|
| 10   | 20 | 2133                  | 3816                    | 22887.6              | 78.12                       | 89.5843                     |
| 20   | 30 | 2133                  | 3816                    | 114480               | 78.12                       | 1221.48                     |
| 30   | 40 | 2133                  | 3816                    | 114480               | 78.12                       | 448.087                     |
| 40   | 50 | 2133                  | 3816                    | 110664               | 78.12                       | 433.151                     |
| 50   | 60 | 2133                  | 3816                    | 22887.6              | 78.12                       | 89.5843                     |

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

*Be sure the wind speed is in accordance with the specified wind design code.*

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### Earthquake Load Calculation Results:

#### Earthquake Load Calculation:

##### Input Values:

|                                     |               |
|-------------------------------------|---------------|
| Seismic Design Code                 | ASCE/SEI 7-16 |
| Seismic Load Reduction Scale Factor | 1.000         |
| Importance Factor                   | 1.250         |
| Table Value Fa                      | 1.600         |
| Table Value Fv                      | 2.400         |
| Max. Mapped Res. Acceleration [Ss]  | 0.147         |
| Max. Eff. Ground Acceleration [S]   | 0.065         |
| Force Modification Factor R         | 3.000         |
| Site Class                          | D             |
| Component Elevation Ratio z/h       | 0.000         |
| Amplification Factor Ap             | 1.000         |
| Force Factor                        | 0.000         |
| Consider Vertical Acceleration      | Yes           |
| Minimum Acceleration Multiplier     | 0.000         |
| User Value of Sds (used if > 0 )    | 0.000         |
| User Value of Sd1 (used if > 0 )    | 0.000         |
| Moment Reduction Factor Tau         | 1.000         |

##### Seismic Analysis Results:

$$\begin{aligned}
 SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\
 SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\
 SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\
 SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104
 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:  
 =  $C_t \cdot h_{nx}$  where  $C_t = 0.020$ ,  $x = 0.75$  and  $h_n$  = Structural Height (ft.)  
 =  $0.020(11.91930.75)$   
 = 0.128 seconds

The Coefficient  $C_u$  from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:  
 = ( 1/Natural Frequency ) = ( 1/33.0 )  
 = 0.030

Check the Value of T which is the smaller of  $C_u \cdot T_a$  and T:  
 = Minimum Value of (  $1.7 \cdot 0.128$ , 0.03 ) per 12.8.2  
 = 0.030

As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:  
 =  $0.3 \cdot Sds \cdot W \cdot I$   
 =  $0.3 \cdot 0.157 \cdot 42802 \cdot 1.25$   
 = 2516.751 Kgf

Vertical load per 12.4-4, [Ev]:  
 =  $0.2 \cdot Sds \cdot W$   
 =  $0.2 \cdot 0.157 \cdot 42802$   
 = 1342.3 Kgf

#### Earthquake Load Calculation:

| From | To | Earthquake Height mm. | Earthquake Weight Kgf | Element Ope Load Kgf |
|------|----|-----------------------|-----------------------|----------------------|
| 10   | 20 | 1500                  | 6114.56               | 359.536              |
| 20   | 30 | 1500                  | 6114.56               | 359.536              |
| 20   | 30 | 1500                  | 6114.56               | 359.536              |
| 30   | 40 | 1500                  | 6114.56               | 359.536              |
| 40   | 0  | 1500                  | 6114.56               | 359.536              |
| 40   | 50 | 1500                  | 6114.56               | 359.536              |



|    |    |      |         |         |
|----|----|------|---------|---------|
| 50 | 60 | 1500 | 6114.56 | 359.536 |
|----|----|------|---------|---------|

The Earthquake Weight does not include the weight of the saddles.

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### Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.  
 Insulation is installed in the Field before being lifted.

Note : The CG is computed from the first Element From Node

|                                                    |              |
|----------------------------------------------------|--------------|
| Center of Gravity of the Saddles                   | 4500.000 mm. |
| Center of Gravity of the Platforms                 | 4550.000 mm. |
| Center of Gravity of the Liquid                    | 4500.000 mm. |
| Center of Gravity of the Insulation                | 4500.000 mm. |
| Center of Gravity of the Nozzles                   | 3245.413 mm. |
| Center of Gravity of the Added Weights (Operating) | 4438.713 mm. |
| Center of Gravity of the Added Weights (Empty)     | 4438.713 mm. |
| Center of Gravity of Bare Shell New and Cold       | 4500.002 mm. |
| Center of Gravity of Bare Shell Corroded           | 4499.998 mm. |
| Vessel CG in the Operating Condition               | 4477.674 mm. |
| Vessel CG in the Fabricated (Shop/Empty) Condition | 4406.557 mm. |
| Vessel CG in the Test Condition                    | 4485.616 mm. |

### Rigging Analysis Results:

• Lifting weight based on the user selected calculated empty weight

|                                                    |                  |
|----------------------------------------------------|------------------|
| Total Effective Length of Vessel for this analysis | 9000.00 mm.      |
| Total vessel weight (No Liquid)                    | Twt 20908.68 Kgf |
| Impact weight multiplication factor                | Imp 1.50         |
| Design lifting weight, DWT = Imp * Twt             | 31363.02 Kgf     |
| Distance to the Lug on the Left End                | 1500.00 mm.      |
| Distance to the Lug on the Right End               | 7500.00 mm.      |
| Design Reaction force at the Left End Lug          | 16169.95 Kgf     |
| Design Reaction force at the Right End Lug         | 15193.07 Kgf     |
| CG distance during field lifting operation         | 4406.56 mm.      |
| CG distance from the left end lug/trunnion         | 2906.56 mm.      |
| CG distance from the nearer lifting lug/trunnion   | 2906.56 mm.      |

Compressive Allowable Increase Factor 1.00

### Critical Values: (N./mm<sup>2</sup>, mm.)

|         | Max Stress | Elevation | Allowables    |
|---------|------------|-----------|---------------|
| Bending | 1.53       | 1850.00   | 79.28 (UG-23) |
| Shear   | 1.24       | 1250.00   | 96.60 (0.7*S) |

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

| Distance<br>mm. | Bending Moment<br>Kg-m. | Bending Stress<br>N./mm <sup>2</sup> | Shear Force<br>Kgf | Shear Stress<br>N./mm <sup>2</sup> |
|-----------------|-------------------------|--------------------------------------|--------------------|------------------------------------|
| 0.00            | 0.0                     | 0.0                                  | 7766.1             | 0.8                                |
| 1850.00         | 11053.1                 | 1.5                                  | 4084.7             | 0.4                                |
| 4250.00         | 3041.9                  | 0.4                                  | 1291.9             | 0.1                                |
| 6630.00         | 4927.7                  | 0.7                                  | 5772.3             | 0.6                                |
| 8950.00         | 1460.6                  | 0.2                                  | 597.4              | 0.1                                |

### Unity Check (Actual Stress / Allowable Stress):

Maximum Unity Check is 0.0193 at elevation 1850.0 mm. - Must be <=1

Note: The rigging analysis is performed using a uniformly distributed load.



--- Plot data successfully generated .....---

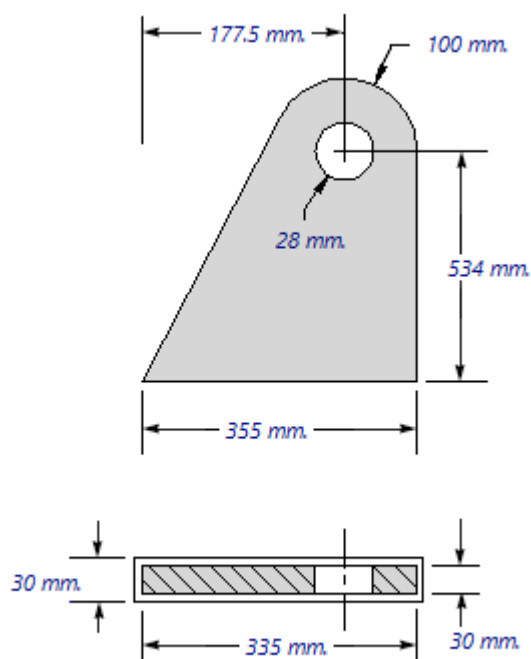
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## Lifting Lug Calculations: Lug(s) on Left End of Vessel

### Input Values:

|                                              |        |               |
|----------------------------------------------|--------|---------------|
| Lifting Lug Material                         |        | SA-516 70     |
| Lifting Lug Yield Stress                     | Yield  | 262.00 N./mm2 |
| Total Height of Lifting Lug                  | w      | 355.0000 mm.  |
| Thickness of Lifting Lug                     | t      | 30.0000 mm.   |
| Diameter of Hole in Lifting Lug              | dh     | 28.0000 mm.   |
| Radius of Semi-Circular Arc of Lifting Lug   | r      | 100.0000 mm.  |
| Height of Lug from Edge to Center of Hole    | h      | 177.5000 mm.  |
| Offset from Vessel Surface to Center of Hole | off    | 534.0000 mm.  |
| Lug Fillet Weld Size                         | tw     | 10.0000 mm.   |
| Length of weld along side of Lifting Lug     | wl     | 335.0000 mm.  |
| Length of Weld along Bottom of Lifting Lug   | wb     | 30.0000 mm.   |
| Thickness of Collar (if any)                 | tc     | 0.0000 mm.    |
| Diameter of Collar (if any)                  | dc     | 0.0000 mm.    |
| Impact Factor                                | Impfac | 1.50          |
| Sling Angle from Horizontal                  |        | 90.0 degrees  |
| Number of Lugs in Group                      |        | 2             |
| Reinforcing Pad Circumferential Width        | wpb    | 450.00 mm.    |
| Reinforcing Pad Longitudinal Length          | wpl    | 700.00 mm.    |
| Reinforcing Pad Thickness                    | thkp   | 10.00 mm.     |
| Reinforcing Pad Fillet Weld Size             | tfp    | 10.00 mm.     |

Lifting Lug Orientation to Vessel: Perpendicular  
 Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

### Computed Results:

|                                            |             |
|--------------------------------------------|-------------|
| Total vessel weight (no liquid)            | 20908.68 Kg |
| Design reaction force at the left end lug  | 16169.95 Kg |
| Design reaction force at the right end lug | 15193.07 Kg |



|                            |     |             |
|----------------------------|-----|-------------|
| Force along vessel axis    | Fax | 0.00 Kgf    |
| Force normal to vessel     | Fn  | 8084.98 Kgf |
| Force tangential to vessel | Ft  | 0.00 Kgf    |

Converting the weld leg dimension (tw) to the weld throat dimension.

#### Weld Group Inertia Calculations:

|                                                     |      |              |
|-----------------------------------------------------|------|--------------|
| Weld group inertia about the longitudinal axis      | Ill  | 5671.070 cm4 |
| Weld group centroid distance in the long. direction | Yll  | 174.570 mm.  |
| Dist. of weld group centroid from lug bottom        | Yllb | 167.500 mm.  |
| Weld group inertia about the circumferential axis   | Ilc  | 167.889 cm4  |
| Weld group centroid distance in the circ. direction | Ylc  | 22.070 mm.   |

*Note: The Impact Factor is applied to the Forces acting on the Lug.*

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2)} / ((2 \cdot (wl + wb)) \cdot tw)$$

$$= \sqrt{(0^2 + 0^2 + 8085^2)} / ((2 \cdot (335.0 + 30.0)) \cdot 7.07)$$

$$= 15.36 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot off \cdot Yll/Ill) + (Ft \cdot off \cdot Ylc/Ilc)$$

$$= (0 \cdot 534.0 \cdot 174.57/5671.07) +$$

$$(0 \cdot 534.0 \cdot 22.07/167.889)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$

$$= 15.363 + 0.0$$

$$= 15.36 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2)} / Sha$$

$$= \sqrt{(8085^2 + 0^2)} / 51.6$$

$$= 15.37 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 30.0 + 2 \cdot 0.0$$

$$= 30.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2)} / (Pbw \cdot dh)$$

$$= \sqrt{(8085^2 + 0^2)} / (30.0 \cdot 28.0)$$

$$= 94.39 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot off / (w \cdot t^2/6) + Fax \cdot off / (w^2 \cdot t/6)$$

$$= 0 \cdot 534.0 / (355.0 \cdot 30.0^2/6) +$$

$$0 \cdot 534.0 / (355.0^2 \cdot 30.0/6)$$

$$= 0.00 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= Fn / (w \cdot t)$$

$$= 8085 / (355.0 \cdot 30.0)$$

$$= 7.44 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 7.4$$

$$= 7.44 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2))^{1/2} / (\pi \cdot \text{Sta}))}$$

$$= \sqrt{(2(8085^2 + 0^2))^{1/2} / (\pi \cdot 105)}$$

$$= 21.9466 \text{ mm.}$$

**WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).**



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

*Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.*

#### Input Echo, WRC107/537 Item 1, Description: Lift Lug

|                                 |        |              |
|---------------------------------|--------|--------------|
| Diameter Basis for Vessel       | Vbasis | ID           |
| Cylindrical or Spherical Vessel | Cylsph | Cylindrical  |
| Internal Corrosion Allowance    | Cas    | 0.0000 mm.   |
| Vessel Diameter                 | Dv     | 3000.000 mm. |
| Vessel Thickness                | Tv     | 10.000 mm.   |

Note: Lifting lug calculations are performed at ambient temperature conditions.

|                              |      |             |
|------------------------------|------|-------------|
| Attachment Type              | Type | Rectangular |
| Parameter C11                | C11  | 30.00 mm.   |
| Parameter C22                | C22  | 120.00 mm.  |
| Thickness of Reinforcing Pad | Tpad | 10.000 mm.  |
| Pad Parameter C11P           | C11p | 470.000 mm. |
| Pad Parameter C22P           | C22p | 710.000 mm. |
| Design Internal Pressure     | Dp   | 0.000 bars  |
| Include Pressure Thrust      |      | No          |

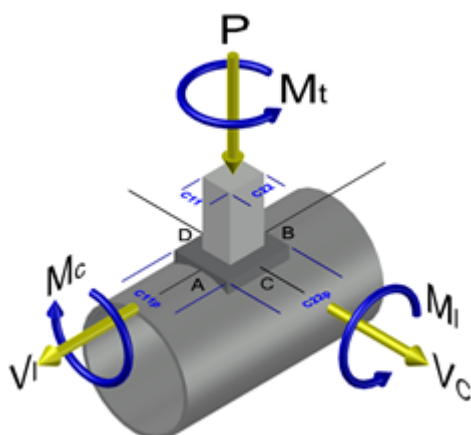
External Forces and Moments in WRC 107/537 Convention:

|                              |    |         |       |
|------------------------------|----|---------|-------|
| Radial Load (SUS)            | P  | -8085.0 | Kgf   |
| Longitudinal Shear (SUS)     | Vl | -0.0    | Kgf   |
| Circumferential Shear (SUS)  | Vc | 0.0     | Kgf   |
| Circumferential Moment (SUS) | Mc | 0.0     | Kg-m. |
| Longitudinal Moment (SUS)    | Ml | -0.0    | Kg-m. |
| Torsional Moment (SUS)       | Mt | 0.0     | Kg-m. |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSained loads:

|                        |    |         |       |
|------------------------|----|---------|-------|
| Radial Load            | P  | -8085.0 | Kgf   |
| Circumferential Shear  | VC | 0.0     | Kgf   |
| Longitudinal Shear     | VL | -0.0    | Kgf   |
| Circumferential Moment | MC | 0.0     | Kg-m. |
| Longitudinal Moment    | ML | -0.0    | Kg-m. |
| Torsional Moment       | MT | 0.0     | Kg-m. |

Dimensionless Parameters used : Gamma = 75.50

#### Dimensionless Loads for Cylindrical Shells at Attachment Junction:

| Curves read for 1979           | Beta  | Figure | Value  | Location  |
|--------------------------------|-------|--------|--------|-----------|
| N(PHI) / ( P/Rm )              | 0.029 | 4C     | 13.998 | (A,B)     |
| N(PHI) / ( P/Rm )              | 0.029 | 3C     | 15.475 | (C,D)     |
| M(PHI) / ( P )                 | 0.017 | 2C1    | 0.221  | (A,B)     |
| M(PHI) / ( P )                 | 0.017 | 1C !   | 0.211  | (C,D)     |
| N(PHI) / ( MC/(Rm**2 * Beta) ) | 0.016 | 3A     | 0.331  | (A,B,C,D) |
| M(PHI) / ( MC/(Rm * Beta) )    | 0.019 | 1A !   | 0.102  | (A,B,C,D) |
| N(PHI) / ( ML/(Rm**2 * Beta) ) | 0.025 | 3B     | 2.628  | (A,B,C,D) |
| M(PHI) / ( ML/(Rm * Beta) )    | 0.021 | 1B !   | 0.062  | (A,B,C,D) |
| N(x) / ( P/Rm )                | 0.024 | 3C     | 16.017 | (A,B)     |
| N(x) / ( P/Rm )                | 0.024 | 4C     | 14.139 | (C,D)     |
| M(x) / ( P )                   | 0.025 | 1C1    | 0.230  | (A,B)     |
| M(x) / ( P )                   | 0.025 | 2C     | 0.183  | (C,D)     |
| N(x) / ( MC/(Rm**2 * Beta) )   | 0.016 | 4A     | 0.313  | (A,B,C,D) |
| M(x) / ( MC/(Rm * Beta) )      | 0.024 | 2A !   | 0.061  | (A,B,C,D) |
| N(x) / ( ML/(Rm**2 * Beta) )   | 0.025 | 4B     | 0.619  | (A,B,C,D) |
| M(x) / ( ML/(Rm * Beta) )      | 0.029 | 2B !   | 0.102  | (A,B,C,D) |

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (N./mm2)

| Type of Stress | Load | Stress Intensity Values at |        |       |        |       |        |       |        |
|----------------|------|----------------------------|--------|-------|--------|-------|--------|-------|--------|
|                |      | Au                         | Al     | Bu    | Bl     | Cu    | Cl     | Du    | Dl     |
| Circ. Memb. P  |      | 36.7                       | 36.7   | 36.7  | 36.7   | 40.6  | 40.6   | 40.6  | 40.6   |
| Circ. Bend. P  |      | 262.3                      | -262.3 | 262.3 | -262.3 | 250.5 | -250.5 | 250.5 | -250.5 |
| Circ. Memb. MC |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |
| Circ. Bend. MC |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |



|                 |       |        |       |        |       |        |       |        |
|-----------------|-------|--------|-------|--------|-------|--------|-------|--------|
| Circ. Memb. ML  | 0.0   | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Circ. Bend. ML  | 0.0   | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Circ. Str. | 299.0 | -225.5 | 299.0 | -225.5 | 291.1 | -209.8 | 291.1 | -209.8 |
| Long. Memb. P   | 42.1  | 42.1   | 42.1  | 42.1   | 37.1  | 37.1   | 37.1  | 37.1   |
| Long. Bend. P   | 273.7 | -273.7 | 273.7 | -273.7 | 218.1 | -218.1 | 218.1 | -218.1 |
| Long. Memb. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |
| Long. Bend. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Long. Memb. ML  | 0.0   | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Long. Bend. ML  | 0.0   | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Long. Str. | 315.7 | -231.6 | 315.7 | -231.6 | 255.2 | -181.0 | 255.2 | -181.0 |
| Shear VC        | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear VL        | -0.0  | -0.0   | -0.0  | -0.0   | 0.0   | 0.0    | -0.0  | -0.0   |
| Shear MT        | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Tot. Shear      | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Str. Int.       | 315.7 | 231.6  | 315.7 | 231.6  | 291.1 | 209.8  | 291.1 | 209.8  |

Dimensionless Parameters used : Gamma = 150.50

#### Dimensionless Loads for Cylindrical Shells at Pad Edge:

| Curves read for 1979           | Beta  | Figure | Value  | Location  |
|--------------------------------|-------|--------|--------|-----------|
| N(PHI) / ( P/Rm )              | 0.233 | 4C     | 12.738 | (A,B)     |
| N(PHI) / ( P/Rm )              | 0.233 | 3C     | 4.206  | (C,D)     |
| M(PHI) / ( P )                 | 0.182 | 2C1    | 0.015  | (A,B)     |
| M(PHI) / ( P )                 | 0.182 | 1C !   | 0.070  | (C,D)     |
| N(PHI) / ( MC/(Rm**2 * Beta) ) | 0.179 | 3A     | 4.660  | (A,B,C,D) |
| M(PHI) / ( MC/(Rm * Beta) )    | 0.185 | 1A     | 0.056  | (A,B,C,D) |
| N(PHI) / ( ML/(Rm**2 * Beta) ) | 0.206 | 3B     | 8.860  | (A,B,C,D) |
| M(PHI) / ( ML/(Rm * Beta) )    | 0.191 | 1B     | 0.010  | (A,B,C,D) |
| N(x) / ( P/Rm )                | 0.209 | 3C     | 5.042  | (A,B)     |
| N(x) / ( P/Rm )                | 0.209 | 4C     | 13.914 | (C,D)     |
| M(x) / ( P )                   | 0.214 | 1C1    | 0.029  | (A,B)     |
| M(x) / ( P )                   | 0.214 | 2C !   | 0.039  | (C,D)     |
| N(x) / ( MC/(Rm**2 * Beta) )   | 0.179 | 4A     | 12.856 | (A,B,C,D) |
| M(x) / ( MC/(Rm * Beta) )      | 0.214 | 2A     | 0.021  | (A,B,C,D) |
| N(x) / ( ML/(Rm**2 * Beta) )   | 0.206 | 4B     | 4.235  | (A,B,C,D) |
| M(x) / ( ML/(Rm * Beta) )      | 0.211 | 2B     | 0.010  | (A,B,C,D) |

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

| Type of Stress  | Load | Stress Intensity Values at |        |       |        |       |        |       |        |
|-----------------|------|----------------------------|--------|-------|--------|-------|--------|-------|--------|
|                 |      | Au                         | Al     | Bu    | Bl     | Cu    | Cl     | Du    | Dl     |
| Circ. Memb. P   |      | 67.1                       | 67.1   | 67.1  | 67.1   | 22.2  | 22.2   | 22.2  | 22.2   |
| Circ. Bend. P   |      | 71.7                       | -71.7  | 71.7  | -71.7  | 334.0 | -334.0 | 334.0 | -334.0 |
| Circ. Memb. MC  |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |
| Circ. Bend. MC  |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Circ. Memb. ML  |      | 0.0                        | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Circ. Bend. ML  |      | 0.0                        | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Circ. Str. |      | 138.8                      | -4.6   | 138.8 | -4.6   | 356.1 | -311.8 | 356.1 | -311.8 |
| Long. Memb. P   |      | 26.6                       | 26.6   | 26.6  | 26.6   | 73.3  | 73.3   | 73.3  | 73.3   |
| Long. Bend. P   |      | 139.6                      | -139.6 | 139.6 | -139.6 | 186.9 | -186.9 | 186.9 | -186.9 |
| Long. Memb. MC  |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |



|                 |       |        |       |        |       |        |       |        |
|-----------------|-------|--------|-------|--------|-------|--------|-------|--------|
| Long. Bend. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Long. Memb. ML  | 0.0   | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Long. Bend. ML  | 0.0   | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Long. Str. | 166.1 | -113.0 | 166.1 | -113.0 | 260.2 | -113.6 | 260.2 | -113.6 |
| Shear VC        | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear VL        | -0.0  | -0.0   | -0.0  | -0.0   | 0.0   | 0.0    | -0.0  | -0.0   |
| Shear MT        | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Tot. Shear      | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Str. Int.       | 166.1 | 113.0  | 166.1 | 113.0  | 356.1 | 311.8  | 356.1 | 311.8  |

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Attachment Junction (N/mm<sup>2</sup>)

| Type of Stress  | Load | Stress Intensity Values at |        |       |        |       |        |       |        |
|-----------------|------|----------------------------|--------|-------|--------|-------|--------|-------|--------|
|                 |      | Au                         | Al     | Bu    | Bl     | Cu    | Cl     | Du    | Dl     |
| Circ. Pm (SUS)  |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Circ. Pl (SUS)  |      | 36.7                       | 36.7   | 36.7  | 36.7   | 40.6  | 40.6   | 40.6  | 40.6   |
| Circ. Q (SUS)   |      | 262.3                      | -262.3 | 262.3 | -262.3 | 250.5 | -250.5 | 250.5 | -250.5 |
| Long. Pm (SUS)  |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Long. Pl (SUS)  |      | 42.1                       | 42.1   | 42.1  | 42.1   | 37.1  | 37.1   | 37.1  | 37.1   |
| Long. Q (SUS)   |      | 273.7                      | -273.7 | 273.7 | -273.7 | 218.1 | -218.1 | 218.1 | -218.1 |
| Shear Pm (SUS)  |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Pl (SUS)  |      | 0.0                        | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Q (SUS)   |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm (SUS)        |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm+Pl (SUS)     |      | 42.1                       | 42.1   | 42.1  | 42.1   | 40.6  | 40.6   | 40.6  | 40.6   |
| Pm+Pl+Q (Total) |      | 315.7                      | 231.6  | 315.7 | 231.6  | 291.1 | 209.8  | 291.1 | 209.8  |

##### Vessel Stress Summation Comparison (N/mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 0.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 42.05     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 315.71    | 414.00         | Passed |

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

#### WRC 107/537 Stress Summations:

##### Vessel Stress Summation at Reinforcing Pad Edge (N/mm<sup>2</sup>)

| Type of Stress | Load | Stress Intensity Values at |       |      |       |       |        |       |        |
|----------------|------|----------------------------|-------|------|-------|-------|--------|-------|--------|
|                |      | Au                         | Al    | Bu   | Bl    | Cu    | Cl     | Du    | Dl     |
| Circ. Pm (SUS) |      | 0.0                        | 0.0   | 0.0  | 0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Circ. Pl (SUS) |      | 67.1                       | 67.1  | 67.1 | 67.1  | 22.2  | 22.2   | 22.2  | 22.2   |
| Circ. Q (SUS)  |      | 71.7                       | -71.7 | 71.7 | -71.7 | 334.0 | -334.0 | 334.0 | -334.0 |



|                 |       |        |       |        |       |        |       |        |
|-----------------|-------|--------|-------|--------|-------|--------|-------|--------|
| Long. Pm (SUS)  | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Long. Pl (SUS)  | 26.6  | 26.6   | 26.6  | 26.6   | 73.3  | 73.3   | 73.3  | 73.3   |
| Long. Q (SUS)   | 139.6 | -139.6 | 139.6 | -139.6 | 186.9 | -186.9 | 186.9 | -186.9 |
| Shear Pm (SUS)  | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Pl (SUS)  | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Q (SUS)   | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm (SUS)        | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm+Pl (SUS)     | 67.1  | 67.1   | 67.1  | 67.1   | 73.3  | 73.3   | 73.3  | 73.3   |
| Pm+Pl+Q (Total) | 166.1 | 113.0  | 166.1 | 113.0  | 356.1 | 311.8  | 356.1 | 311.8  |

#### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 0.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 73.30     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 356.14    | 414.00         | Passed |

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

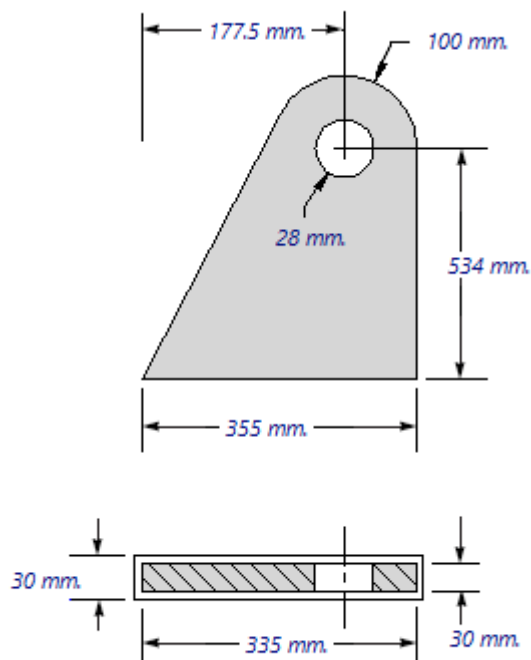
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## Lifting Lug Calculations: Lug(s) on Right End of Vessel

### Input Values:

|                                              |        |               |
|----------------------------------------------|--------|---------------|
| Lifting Lug Material                         |        | SA-516 70     |
| Lifting Lug Yield Stress                     | Yield  | 262.00 N./mm2 |
| Total Height of Lifting Lug                  | w      | 355.0000 mm.  |
| Thickness of Lifting Lug                     | t      | 30.0000 mm.   |
| Diameter of Hole in Lifting Lug              | dh     | 28.0000 mm.   |
| Radius of Semi-Circular Arc of Lifting Lug   | r      | 100.0000 mm.  |
| Height of Lug from Edge to Center of Hole    | h      | 177.5000 mm.  |
| Offset from Vessel Surface to Center of Hole | off    | 534.0000 mm.  |
| Lug Fillet Weld Size                         | tw     | 10.0000 mm.   |
| Length of weld along side of Lifting Lug     | wl     | 335.0000 mm.  |
| Length of Weld along Bottom of Lifting Lug   | wb     | 30.0000 mm.   |
| Thickness of Collar (if any)                 | tc     | 0.0000 mm.    |
| Diameter of Collar (if any)                  | dc     | 0.0000 mm.    |
| Impact Factor                                | Impfac | 1.50          |
| Number of Lugs in Group                      |        | 2             |
| Reinforcing Pad Circumferential Width        | wpb    | 450.00 mm.    |
| Reinforcing Pad Longitudinal Length          | wpl    | 700.00 mm.    |
| Reinforcing Pad Thickness                    | thkp   | 10.00 mm.     |
| Reinforcing Pad Fillet Weld Size             | tfp    | 10.00 mm.     |

Lifting Lug Orientation to Vessel: Perpendicular  
 Lift Orientation : Horizontal Lift



PV Elite does not compute weak axis bending forces on the lugs. It is assumed that a spreader bar is used.

### Computed Results:

|                                            |     |             |
|--------------------------------------------|-----|-------------|
| Total vessel weight (no liquid)            |     | 20908.68 Kg |
| Design reaction force at the left end lug  |     | 16169.95 Kg |
| Design reaction force at the right end lug |     | 15193.07 Kg |
| Force along vessel axis                    | Fax | 0.00 Kg     |



|                             |    |             |
|-----------------------------|----|-------------|
| Force normal to vessel      | Fn | 7596.53 Kgf |
| Force tangential to vessel  | Ft | 0.00 Kgf    |
| Sling angle from horizontal |    | 0.0 deg     |

Converting the weld leg dimension (tw) to the weld throat dimension.

#### Weld Group Inertia Calculations:

|                                                     |      |                          |
|-----------------------------------------------------|------|--------------------------|
| Weld group inertia about the longitudinal axis      | Ill  | 5671.070 cm <sup>4</sup> |
| Weld group centroid distance in the long. direction | Yll  | 174.570 mm.              |
| Dist. of weld group centroid from lug bottom        | Yllb | 167.500 mm.              |
| Weld group inertia about the circumferential axis   | Ilc  | 167.889 cm <sup>4</sup>  |
| Weld group centroid distance in the circ. direction | Ylc  | 22.070 mm.               |

*Note: The Impact Factor is applied to the Forces acting on the Lug.*

Primary Shear Stress in the Welds due to Shear Loads [Ssll]:

$$= \sqrt{(Fax^2 + Ft^2 + Fn^2)} / ((2 \cdot (wl + wb)) \cdot tw)$$

$$= \sqrt{(0^2 + 0^2 + 7597^2)} / ((2 \cdot (335.0 + 30.0)) \cdot 7.07)$$

$$= 14.43 \text{ N./mm}^2$$

Shear Stress in the Welds due to Bending Loads [Sblf]:

$$= (Fax \cdot off \cdot Yll/Ill) + (Ft \cdot off \cdot Ylc/Ilc)$$

$$= (0 \cdot 534.0 \cdot 174.57/5671.07) +$$

$$(0 \cdot 534.0 \cdot 22.07/167.889)$$

$$= 0.00 \text{ N./mm}^2$$

Total Shear Stress for Combined Loads [St]:

$$= Ssll + Sblf$$

$$= 14.435 + 0.0$$

$$= 14.43 \text{ N./mm}^2$$

Allowable Shear Stress for Combined Loads [Sta]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac (AISC Shear Allowable)}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Shear Stress in Lug above Hole [Shs]:

$$= \sqrt{(Pl^2 + Fax^2)} / Sha$$

$$= \sqrt{(7597^2 + 0^2)} / 51.6$$

$$= 14.44 \text{ N./mm}^2$$

Allowable Shear Stress in Lug above Hole [Sas]:

$$= 0.4 \cdot \text{Yield} \cdot \text{Occfac}$$

$$= 0.4 \cdot 262 \cdot 1.0$$

$$= 104.80 \text{ N./mm}^2$$

Pin Hole Bearing Width [Pbw]:

$$= \text{Lug Thickness} + 2 \cdot \text{Collar Thickness}$$

$$= 30.0 + 2 \cdot 0.0$$

$$= 30.000 \text{ mm.}$$

Pin Hole Bearing Stress [Pbs]:

$$= \sqrt{(Pl^2 + Faxial^2)} / (Pbw \cdot dh)$$

$$= \sqrt{(7597^2 + 0^2)} / (30.0 \cdot 28.0)$$

$$= 88.69 \text{ N./mm}^2$$

Allowable Bearing Stress [Pba]:

$$= \min(0.75 \cdot \text{Yield} \cdot \text{Occfac}, 0.9 \cdot \text{Yield}) \text{ AISC Bearing All.}$$

$$= \min(0.75 \cdot 262 \cdot 1.0, 235.8)$$

$$= 196.50 \text{ N./mm}^2$$

Bending Stress at the Base of the Lug [Fbs]:

$$= Ft \cdot off / (w \cdot t^2/6) + Fax \cdot off / (w^2 \cdot t/6)$$

$$= 0 \cdot 534.0 / (355.0 \cdot 30.0^2/6) +$$

$$0 \cdot 534.0 / (355.0^2 \cdot 30.0/6)$$

$$= 0.00 \text{ N./mm}^2$$

Tensile Stress at the Base of the Lug [Fa]:

$$= F_n / (w \cdot t)$$

$$= 7597 / (355.0 \cdot 30.0)$$

$$= 7.00 \text{ N./mm}^2$$

Total Combined Stress at the Base of the Lug:

$$= F_{bs} + F_a$$

$$= 0.0 + 7.0$$

$$= 7.00 \text{ N./mm}^2$$

Lug Allowable Stress for Bending and Tension:

$$= \min(0.66 \cdot \text{Yield} \cdot \text{Occfac}, 0.75 \cdot \text{Yield})$$

$$= \min(0.66 \cdot 262 \cdot 1.0, 196.5)$$

$$= 172.92 \text{ N./mm}^2$$

Required Shackle Pin Diameter [Spd]:

$$= \sqrt{((2(F_n^2 + F_{ax}^2)^{1/2} / (\pi \cdot \text{Sta})))}$$

$$= \sqrt{(2(7597^2 + 0^2)^{1/2} / (\pi \cdot 105))}$$

$$= 21.2733 \text{ mm.}$$

**WRC 107/537 Stress Analysis for the Lifting Lug to Shell Junction in the new and Cold Condition (no corrosion applied).**



The ratio Beta1/Beta2 for rectangular attachments was outside the range of values in tables 7 & 8 and was set to be an acceptable (maximum or minimum) value. This generally results in one dimension being altered so that the aspect ratio does not exceed 4 to 1.

*Attachment dimensions (C11P, C22P, etc.) may be modified accounting for attachment welds where applicable.*

**Input Echo, WRC107/537 Item 1, Description: Lift Lug**

|                                 |        |              |
|---------------------------------|--------|--------------|
| Diameter Basis for Vessel       | Vbasis | ID           |
| Cylindrical or Spherical Vessel | Cylsph | Cylindrical  |
| Internal Corrosion Allowance    | Cas    | 0.0000 mm.   |
| Vessel Diameter                 | Dv     | 3000.000 mm. |
| Vessel Thickness                | Tv     | 10.000 mm.   |

Note: Lifting lug calculations are performed at ambient temperature conditions.

|                              |      |             |
|------------------------------|------|-------------|
| Attachment Type              | Type | Rectangular |
| Parameter C11                | C11  | 30.00 mm.   |
| Parameter C22                | C22  | 120.00 mm.  |
| Thickness of Reinforcing Pad | Tpad | 10.000 mm.  |
| Pad Parameter C11P           | C11p | 470.000 mm. |
| Pad Parameter C22P           | C22p | 710.000 mm. |
| Design Internal Pressure     | Dp   | 0.000 bars  |
| Include Pressure Thrust      |      | No          |

External Forces and Moments in WRC 107/537 Convention:

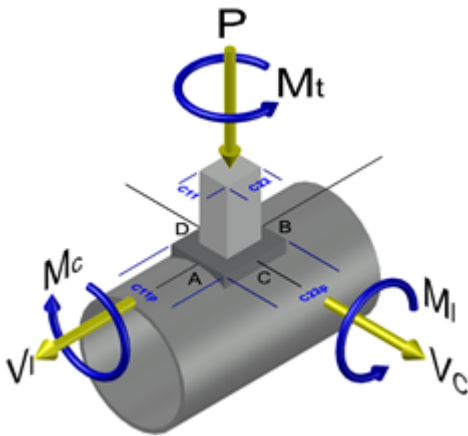
|                              |    |         |       |
|------------------------------|----|---------|-------|
| Radial Load (SUS)            | P  | -7596.5 | Kgf   |
| Longitudinal Shear (SUS)     | Vl | -0.0    | Kgf   |
| Circumferential Shear (SUS)  | Vc | 0.0     | Kgf   |
| Circumferential Moment (SUS) | Mc | 0.0     | Kg-m. |
| Longitudinal Moment (SUS)    | Ml | -0.0    | Kg-m. |
| Torsional Moment (SUS)       | Mt | 0.0     | Kg-m. |

|                         |                    |
|-------------------------|--------------------|
| Use Interactive Control | No                 |
| WRC107 Version          | Version March 1979 |

|                                                 |    |
|-------------------------------------------------|----|
| Include Pressure Stress Indices per Div. 2      | No |
| Compute Pressure Stress per WRC-368             | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the*

results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

WRC 107 Stress Calculation for SUSTAINED loads:

|                        |    |         |       |
|------------------------|----|---------|-------|
| Radial Load            | P  | -7596.5 | Kgf   |
| Circumferential Shear  | VC | 0.0     | Kgf   |
| Longitudinal Shear     | VL | -0.0    | Kgf   |
| Circumferential Moment | MC | 0.0     | Kg-m. |
| Longitudinal Moment    | ML | -0.0    | Kg-m. |
| Torsional Moment       | MT | 0.0     | Kg-m. |

Dimensionless Parameters used : Gamma = 75.50

#### Dimensionless Loads for Cylindrical Shells at Attachment Junction:

| Curves read for 1979           | Beta  | Figure | Value  | Location  |
|--------------------------------|-------|--------|--------|-----------|
| N(PHI) / ( P/Rm )              | 0.029 | 4C     | 13.998 | (A,B)     |
| N(PHI) / ( P/Rm )              | 0.029 | 3C     | 15.475 | (C,D)     |
| M(PHI) / ( P )                 | 0.017 | 2C1    | 0.221  | (A,B)     |
| M(PHI) / ( P )                 | 0.017 | 1C !   | 0.211  | (C,D)     |
| N(PHI) / ( MC/(Rm**2 * Beta) ) | 0.016 | 3A     | 0.331  | (A,B,C,D) |
| M(PHI) / ( MC/(Rm * Beta) )    | 0.019 | 1A !   | 0.102  | (A,B,C,D) |
| N(PHI) / ( ML/(Rm**2 * Beta) ) | 0.025 | 3B     | 2.628  | (A,B,C,D) |
| M(PHI) / ( ML/(Rm * Beta) )    | 0.021 | 1B !   | 0.062  | (A,B,C,D) |
| N(x) / ( P/Rm )                | 0.024 | 3C     | 16.017 | (A,B)     |
| N(x) / ( P/Rm )                | 0.024 | 4C     | 14.139 | (C,D)     |
| M(x) / ( P )                   | 0.025 | 1C1    | 0.230  | (A,B)     |
| M(x) / ( P )                   | 0.025 | 2C     | 0.183  | (C,D)     |
| N(x) / ( MC/(Rm**2 * Beta) )   | 0.016 | 4A     | 0.313  | (A,B,C,D) |
| M(x) / ( MC/(Rm * Beta) )      | 0.024 | 2A !   | 0.061  | (A,B,C,D) |
| N(x) / ( ML/(Rm**2 * Beta) )   | 0.025 | 4B     | 0.619  | (A,B,C,D) |
| M(x) / ( ML/(Rm * Beta) )      | 0.029 | 2B !   | 0.102  | (A,B,C,D) |

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Attachment Junction (N./mm2)

| Type of Stress | Load | Stress Intensity Values at |        |       |        |       |        |       |        |
|----------------|------|----------------------------|--------|-------|--------|-------|--------|-------|--------|
|                |      | Au                         | Al     | Bu    | Bl     | Cu    | Cl     | Du    | Dl     |
| Circ. Memb.    | P    | 34.5                       | 34.5   | 34.5  | 34.5   | 38.2  | 38.2   | 38.2  | 38.2   |
| Circ. Bend.    | P    | 246.4                      | -246.4 | 246.4 | -246.4 | 235.3 | -235.3 | 235.3 | -235.3 |
| Circ. Memb.    | MC   | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |



|                 |       |        |       |        |       |        |       |        |
|-----------------|-------|--------|-------|--------|-------|--------|-------|--------|
| Circ. Bend. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Circ. Memb. ML  | 0.0   | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Circ. Bend. ML  | 0.0   | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Circ. Str. | 281.0 | -211.9 | 281.0 | -211.9 | 273.5 | -197.2 | 273.5 | -197.2 |
| Long. Memb. P   | 39.5  | 39.5   | 39.5  | 39.5   | 34.9  | 34.9   | 34.9  | 34.9   |
| Long. Bend. P   | 257.1 | -257.1 | 257.1 | -257.1 | 204.9 | -204.9 | 204.9 | -204.9 |
| Long. Memb. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |
| Long. Bend. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Long. Memb. ML  | 0.0   | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Long. Bend. ML  | 0.0   | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Long. Str. | 296.6 | -217.6 | 296.6 | -217.6 | 239.8 | -170.0 | 239.8 | -170.0 |
| Shear VC        | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear VL        | -0.0  | -0.0   | -0.0  | -0.0   | 0.0   | 0.0    | -0.0  | -0.0   |
| Shear MT        | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Tot. Shear      | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Str. Int.       | 296.6 | 217.6  | 296.6 | 217.6  | 273.5 | 197.2  | 273.5 | 197.2  |

Dimensionless Parameters used : Gamma = 150.50

#### Dimensionless Loads for Cylindrical Shells at Pad Edge:

| Curves read for 1979           | Beta  | Figure | Value  | Location  |
|--------------------------------|-------|--------|--------|-----------|
| N(PHI) / ( P/Rm )              | 0.233 | 4C     | 12.738 | (A,B)     |
| N(PHI) / ( P/Rm )              | 0.233 | 3C     | 4.206  | (C,D)     |
| M(PHI) / ( P )                 | 0.182 | 2C1    | 0.015  | (A,B)     |
| M(PHI) / ( P )                 | 0.182 | 1C !   | 0.070  | (C,D)     |
| N(PHI) / ( MC/(Rm**2 * Beta) ) | 0.179 | 3A     | 4.660  | (A,B,C,D) |
| M(PHI) / ( MC/(Rm * Beta) )    | 0.185 | 1A     | 0.056  | (A,B,C,D) |
| N(PHI) / ( ML/(Rm**2 * Beta) ) | 0.206 | 3B     | 8.860  | (A,B,C,D) |
| M(PHI) / ( ML/(Rm * Beta) )    | 0.191 | 1B     | 0.010  | (A,B,C,D) |
| N(x) / ( P/Rm )                | 0.209 | 3C     | 5.042  | (A,B)     |
| N(x) / ( P/Rm )                | 0.209 | 4C     | 13.914 | (C,D)     |
| M(x) / ( P )                   | 0.214 | 1C1    | 0.029  | (A,B)     |
| M(x) / ( P )                   | 0.214 | 2C !   | 0.039  | (C,D)     |
| N(x) / ( MC/(Rm**2 * Beta) )   | 0.179 | 4A     | 12.856 | (A,B,C,D) |
| M(x) / ( MC/(Rm * Beta) )      | 0.214 | 2A     | 0.021  | (A,B,C,D) |
| N(x) / ( ML/(Rm**2 * Beta) )   | 0.206 | 4B     | 4.235  | (A,B,C,D) |
| M(x) / ( ML/(Rm * Beta) )      | 0.211 | 2B     | 0.010  | (A,B,C,D) |

Note - The ! mark next to the figure name denotes curve value exceeded.

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

#### Stresses in the Vessel at the Edge of Reinforcing Pad (N./mm2)

| Type of Stress  | Load | Stress Intensity Values at |        |       |        |       |        |       |        |
|-----------------|------|----------------------------|--------|-------|--------|-------|--------|-------|--------|
|                 |      | Au                         | Al     | Bu    | Bl     | Cu    | Cl     | Du    | Dl     |
| Circ. Memb. P   |      | 63.1                       | 63.1   | 63.1  | 63.1   | 20.8  | 20.8   | 20.8  | 20.8   |
| Circ. Bend. P   |      | 67.4                       | -67.4  | 67.4  | -67.4  | 313.8 | -313.8 | 313.8 | -313.8 |
| Circ. Memb. MC  |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |
| Circ. Bend. MC  |      | 0.0                        | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Circ. Memb. ML  |      | 0.0                        | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Circ. Bend. ML  |      | 0.0                        | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Circ. Str. |      | 130.4                      | -4.3   | 130.4 | -4.3   | 334.6 | -293.0 | 334.6 | -293.0 |
| Long. Memb. P   |      | 25.0                       | 25.0   | 25.0  | 25.0   | 68.9  | 68.9   | 68.9  | 68.9   |
| Long. Bend. P   |      | 131.1                      | -131.1 | 131.1 | -131.1 | 175.6 | -175.6 | 175.6 | -175.6 |



|                 |       |        |       |        |       |        |       |        |
|-----------------|-------|--------|-------|--------|-------|--------|-------|--------|
| Long. Memb. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    |
| Long. Bend. MC  | 0.0   | 0.0    | 0.0   | 0.0    | -0.0  | 0.0    | 0.0   | -0.0   |
| Long. Memb. ML  | 0.0   | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   | -0.0  | -0.0   |
| Long. Bend. ML  | 0.0   | -0.0   | -0.0  | 0.0    | -0.0  | -0.0   | -0.0  | -0.0   |
| Tot. Long. Str. | 156.1 | -106.2 | 156.1 | -106.2 | 244.5 | -106.8 | 244.5 | -106.8 |
| Shear VC        | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear VL        | -0.0  | -0.0   | -0.0  | -0.0   | 0.0   | 0.0    | -0.0  | -0.0   |
| Shear MT        | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Tot. Shear      | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Str. Int.       | 156.1 | 106.2  | 156.1 | 106.2  | 334.6 | 293.0  | 334.6 | 293.0  |

### WRC 107/537 Stress Summations:

#### Vessel Stress Summation at Attachment Junction (N./mm2)

| Type of Stress  | Load | Stress Intensity Values at |        |       |        |       |        |       |        |
|-----------------|------|----------------------------|--------|-------|--------|-------|--------|-------|--------|
|                 |      | Au                         | Al     | Bu    | Bl     | Cu    | Cl     | Du    | Dl     |
| Circ. Pm (SUS)  |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Circ. Pl (SUS)  |      | 34.5                       | 34.5   | 34.5  | 34.5   | 38.2  | 38.2   | 38.2  | 38.2   |
| Circ. Q (SUS)   |      | 246.4                      | -246.4 | 246.4 | -246.4 | 235.3 | -235.3 | 235.3 | -235.3 |
| Long. Pm (SUS)  |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Long. Pl (SUS)  |      | 39.5                       | 39.5   | 39.5  | 39.5   | 34.9  | 34.9   | 34.9  | 34.9   |
| Long. Q (SUS)   |      | 257.1                      | -257.1 | 257.1 | -257.1 | 204.9 | -204.9 | 204.9 | -204.9 |
| Shear Pm (SUS)  |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Pl (SUS)  |      | 0.0                        | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Q (SUS)   |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm (SUS)        |      | 0.0                        | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm+Pl (SUS)     |      | 39.5                       | 39.5   | 39.5  | 39.5   | 38.2  | 38.2   | 38.2  | 38.2   |
| Pm+Pl+Q (Total) |      | 296.6                      | 217.6  | 296.6 | 217.6  | 273.5 | 197.2  | 273.5 | 197.2  |

#### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 0.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 39.51     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 296.63    | 414.00         | Passed |

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

### WRC 107/537 Stress Summations:

#### Vessel Stress Summation at Reinforcing Pad Edge (N./mm2)

| Type of Stress | Load | Stress Intensity Values at |       |      |       |       |        |       |        |
|----------------|------|----------------------------|-------|------|-------|-------|--------|-------|--------|
|                |      | Au                         | Al    | Bu   | Bl    | Cu    | Cl     | Du    | Dl     |
| Circ. Pm (SUS) |      | 0.0                        | 0.0   | 0.0  | 0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Circ. Pl (SUS) |      | 63.1                       | 63.1  | 63.1 | 63.1  | 20.8  | 20.8   | 20.8  | 20.8   |
| Circ. Q (SUS)  |      | 67.4                       | -67.4 | 67.4 | -67.4 | 313.8 | -313.8 | 313.8 | -313.8 |



|                 |       |        |       |        |       |        |       |        |
|-----------------|-------|--------|-------|--------|-------|--------|-------|--------|
| Long. Pm (SUS)  | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Long. Pl (SUS)  | 25.0  | 25.0   | 25.0  | 25.0   | 68.9  | 68.9   | 68.9  | 68.9   |
| Long. Q (SUS)   | 131.1 | -131.1 | 131.1 | -131.1 | 175.6 | -175.6 | 175.6 | -175.6 |
| Shear Pm (SUS)  | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Pl (SUS)  | 0.0   | 0.0    | -0.0  | -0.0   | 0.0   | 0.0    | 0.0   | 0.0    |
| Shear Q (SUS)   | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm (SUS)        | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    | 0.0   | 0.0    |
| Pm+Pl (SUS)     | 63.1  | 63.1   | 63.1  | 63.1   | 68.9  | 68.9   | 68.9  | 68.9   |
| Pm+Pl+Q (Total) | 156.1 | 106.2  | 156.1 | 106.2  | 334.6 | 293.0  | 334.6 | 293.0  |

#### Vessel Stress Summation Comparison (N./mm<sup>2</sup>):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 0.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 68.87     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 334.63    | 414.00         | Passed |

The Pm+Pl+Q allowable above was based on the rules of ASME VIII-2 Part 4, 4.1.6.3 (a)&(b) or per ASME VIII-2, 5.5.6.1(1)&(2) Part 5 as applicable.

The Pm+Pl+Q allowable was based on a temperature range cycling from ambient to design temperature.

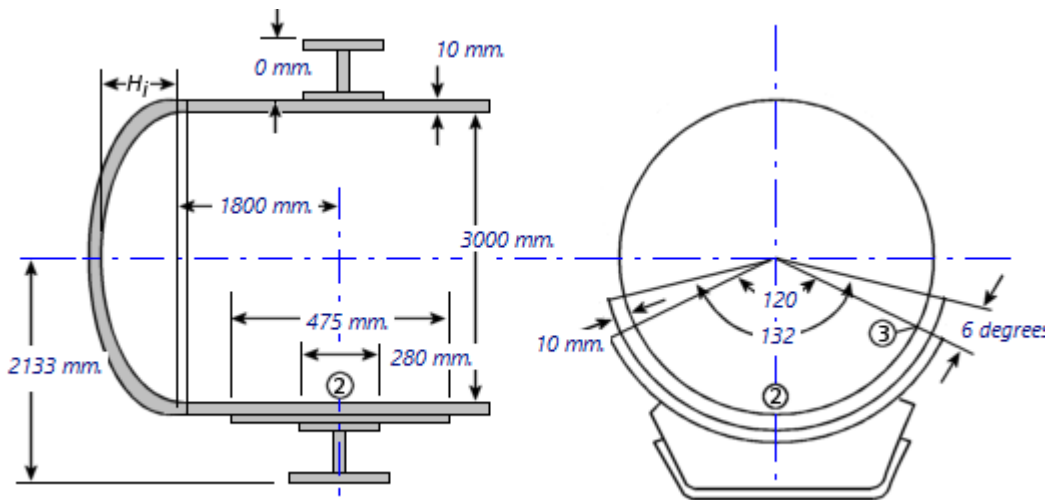
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### ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

#### Input and Calculated Values:

|                                               |                  |         |                    |
|-----------------------------------------------|------------------|---------|--------------------|
| Vessel Mean Radius                            | Rm               | 1506.50 | mm.                |
| Shell Thickness used in this Case             | t                | 10.000  | mm.                |
| Stiffened Vessel Length per 4.15.6            | L                | 9000.00 | mm.                |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1800.00 | mm.                |
| Saddle Width                                  | b1 or b          | 280.00  | mm.                |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees            |
| Wear Plate Width                              | b2 or b1         | 475.00  | mm.                |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees            |
| Wear Plate Thickness                          | e2 or tr         | 10.0    | mm.                |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm <sup>2</sup> |
| Inside or Mean Depth of Head                  | Hi or hm         | 756.50  | mm.                |
| Shell Allowable Stress used in Calculation    |                  | 138.00  | N./mm <sup>2</sup> |
| Head Allowable Stress used in Calculation     |                  | 138.00  | N./mm <sup>2</sup> |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |                    |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |                    |
| Distance from Saddle Base to Centerline       | B                | 2133.00 | mm.                |
| Coefficient of Friction                       | mu               | 0.00    |                    |



#### Results for Horizontal Vessel Analysis:

|                                |          |      |
|--------------------------------|----------|------|
| Saddle Force Q, Operating Case | 25305.74 | Kgf  |
| Pressure used in Analysis      | P 4.500  | bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 46.89                        | 138.00                          |
| Long. Stress at Bottom of Midspan   | 49.95                        | 138.00                          |
| Long. Stress at Top of Saddles      | 68.03                        | 138.00                          |
| Long. Stress at Bottom of Saddles   | 37.56                        | 138.00                          |
| Tangential Shear in Shell           | 14.86                        | 110.40                          |
| Circ. Stress at Horn of Saddle      | 99.45                        | 172.50                          |
| Circ. Compressive Stress in Shell   | 2.52                         | 138.00                          |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$   
 $= 3.0(2281.9/2 + 0) \cdot 2133.0/2632.7173$   
 $= 2773.1 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$   
 $= \max(1244.89, 0) \cdot 2133.0/5400.0$   
 $= 491.7 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$   
 $= \max(2516.75, 0) \cdot 2133.0/5400.0$   
 $= 994.1 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:  
 $= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$   
 $= 3.0(2517/2 + 0) \cdot 2133.0/2632.7173$   
 $= 3058.6 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$   
 $= 22247 + \max(492, 2773, 994, 3059)$   
 $= 25305.7 \text{ Kg}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 766.08(3.05 + \pi/4(3.18 \cdot 1.2)^2)$   
 $= 12207.991 \text{ N}$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 26405.72 | Kgf |
| Transverse Shear Load                   | 1258.38  | Kgf |
| Longitudinal Shear Load                 | 2516.75  | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$   
 $= \min(2(3000.0/2 + 10.0 + 10.0)\sin(120.0/2), 2 \cdot 1506.5)$   
 $= 2632.717 \text{ mm}$

#### The Computed K values from Table 4.15.1:

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to  $\text{Rm}/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$   
 $= -25306 \cdot 1800.0 [1 - (1 - 1800.0/9000.0 + (1506.5^2 - 756.5^2)/(2 \cdot 1800.0 \cdot 9000.0))/(1 + (4 \cdot 756.5)/(3 \cdot 9000.0))]$   
 $= -10637.1 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$   
 $= 25306 \cdot 9000/4((1 + 2(1507^2 - 757^2)/(9000^2))/(1 + (4 \cdot 757)/(3 \cdot 9000)) - 4 \cdot 1800/9000)$   
 $= 7795.2 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$   
 $= 4.5 \cdot 1506.5/(2 \cdot 7.0) - 7795.2/(\pi \cdot 1506.5^2 \cdot 7.0)$   
 $= 46.89 \text{ N./mm}^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 4.5 \cdot 1506.5 / (2 \cdot 7.0) + 7795.2 / (\pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 49.95 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma_{*3}$ ]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4.5 \cdot 1506.5 / (2 \cdot 7.0) - 10637.1 / (0.1066 \cdot \pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 68.03 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma_{*4}$ ]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 4.5 \cdot 1506.5 / (2 \cdot 7.0) + 10637.1 / (0.1923 \cdot \pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 37.56 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 25306(9000.0 - 2 \cdot 1800.0) / (9000.0 + (4 \cdot 756.5 / 3))$$

$$= 13653.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 13653.27 / (1506.5 \cdot 7.0)$$

$$= 14.86 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506.5 \cdot 7.0)}$$

$$= 80.099 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{(1506.5 \cdot 7.0)}, 2 \cdot 1800.0)$$

$$= 440.20 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [ $\eta$ ]:

$$= \min(S_r / S, 1)$$

$$= \min(138.0 / 138.0, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 25306 \cdot 0.1 / (440.198(7.0 + 1.0 \cdot 10.0))$$

$$= -2.52 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L < 8R_m$  (4.15.29) [ $\sigma_7, r_{*}$ ]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -25306 / (4(7.0 + 1.0 \cdot 10.0)440.198) -$$

$$12 \cdot 0.053 \cdot 25306 \cdot 1506.5 / (9000.0(7.0 + 1.0 \cdot 10.0)^2)$$

$$= -99.45 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 5400.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 5400.0(80.0 - 21.1)$$

$$= 3.807 \text{ mm.}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |                    |
|--------------------------------------|--------|-----------|--------------------|
| Baseplate Length                     | Bplen  | 2790.0000 | mm.                |
| Baseplate Thickness                  | Bpthk  | 26.0000   | mm.                |
| Baseplate Width                      | Bpwid  | 310.0000  | mm.                |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 6         |                    |
| Rib Thickness                        | Ribtk  | 20.0000   | mm.                |
| Web Thickness                        | Webtk  | 20.0000   | mm.                |
| Web Location                         | Webloc | Center    |                    |
| Saddle Design Temperature            |        | 21.1      | °C                 |
| Saddle Yield Stress                  | Sy     | 243.0     | N./mm <sup>2</sup> |
| Height of Web at Center              | Hw,c   | 607.0     | mm.                |

Friction Coefficient mu 0.000

### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$   
 $= ((1.56 \cdot \sqrt{(1503.0 \cdot 7.0)} + 475.0)^3) / 12$   
 $= 1.815 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:  
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$   
 $= 1.815 + 44.451(299.735 - 3.5)^2$   
 $= 39009.758 \text{ cm}^4$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 635.0    | 7.0      | 3.5      | 44.5                 | 15557.8                | 0.390E+05                                |
| Wearplate | 475.0    | 10.0     | 12.0     | 47.5                 | 57000.0                | 0.393E+05                                |
| Web       | 20.0     | 587.0    | 310.5    | 117.4                | 3645270.5              | 0.338E+05                                |
| BasePlate | 310.0    | 26.0     | 617.0    | 80.6                 | 4973020.0              | 0.812E+05                                |
| Totals    | ...      | ...      | ...      | 290.0                | 8690848.0              | 0.193E+06                                |

Distance to Centroid [C1]:  
 $= AY / A$   
 $= 8690848 / 289.951$   
 $= 299.735 \text{ mm.}$

Angle [β]:  
 $= 180 - \text{Saddle Angle} / 2$   
 $= 180 - 120.0 / 2$   
 $= 120.0$

Saddle Splitting Coefficient [K1]:  
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$   
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$   
 $= 0.2035$

Saddle Splitting Force [Fh]:  
 $= K1 \cdot Q$   
 $= 0.204 \cdot 25305.74$   
 $= 5150.2671 \text{ Kgf}$

Tension Stress, St = ( Fh / As ) = 2.0573 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 0.6 \* Yield Str = 145.7983 N./mm<sup>2</sup>

Saddle Splitting Dimension [d]:  
 $= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$   
 $= 2133.0 - 1503.0 \cdot \sin(120.0 / 2) / 1.0472$   
 $= 890.029 \text{ mm.}$

Bending Moment, M = Fh \* d = 4583.9775 Kg-m.

Bending Stress, Sb = ( M \* C1 / I ) = 6.9684 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 2/3 \* Yield Str = 161.9981 N./mm<sup>2</sup>

### Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$   
 $= \sqrt{(3(25306 + 1100)310.0 / (4 \cdot 2790.0 \cdot 161.998))}$   
 $= 11.542 \text{ mm.}$

### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [ Web Length ]:  
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$   
 $= 2 \cdot \cos(90 - 120.0 / 2) (1500.0 + 10.0 + 10.0)$   
 $= 2632.717 \text{ mm.}$

Distance between Ribs [e]:  
 $= \text{Web Length} / (\text{Nr ribs} - 1)$

$$= 2632.717 / (6 - 1)$$

$$= 526.543 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 526.5434 \cdot 310.0 / 2$$

$$= 816.142 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 26405.725 / (2790.0 \cdot 310.0)$$

$$= 3.053 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 816.1 \cdot 3.05$$

$$= 2491.713 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 52.0 + 52.654$$

$$= 104.654 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 2491.7 / 104.6544$$

$$= 2.335 \text{ N/mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|         | B    | D     | Y   | A    | AY  | Io        |
|---------|------|-------|-----|------|-----|-----------|
| Rib+Web | 20.0 | 280.0 | ... | 56.0 | ... | 0.366E+04 |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 280.0 - 20.0$$

$$= 260.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 104.654$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0 / 2$$

$$= 140.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3658.7 / 104.654}$$

$$= 59.127 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133.0 - \cos(120.0/2) (1500.0 + 10.0 + 10.0) - 26.0$$

$$= 1347.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 243.0}$$

$$= 127.447$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1347.0 / 59.127$$

$$= 22.782$$

**Bending Moment [Rm]:**

$$= FI / ( 2 \cdot B_{pln} ) \cdot e \cdot L / 2$$

$$= 2516.8 / ( 2 \cdot 2790.0 ) \cdot 526.543 \cdot 1347.0 / 2$$

$$= 159.951 \text{ Kg-m.}$$

**Compressive Allowable,  $KL/r < C_c$  ( 22.7816 < 127.4472 ) per AISC E2-1 [Sca]:**

$$= ( 1 - (Kl/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3) )$$

$$= ( 1 - ( 22.78 )^2 / ( 2 \cdot 127.45^2 ) ) 243 /$$

$$( 5/3 + 3 \cdot (22.78) / (8 \cdot 127.45) - ( 22.78^3 ) / (8 \cdot 127.45^3) )$$

$$= 138 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + ( R_m \cdot C_1 / I ) / S_{ba}$$

$$= 2.33 / 137.98 + ( 159.95 \cdot 140.0 / 36586676 ) / 162.0$$

$$= 0.054$$

**Check of Inside Ribs:**

**Inertia of Saddle, Inner Ribs - Axial Direction**

|        | B     | D     | Y   | A     | AY  | Io        |
|--------|-------|-------|-----|-------|-----|-----------|
| Rib    | 20.0  | 260.0 | 0.0 | 52.0  | 0.0 | 0.366E+04 |
| Web    | 526.5 | 20.0  | 0.0 | 105.3 | 0.0 | 35.1      |
| Totals | ...   | ...   | ... | 157.3 | ... | 0.369E+04 |

**Distance to Centroid from Datum [ytot]:**

$$= AY / A$$

$$= 0.0 / 157.309$$

$$= 0.000 \text{ mm.}$$

**Distance to Centroid [C1]:**

$$= \text{Saddle Width} / 2$$

$$= 280.0 / 2$$

$$= 140.000 \text{ mm.}$$

**Length of Inner Rib [L]:**

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 2133.0 - \sqrt{ (1520.0 + 10.0)^2 - (526.543/2)^2 } - 26.0$$

$$= 609.974 \text{ mm.}$$

**Radius of Gyration [r]:**

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3692.4 / 157.309 }$$

$$= 48.448 \text{ mm.}$$

**Slenderness ratio [KL/r]:**

$$= KL/r$$

$$= 1 \cdot 609.974 / 48.448$$

$$= 12.590$$

**Unit Force [Force,u]:**

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 2516.751 / ( 2 \cdot 2790.0 )$$

$$= 0.451 \text{ Kg/mm.}$$

**Moment at base of inner Rib [Mbase,c]:**

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.451 \cdot 526.543 \cdot 609.974$$

$$= 144.864 \text{ Kg-m.}$$

**Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:**

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 144.864 / 263745.469$$

$$= 5.386 \text{ N./mm}^2$$

**Compressive Allowable,  $KL/r < C_c$  ( 12.5901 < 127.4472 ) per AISC E2-1 [Sca]:**

$$= ( 1 - (Kl/r)^2 / (2 \cdot C_c^2) ) F_y / ( 5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3) )$$

$$= ( 1 - ( 12.59 )^2 / ( 2 \cdot 127.45^2 ) ) 243 /$$

$$( 5/3 + 3 \cdot (12.59) / (8 \cdot 127.45) - ( 12.59^3 ) / (8 \cdot 127.45^3) )$$

$$= 141.9 \text{ N./mm}^2$$

#### AISC Unity Check of Inside Ribs ( must be $\leq 1$ )

$$= Sc/Sca + ( Mbase,c \cdot C1/I )/Sba$$

$$= 3.09/141.94 + ( 144.86 \cdot 140.0/3692.434 )/162.0$$

$$= 0.055$$

#### Input Data for Base Plate Bolting Calculations:

|                                            |              |               |
|--------------------------------------------|--------------|---------------|
| Total Number of Bolts per BasePlate        | Nbolts       | 10            |
| Total Number of Bolts in Tension/Baseplate | Nbt          | 5             |
| Bolt Design Temperature                    |              | 21.1 °C       |
| Bolt Material Specification                |              | SA-307 A      |
| Bolt Allowable Stress                      | Stba         | 138.00 N./mm2 |
| Bolt Corrosion Allowance                   | Bca          | 1.5 mm.       |
| Distance from Bolts to Edge                | Edgedis      | 50.8 mm.      |
| User Defined Area in a Single Bolt         | UserBoltArea | 2.4000 cm2    |
| BasePlate Allowable Stress                 | S            | 138.00 N./mm2 |
| Area Available in a Single Bolt            | BltArea      | 1.6476 cm2    |
| Saddle Load QO (Weight)                    | QO           | 23347.2 Kg    |
| Saddle Load QL (Wind/Seismic contribution) | QL           | 994.1 Kg      |
| Maximum Transverse Force                   | Ft           | 1258.4 Kg     |
| Maximum Longitudinal Force                 | Fl           | 2516.8 Kg     |
| Saddle Bolted to Steel Foundation          |              | No            |

Friction force due to thermal growth, [Ff]:

$$= 9198.8 \text{ Kg}$$

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= \max( Ff, Fl ) / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 9198.8 / ( 1.648 \cdot 10 )$$

$$= 54.8 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$$

$$= 1258.4 / ( 1.648 \cdot 10 )$$

$$= 7.5 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

#### Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / ( \text{BoltShearAllowable} \cdot \text{Nbolts} )$$

$$= 2516.75 / ( 82.8 \cdot 10.0 )$$

$$= 0.2981 \text{ cm}^2$$

#### Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 2133.0 \cdot 1258.38 + 0.0$$

$$= 2684.17 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 2684.17 / 23347.15$$

$$= 114.97 \text{ mm.} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[ \text{Bltarearl}, \text{Bltarears}, \text{Bltareart} ]$$

$$= \max[ 0.0, 0.2981, 0.0 ]$$

$$= 0.2981 \text{ cm}^2$$



Govrn. thk, tg = 10.0, tr = 7.0, c = 3.0 mm., E\* = 0.85  
 Thickness Ratio = tr \* E\*/(tg - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C  
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -48 °C

### ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

#### Input and Calculated Values:

|                                               |                  |         |         |
|-----------------------------------------------|------------------|---------|---------|
| Vessel Mean Radius                            | Rm               | 1506.50 | mm.     |
| Shell Thickness used in this Case             | t                | 10.000  | mm.     |
| Stiffened Vessel Length per 4.15.6            | L                | 9000.00 | mm.     |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1800.00 | mm.     |
| Saddle Width                                  | b1 or b          | 280.00  | mm.     |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees |
| Wear Plate Width                              | b2 or b1         | 475.00  | mm.     |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees |
| Wear Plate Thickness                          | e2 or tr         | 10.0    | mm.     |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm2  |
| Inside or Mean Depth of Head                  | Hi or hm         | 756.50  | mm.     |
| Shell Allowable Stress used in Calculation    |                  | 138.00  | N./mm2  |
| Head Allowable Stress used in Calculation     |                  | 138.00  | N./mm2  |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |         |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |         |
| Distance from Saddle Base to Centerline       | B                | 2133.00 | mm.     |
| Coefficient of Friction                       | mu               | 0.40    |         |

#### Results for Horizontal Vessel Analysis:

|                                |          |            |
|--------------------------------|----------|------------|
| Saddle Force Q, Operating Case | 24955.56 | Kgf        |
| Pressure used in Analysis      | P        | 4.500 bars |

| Horizontal Vessel Analysis Results: | Actual<br>N./mm^2 | Allowable<br>N./mm^2 |
|-------------------------------------|-------------------|----------------------|
| Long. Stress at Top of Midspan      | 46.91             | 138.00               |
| Long. Stress at Bottom of Midspan   | 49.93             | 138.00               |
| Long. Stress at Top of Saddles      | 67.76             | 138.00               |
| Long. Stress at Bottom of Saddles   | 37.71             | 138.00               |
| Tangential Shear in Shell           | 14.66             | 110.40               |
| Circ. Stress at Horn of Saddle      | 98.08             | 172.50               |
| Circ. Compressive Stress in Shell   | 2.49              | 138.00               |

#### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 = Ftr( Ft/Num of Saddles + Z Force Load ) \* B / E  
 = 3.0( 2281.9/2 + 0 ) \* 2133.0/2632.7173  
 = 2773.1 Kgf

Longitudinal Saddle Reaction Force [Fwl]:  
 = max( Fl,wind; Sum of X Forces ) \* B / Ls  
 = max( 1244.89, 0 ) \* 2133.0/5400.0  
 = 491.7 Kgf

Saddle Reaction Force due to Earthquake Fl [Fsl]:  
 = max( Fl,eq; Sum of X Forces ) \* B / Ls  
 = max( 2516.75, 0 ) \* 2133.0/5400.0  
 = 994.1 Kgf

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(2517/2 + 0) \cdot 2133.0/2632.7173$$

$$= 3058.6 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 21897 + \max(492, 2773, 994, 3059)$$

$$= 24955.6 \text{ Kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(3.05 + \pi/4(3.18 \cdot 1.2)^2)$$

$$= 12207.991 \text{ N}$$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 26055.54 | Kgf |
| Transverse Shear Load                   | 1258.38  | Kgf |
| Longitudinal Shear Load                 | 2516.75  | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(3000.0/2 + 10.0 + 10.0)\sin(120.0/2), 2 \cdot 1506.5)$$

$$= 2632.717 \text{ mm.}$$

#### The Computed K values from Table 4.15.1:

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to  $\text{Rm}/2$ .

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/3L)]$$

$$= -24956 \cdot 1800.0 [1 - (1 - 1800.0/9000.0 + (1506.5^2 - 756.5^2)/(2 \cdot 1800.0 \cdot 9000.0))/(1 + (4 \cdot 756.5)/(3 \cdot 9000.0))]$$

$$= -10489.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$$

$$= 24956 \cdot 9000/4((1 + 2(1507^2 - 757^2)/(9000^2))/(1 + (4 \cdot 757)/(3 \cdot 9000)) - 4 \cdot 1800/9000)$$

$$= 7687.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$= P \cdot \text{Rm}/(2t) - M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 4.5 \cdot 1506.5/(2 \cdot 7.0) - 7687.3/(\pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 46.91 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$= P \cdot \text{Rm}/(2t) + M2/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 4.5 \cdot 1506.5/(2 \cdot 7.0) + 7687.3/(\pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 49.93 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma_{*3}$ ]:

$$= P \cdot \text{Rm}/(2t) - M1/(K1 \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 4.5 \cdot 1506.5/(2 \cdot 7.0) - 10489.9/(0.1066 \cdot \pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 67.76 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma_{*4}$ ]:

$$= P \cdot \text{Rm}/(2t) + M1/(K1 \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 4.5 \cdot 1506.5/(2 \cdot 7.0) + 10489.9/(0.1923 \cdot \pi \cdot 1506.5^2 \cdot 7.0)$$

$$= 37.71 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot \text{hm}/3))$$



$$= 24956(9000.0 - 2 \cdot 1800.0)/(9000.0 + (4 \cdot 756.5/3))$$

$$= 13464.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 13464.33 / (1506.5 \cdot 7.0)$$

$$= 14.66 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1506.5 \cdot 7.0)}$$

$$= 80.099 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{(1506.5 \cdot 7.0)}, 2 \cdot 1800.0)$$

$$= 440.20 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [ $\eta$ ]:

$$= \min(S_r/S, 1)$$

$$= \min(138.0/138.0, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 24956 \cdot 0.1 / (440.198(7.0 + 1.0 \cdot 10.0))$$

$$= -2.49 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L < 8R_m$  (4.15.29) [ $\sigma_7, r^*$ ]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -24956 / (4(7.0 + 1.0 \cdot 10.0)440.198) -$$

$$12 \cdot 0.053 \cdot 24956 \cdot 1506.5 / (9000.0(7.0 + 1.0 \cdot 10.0)^2)$$

$$= -98.08 \text{ N./mm}^2$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |        |
|--------------------------------------|--------|-----------|--------|
| Baseplate Length                     | Bplen  | 2790.0000 | mm.    |
| Baseplate Thickness                  | Bpthk  | 26.0000   | mm.    |
| Baseplate Width                      | Bpwid  | 310.0000  | mm.    |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 6         |        |
| Rib Thickness                        | Ribtk  | 20.0000   | mm.    |
| Web Thickness                        | Webtk  | 20.0000   | mm.    |
| Web Location                         | Webloc | Center    |        |
| Saddle Design Temperature            |        | 21.1      | °C     |
| Saddle Yield Stress                  | Sy     | 243.0     | N./mm2 |
| Height of Web at Center              | Hw,c   | 607.0     | mm.    |
| Friction Coefficient                 | mu     | 0.400     |        |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1503.0 \cdot 7.0)} + 475.0)7.0^3)/12$$

$$= 1.815 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 1.815 + 44.451(299.735 - 3.5)^2$$

$$= 39009.758 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 635.0    | 7.0      | 3.5      | 44.5                 | 15557.8                | 0.390E+05                                |
| Wearplate | 475.0    | 10.0     | 12.0     | 47.5                 | 57000.0                | 0.393E+05                                |
| Web       | 20.0     | 587.0    | 310.5    | 117.4                | 3645270.5              | 0.338E+05                                |
| BasePlate | 310.0    | 26.0     | 617.0    | 80.6                 | 4973020.0              | 0.812E+05                                |
| Totals    | ...      | ...      | ...      | 290.0                | 8690848.0              | 0.193E+06                                |

Distance to Centroid [C1]:



$$\begin{aligned}
 &= AY / A \\
 &= 8690848/289.951 \\
 &= 299.735 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 120.0/2 \\
 &= 120.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= ( 1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2 ) / (\pi - \beta + \sin(\beta) \cos(\beta) ) \\
 &= ( 1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2 ) / (\pi - 2.094 + \sin(120.0) \cos(120.0) ) \\
 &= 0.2035
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.204 \cdot 24955.559 \\
 &= 5078.9971 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= ( Fh/As ) = 2.0289 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 145.7983 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\
 &= 2133.0 - 1503.0 \cdot \sin(120.0/2) / 1.0472 \\
 &= 890.029 \text{ mm.}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 4520.5444 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= ( M \cdot C1 / I ) = 6.8719 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 161.9981 \text{ N./mm}^2
 \end{aligned}$$

#### Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{ 3( Q + \text{SaddleWt} ) \text{BasePlateWidth} / ( 4 \cdot \text{BasePlateLength} \cdot \text{AllStress} ) } \\
 &= \sqrt{ 3(24956 + 1100)310.0 / ( 4 \cdot 2790.0 \cdot 161.998 ) } \\
 &= 11.465 \text{ mm.}
 \end{aligned}$$

#### Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [ Web Length ]:} \\
 &= 2 \cdot \cos( 90 - \text{Saddle Angle}/2 ) ( \text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk} ) \\
 &= 2 \cdot \cos( 90 - 120.0/2 ) ( 1500.0 + 10.0 + 10.0 ) \\
 &= 2632.717 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / ( \text{Nr ribs} - 1 ) \\
 &= 2632.717 / ( 6 - 1 ) \\
 &= 526.543 \text{ mm.}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot \text{Bpwid} / 2 \\
 &= 526.5434 \cdot 310.0/2 \\
 &= 816.142 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= ( Q + \text{SaddleWt} ) / ( \text{BasePlateLength} \cdot \text{BasePlateWidth} ) \\
 &= 26055.539 / ( 2790.0 \cdot 310.0 ) \\
 &= 3.013 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 816.1 \cdot 3.01 \\
 &= 2458.669 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Area of the Rib and Web [Ar]:} \\
 &= \text{Rib Area} + \text{Web Area} \\
 &= 52.0 + 52.654 \\
 &= 104.654 \text{ cm}^2
 \end{aligned}$$

$$\text{Compressive Stress [Sc]:}$$

$$\begin{aligned}
 &= P/Ar \\
 &= 2458.7/104.6544 \\
 &= 2.304 \text{ N./mm}^2
 \end{aligned}$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|         | B    | D     | Y   | A    | AY  | Io        |
|---------|------|-------|-----|------|-----|-----------|
| Rib+Web | 20.0 | 280.0 | ... | 56.0 | ... | 0.366E+04 |

Rib dimension [D]:

$$\begin{aligned}
 &= \text{Saddle Width} - \text{Web Thickness} \\
 &= 280.0 - 20.0 \\
 &= 260.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned}
 &= AY / A \\
 &= 0.0/104.654 \\
 &= 0.000 \text{ mm.}
 \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned}
 &= \text{Saddle Width} / 2 \\
 &= 280.0/2 \\
 &= 140.000 \text{ mm.}
 \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned}
 &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\
 &= \sqrt{(3658.7/104.654)} \\
 &= 59.127 \text{ mm.}
 \end{aligned}$$

Length of Outer Rib [L]:

$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 2133.0 - \cos(120.0/2) (1500.0 + 10.0 + 10.0) - 26.0 \\
 &= 1347.000 \text{ mm.}
 \end{aligned}$$

Intermediate Term [Cc]:

$$\begin{aligned}
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/243.0)} \\
 &= 127.447
 \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned}
 &= KL/r \\
 &= 1 \cdot 1347.0/59.127 \\
 &= 22.782
 \end{aligned}$$

Bending Moment [Rm]:

$$\begin{aligned}
 &= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 2516.8 / (2 \cdot 2790.0) \cdot 526.543 \cdot 1347.0/2 \\
 &= 159.951 \text{ Kg-m.}
 \end{aligned}$$

Compressive Allowable,  $KL/r < Cc$  (  $22.7816 < 127.4472$  ) per AISC E2-1 [Sca]:

$$\begin{aligned}
 &= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3)) \\
 &= (1 - (22.78)^2 / (2 \cdot 127.45^2)) 243 / \\
 &\quad (5/3 + 3 \cdot (22.78) / (8 \cdot 127.45) - (22.78^3) / (8 \cdot 127.45^3)) \\
 &= 138 \text{ N./mm}^2
 \end{aligned}$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$\begin{aligned}
 &= Sc/Sca + (Rm \cdot C1 / I) / Sba \\
 &= 2.3/137.98 + (159.95 \cdot 140.0/36586676) / 162.0 \\
 &= 0.054
 \end{aligned}$$

#### Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

|     | B     | D     | Y   | A     | AY  | Io        |
|-----|-------|-------|-----|-------|-----|-----------|
| Rib | 20.0  | 260.0 | 0.0 | 52.0  | 0.0 | 0.366E+04 |
| Web | 526.5 | 20.0  | 0.0 | 105.3 | 0.0 | 35.1      |



|        |     |     |     |       |     |           |
|--------|-----|-----|-----|-------|-----|-----------|
| Totals | ... | ... | ... | 157.3 | ... | 0.369E+04 |
|--------|-----|-----|-----|-------|-----|-----------|

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/157.309$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2133.0 - \sqrt{(1520.0 + 10.0)^2 - (526.543/2)^2} - 26.0$$

$$= 609.974 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3692.4/157.309)}$$

$$= 48.448 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 609.974/48.448$$

$$= 12.590$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2516.751/(2 \cdot 2790.0)$$

$$= 0.451 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.451 \cdot 526.543 \cdot 609.974$$

$$= 144.864 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 144.864/263745.469$$

$$= 5.386 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < C_c$  ( 12.5901 < 127.4472 ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.59)^2 / (2 \cdot 127.45^2)) 243 /$$

$$(5/3 + 3 \cdot (12.59) / (8 \cdot 127.45) - (12.59)^3 / (8 \cdot 127.45^3))$$

$$= 141.9 \text{ N./mm}^2$$

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 3.05/141.94 + (144.86 \cdot 140.0/3692.434) / 162.0$$

$$= 0.055$$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |              |               |
|--------------------------------------------|--------------|---------------|
| Total Number of Bolts per BasePlate        | Nbolts       | 10            |
| Total Number of Bolts in Tension/Baseplate | Nbt          | 5             |
| Bolt Design Temperature                    |              | 21.1 °C       |
| Bolt Material Specification                |              | SA-307 A      |
| Bolt Allowable Stress                      | Stba         | 138.00 N./mm2 |
| Bolt Corrosion Allowance                   | Bca          | 1.5 mm.       |
| Distance from Bolts to Edge                | Edgedis      | 50.8 mm.      |
| User Defined Area in a Single Bolt         | UserBoltArea | 2.4000 cm2    |
| BasePlate Allowable Stress                 | S            | 138.00 N./mm2 |
| Area Available in a Single Bolt            | BltArea      | 1.6476 cm2    |
| Saddle Load Q0 (Weight)                    | Q0           | 22997.0 Kg    |
| Saddle Load QL (Wind/Seismic contribution) | QL           | 994.1 Kg      |
| Maximum Transverse Force                   | Ft           | 1258.4 Kg     |
| Maximum Longitudinal Force                 | Fl           | 2516.8 Kg     |
| Saddle Bolted to Steel Foundation          | No           |               |

Shear Stress in a Single Bolt, Longitudinal Direction [ $\tau_b, l$ ]:  
=  $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$   
=  $2516.8 / (1.648 \cdot 10)$   
=  $15.0 \text{ N./mm}^2$ . Must be less than  $82.8 \text{ N./mm}^2$ .

Shear Stress in a Single Bolt, Transverse Direction [ $\tau_b, t$ ]:  
=  $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$   
=  $1258.4 / (1.648 \cdot 10)$   
=  $7.5 \text{ N./mm}^2$ . Must be less than  $82.8 \text{ N./mm}^2$ .

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [ $Bltarearl$ ]:  
=  $0.0$  ( $QO > QL \rightarrow$  No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [ $Bltarears$ ]:  
=  $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$   
=  $2516.75 / (82.8 \cdot 10.0)$   
=  $0.2981 \text{ cm}^2$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [ $Rmom$ ]:  
=  $B \cdot F_t + \text{Sum of X Moments}$   
=  $2133.0 \cdot 1258.38 + 0.0$   
=  $2684.17 \text{ Kg-m}$ .

Eccentricity ( $e$ ):  
=  $Rmom / QO$   
=  $2684.17 / 22996.97$   
=  $116.72 \text{ mm}$ .  $< B_{plen}/6 \rightarrow$  No Uplift in Transverse direction

Bolt Area due to Transverse Load [ $Bltareart$ ]:  
=  $0$  (No Uplift)

Required Area of a Single Bolt [ $Bltarear$ ]:  
=  $\max[Bltarearl, Bltarears, Bltareart]$   
=  $\max[0.0, 0.2981, 0.0]$   
=  $0.2981 \text{ cm}^2$

#### **MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: D**

Govrn.  $thk, t_g = 10.0, t_r = 7.0, c = 3.0 \text{ mm.}, E^* = 0.85$   
Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.85$ , Temp. Reduction =  $8^\circ \text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve D -48  $^\circ \text{C}$   
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -48  $^\circ \text{C}$

*Review notes about nozzle loadings and supports in the Warnings report.*

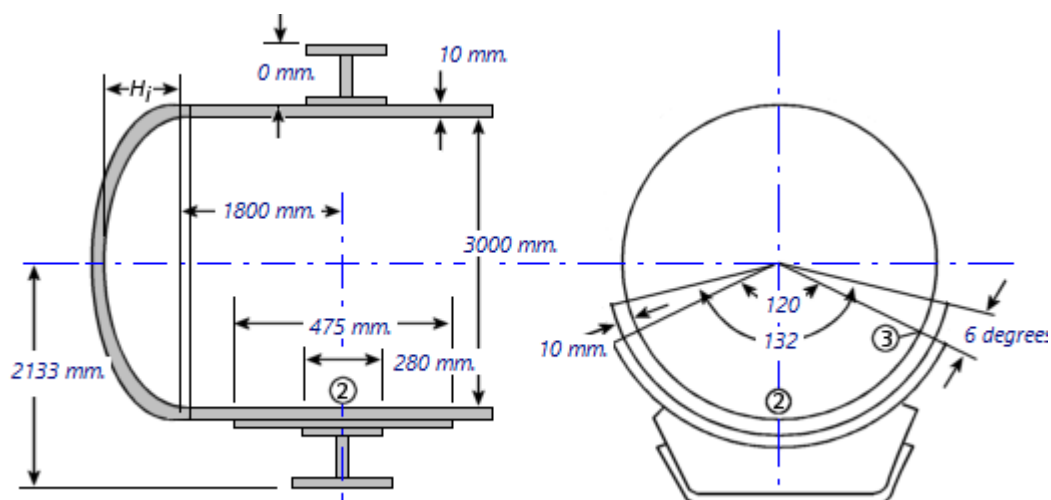
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## ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

### Input and Calculated Values:

|                                               |                  |         |         |
|-----------------------------------------------|------------------|---------|---------|
| Vessel Mean Radius                            | Rm               | 1505.00 | mm.     |
| Shell Thickness used in this Case             | t                | 10.000  | mm.     |
| Stiffened Vessel Length per 4.15.6            | L                | 9000.00 | mm.     |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1800.00 | mm.     |
| Saddle Width                                  | b1 or b          | 280.00  | mm.     |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees |
| Wear Plate Width                              | b2 or b1         | 475.00  | mm.     |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees |
| Wear Plate Thickness                          | e2 or tr         | 10.0    | mm.     |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm2  |
| Inside or Mean Depth of Head                  | Hi or hm         | 755.00  | mm.     |
| Shell Allowable Stress used in Calculation    |                  | 234.00  | N./mm2  |
| Head Allowable Stress used in Calculation     |                  | 234.00  | N./mm2  |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |         |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |         |
| Distance from Saddle Base to Centerline       | B                | 2133.00 | mm.     |
| Coefficient of Friction                       | mu               | 0.00    |         |



### Results for Horizontal Vessel Analysis:

|                                           |          |      |
|-------------------------------------------|----------|------|
| Saddle Force Q, Test Case, no Ext. Forces | 46243.68 | Kgf  |
| Pressure used in Analysis                 | P 7.099  | bars |

| Horizontal Vessel Analysis Results: |  | Actual<br>N./mm^2 | Allowable<br>N./mm^2 |
|-------------------------------------|--|-------------------|----------------------|
| Long. Stress at Top of Midspan      |  | 51.46             | 234.00               |
| Long. Stress at Bottom of Midspan   |  | 55.39             | 234.00               |
| Long. Stress at Top of Saddles      |  | 78.54             | 234.00               |
| Long. Stress at Bottom of Saddles   |  | 39.50             | 234.00               |
| Tangential Shear in Shell           |  | 19.04             | 138.58               |
| Circ. Stress at Horn of Saddle      |  | 132.27            | 350.99               |
| Circ. Compressive Stress in Shell   |  | 3.66              | 234.00               |

### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$   
 $= 3.0 (1140.9/2 + 0) \cdot 2133.0/2632.7173$   
 $= 1386.6 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$   
 $= \max(622.44, 0) \cdot 2133.0/5400.0$   
 $= 245.9 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$   
 $= 44857 + \max(246, 1387, 0, 0)$   
 $= 46243.7 \text{ Kg}$

Longitudinal Wind Force [F]:  
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 766.08 (3.05 + \pi/4 (3.18 \cdot 1.2)^2)$   
 $= 12207.991 \text{ N}$

#### Summary of Loads at the Base of this Saddle:

|                                         |          |     |
|-----------------------------------------|----------|-----|
| Vertical Load (including saddle weight) | 47343.66 | Kgf |
| Transverse Shear Load                   | 570.47   | Kgf |
| Longitudinal Shear Load                 | 622.44   | Kgf |

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$   
 $= \min(2(3000.0/2 + 10.0 + 10.0) \sin(120.0/2), 2 \cdot 1505.0)$   
 $= 2632.717 \text{ mm}$

#### The Computed K values from Table 4.15.1:

|              |              |              |              |
|--------------|--------------|--------------|--------------|
| K1 = 0.1066  | K2 = 1.1707  | K3 = 0.8799  | K4 = 0.4011  |
| K5 = 0.7603  | K6 = 0.0529  | K7 = 0.0529  | K8 = 0.3405  |
| K9 = 0.2711  | K10 = 0.0581 | K1* = 0.1923 | K6p = 0.0434 |
| K7p = 0.0434 |              |              |              |

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to  $R_m/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$   
 $= -46244 \cdot 1800.0 [1 - (1 - 1800.0/9000.0 + (1505.0^2 - 755.0^2)/(2 \cdot 1800.0 \cdot 9000.0)) / (1 + (4 \cdot 755.0)/(3 \cdot 9000.0))]$   
 $= -19430.6 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:  
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m)/3L) - 4a/L)$   
 $= 46244 \cdot 9000/4 ((1 + 2(1505^2 - 755^2)/(9000^2)) / (1 + (4 \cdot 755)/(3 \cdot 9000)) - 4 \cdot 1800/9000)$   
 $= 14259.3 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:  
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$   
 $= 7.099 \cdot 1505.0/(2 \cdot 10.0) - 14259.3/(\pi \cdot 1505.0^2 \cdot 10.0)$   
 $= 51.46 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:  
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$   
 $= 7.099 \cdot 1505.0/(2 \cdot 10.0) + 14259.3/(\pi \cdot 1505.0^2 \cdot 10.0)$   
 $= 55.39 \text{ N/mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma_{*3}$ ]:  
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$   
 $= 7.099 \cdot 1505.0/(2 \cdot 10.0) - 19430.6/(0.1066 \cdot \pi \cdot 1505.0^2 \cdot 10.0)$   
 $= 78.54 \text{ N/mm}^2$



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma_{*4}$ ]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 7.099 \cdot 1505.0 / (2 \cdot 10.0) + 19430.6 / (0.1923 \cdot \pi \cdot 1505.0^2 \cdot 10.0)$$

$$= 39.50 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 46244(9000.0 - 2 \cdot 1800.0) / (9000.0 + (4 \cdot 755.0 / 3))$$

$$= 24955.0 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 24954.95 / (1505.0 \cdot 10.0)$$

$$= 19.04 \text{ N./mm}^2$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)}$$

$$= 95.689 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(280.0 + 1.56 \cdot \sqrt{(1505.0 \cdot 10.0)}, 2 \cdot 1800.0)$$

$$= 471.38 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 46244 \cdot 0.1 / (471.378(10.0 + 1.0 \cdot 10.0))$$

$$= -3.66 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle,  $L < 8R_m$  (4.15.29) [ $\sigma_7, r$ ]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -46244 / (4(10.0 + 1.0 \cdot 10.0)471.378) -$$

$$12 \cdot 0.053 \cdot 46244 \cdot 1505.0 / (9000.0(10.0 + 1.0 \cdot 10.0)^2)$$

$$= -132.27 \text{ N./mm}^2$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |        |
|--------------------------------------|--------|-----------|--------|
| Baseplate Length                     | Bplen  | 2790.0000 | mm.    |
| Baseplate Thickness                  | Bpthk  | 26.0000   | mm.    |
| Baseplate Width                      | Bpwid  | 310.0000  | mm.    |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 6         |        |
| Rib Thickness                        | Ribtk  | 20.0000   | mm.    |
| Web Thickness                        | Webtk  | 20.0000   | mm.    |
| Web Location                         | Webloc | Center    |        |
| Saddle Yield Stress                  | Sy     | 260.0     | N./mm2 |
| Height of Web at Center              | Hw,c   | 607.0     | mm.    |
| Friction Coefficient                 | mu     | 0.000     |        |

#### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 475.0)10.0^3) / 12$$

$$= 5.550 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 5.55 + 66.606(281.387 - 5.0)^2$$

$$= 50885.516 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 666.1    | 10.0     | 5.0      | 66.6                 | 33303.0                | 0.509E+05                                |
| Wearplate | 475.0    | 10.0     | 15.0     | 47.5                 | 71250.0                | 0.337E+05                                |
| Web       | 20.0     | 587.0    | 313.5    | 117.4                | 3680490.2              | 0.349E+05                                |
| BasePlate | 310.0    | 26.0     | 620.0    | 80.6                 | 4997200.0              | 0.925E+05                                |

|        |     |     |     |       |           |           |
|--------|-----|-----|-----|-------|-----------|-----------|
| Totals | ... | ... | ... | 312.1 | 8782243.0 | 0.212E+06 |
|--------|-----|-----|-----|-------|-----------|-----------|

Distance to Centroid [C1]:

$$= AY / A$$

$$= 8782243/312.106$$

$$= 281.387 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 46243.68$$

$$= 9411.5908 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.7596 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2133.0 - 1500.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 892.510 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 8400.1055 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 10.9350 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

**Minimum Thickness of Baseplate per Moss:**

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(46244 + 1100) \cdot 310.0) / (4 \cdot 2790.0 \cdot 173.33)}$$

$$= 14.941 \text{ mm.}$$

**Calculation of Axial Load, Intermediate Values and Compressive Stress:**

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1500.0 + 10.0 + 10.0)$$

$$= 2632.717 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2632.717 / (6 - 1)$$

$$= 526.543 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 526.5434 \cdot 310.0/2$$

$$= 816.142 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 47343.664 / (2790.0 \cdot 310.0)$$

$$= 5.474 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 816.1 \cdot 5.47$$

$$= 4467.472 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 52.0 + 52.654$$

$$= 104.654 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 4467.5/104.6544$$

$$= 4.186 \text{ N./mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|         | B    | D     | Y   | A    | AY  | Io        |
|---------|------|-------|-----|------|-----|-----------|
| Rib+Web | 20.0 | 280.0 | ... | 56.0 | ... | 0.366E+04 |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 280.0 - 20.0$$

$$= 260.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/104.654$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3658.7/104.654)}$$

$$= 59.127 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133.0 - \cos(120.0/2) (1500.0 + 10.0 + 10.0) - 26.0$$

$$= 1347.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1347.0/59.127$$

$$= 22.782$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 622.4 / (2 \cdot 2790.0) \cdot 526.543 \cdot 1347.0/2$$

$$= 39.559 \text{ Kg-m.}$$

Compressive Allowable,  $KL/r < C_c$  (  $22.7816 < 123.2107$  ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.78)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (22.78) / (8 \cdot 123.21) - (22.78^3) / (8 \cdot 123.21^3))$$

$$= 147.3 \text{ N./mm}^2$$

#### AISC Unity Check of Outside Ribs ( must be $\leq 1$ )

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.19/147.27 + (39.56 \cdot 140.0/36586676) / 173.33$$

$$= 0.037$$

#### Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

|  | B | D | Y | A | AY | Io |
|--|---|---|---|---|----|----|
|--|---|---|---|---|----|----|



|        |       |       |     |       |     |           |
|--------|-------|-------|-----|-------|-----|-----------|
| Rib    | 20.0  | 260.0 | 0.0 | 52.0  | 0.0 | 0.366E+04 |
| Web    | 526.5 | 20.0  | 0.0 | 105.3 | 0.0 | 35.1      |
| Totals | ...   | ...   | ... | 157.3 | ... | 0.369E+04 |

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/157.309$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0/2$$

$$= 140.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 2133.0 - \sqrt{(1520.0 + 10.0)^2 - (526.543/2)^2} - 26.0$$

$$= 609.974 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3692.4/157.309)}$$

$$= 48.448 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 609.974/48.448$$

$$= 12.590$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 622.444/(2 \cdot 2790.0)$$

$$= 0.112 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.112 \cdot 526.543 \cdot 609.974$$

$$= 35.828 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 35.828/263745.469$$

$$= 1.332 \text{ N./mm}^2$$

Compressive Allowable,  $KL/r < C_c$  ( 12.5901 < 123.2107 ) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.59)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (12.59) / (8 \cdot 123.21) - (12.59^3) / (8 \cdot 123.21^3))$$

$$= 151.7 \text{ N./mm}^2$$

**AISC Unity Check of Inside Ribs ( must be  $\leq 1$  )**

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 5.54/151.71 + (35.83 \cdot 140.0/3692.434) / 173.33$$

$$= 0.044$$

#### **Input Data for Base Plate Bolting Calculations:**

|                                            |              |               |
|--------------------------------------------|--------------|---------------|
| Total Number of Bolts per BasePlate        | Nbolts       | 10            |
| Total Number of Bolts in Tension/Baseplate | Nbt          | 5             |
| Bolt Material Specification                |              | SA-307 A      |
| Bolt Allowable Stress                      | Stba         | 48.30 N./mm2  |
| Bolt Corrosion Allowance                   | Bca          | 1.5 mm.       |
| Distance from Bolts to Edge                | Edgedis      | 50.8 mm.      |
| User Defined Area in a Single Bolt         | UserBoltArea | 2.4000 cm2    |
| BasePlate Allowable Stress                 | S            | 138.00 N./mm2 |
| Area Available in a Single Bolt            | BltArea      | 1.6476 cm2    |
| Saddle Load Q0 (Weight)                    | Q0           | 45957.1 Kg    |
| Saddle Load QL (Wind/Seismic contribution) | QL           | 245.9 Kg      |
| Maximum Transverse Force                   | Ft           | 570.5 Kg      |



Maximum Longitudinal Force F1 622.4 Kg  
 Saddle Bolted to Steel Foundation No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:  
 $= F_l / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 622.4 / ( 1.648 \cdot 10 )$   
 $= 3.7 \text{ N./mm}^2$ . Must be less than 29.0 N./mm<sup>2</sup>.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:  
 $= F_t / ( \text{Bolt Area} \cdot \text{Number of Bolts} )$   
 $= 570.5 / ( 1.648 \cdot 10 )$   
 $= 3.4 \text{ N./mm}^2$ . Must be less than 29.0 N./mm<sup>2</sup>.

#### **Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:  
 $= 0.0$  (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:  
 $= F_l / ( \text{BoltShearAllowable} \cdot \text{Nbols} )$   
 $= 622.44 / ( 28.98 \cdot 10.0 )$   
 $= 0.2106 \text{ cm}^2$

#### **Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:  
 $= B \cdot F_t + \text{Sum of X Moments}$   
 $= 2133.0 \cdot 570.47 + 0.0$   
 $= 1216.84 \text{ Kg-m.}$

Eccentricity (e):  
 $= R_{mom} / QO$   
 $= 1216.84 / 45957.1$   
 $= 26.48 \text{ mm.} < B_{plen} / 6$  --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:  
 $= 0$  (No Uplift)

Required Area of a Single Bolt [Bltarear]:  
 $= \max[Bltarearl, Bltarears, Bltareart]$   
 $= \max[0.0, 0.2106, 0.0]$   
 $= 0.2106 \text{ cm}^2$

#### **ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:**

##### **Input and Calculated Values:**

|                                               |                  |         |         |
|-----------------------------------------------|------------------|---------|---------|
| Vessel Mean Radius                            | Rm               | 1505.00 | mm.     |
| Shell Thickness used in this Case             | t                | 10.000  | mm.     |
| Stiffened Vessel Length per 4.15.6            | L                | 9000.00 | mm.     |
| Distance from Saddle to Vessel tangent        | a1 or a          | 1800.00 | mm.     |
| Saddle Width                                  | b1 or b          | 280.00  | mm.     |
| Saddle Bearing Angle                          | delta or theta   | 120.00  | degrees |
| Wear Plate Width                              | b2 or b1         | 475.00  | mm.     |
| Wear Plate Bearing Angle                      | delta2 or theta1 | 132.00  | degrees |
| Wear Plate Thickness                          | e2 or tr         | 10.0    | mm.     |
| Wear Plate Allowable Stress                   | fw or Sr         | 138.00  | N./mm2  |
| Inside or Mean Depth of Head                  | Hi or hm         | 755.00  | mm.     |
| Shell Allowable Stress used in Calculation    |                  | 234.00  | N./mm2  |
| Head Allowable Stress used in Calculation     |                  | 234.00  | N./mm2  |
| Circumferential Efficiency in Plane of Saddle |                  | 1.00    |         |
| Circumferential Efficiency at Mid-Span        |                  | 1.00    |         |
| Distance from Saddle Base to Centerline       | B                | 2133.00 | mm.     |
| Coefficient of Friction                       | mu               | 0.40    |         |

### Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 45893.52 Kgf  
 Pressure used in Analysis P 7.099 bars

| Horizontal Vessel Analysis Results: | Actual<br>N./mm <sup>2</sup> | Allowable<br>N./mm <sup>2</sup> |
|-------------------------------------|------------------------------|---------------------------------|
| Long. Stress at Top of Midspan      | 51.47                        | 234.00                          |
| Long. Stress at Bottom of Midspan   | 55.37                        | 234.00                          |
| Long. Stress at Top of Saddles      | 78.35                        | 234.00                          |
| Long. Stress at Bottom of Saddles   | 39.60                        | 234.00                          |
| Tangential Shear in Shell           | 18.89                        | 138.58                          |
| Circ. Stress at Horn of Saddle      | 131.27                       | 350.99                          |
| Circ. Compressive Stress in Shell   | 3.63                         | 234.00                          |

### Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:  
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$   
 $= 3.0(1140.9/2 + 0) \cdot 2133.0/2632.7173$   
 $= 1386.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:  
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$   
 $= \max(622.44, 0) \cdot 2133.0/5400.0$   
 $= 245.9 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:  
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$   
 $= 44507 + \max(246, 1387, 0, 0)$   
 $= 45893.5 \text{ Kgf}$

Longitudinal Wind Force [Fl]:  
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$   
 $= 1.1 \cdot 766.08(3.05 + \pi/4(3.18 \cdot 1.2)^2)$   
 $= 12207.991 \text{ N}$

### Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 46993.50 Kgf  
 Transverse Shear Load 570.47 Kgf  
 Longitudinal Shear Load 622.44 Kgf

Note: Wear Plate is Welded to the Shell,  $k = 0.1$

Saddle Dimension [E]:  
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$   
 $= \min(2(3000.0/2 + 10.0 + 10.0)\sin(120.0/2), 2 \cdot 1505.0)$   
 $= 2632.717 \text{ mm.}$

### The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011  
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405  
 K9 = 0.2711 K10 = 0.0581 K1\* = 0.1923 K6p = 0.0434  
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to  $Rm/2$ .

Moment per Equation 4.15.1 [M1]:  
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$   
 $= -45894 \cdot 1800.0 [1 - (1 - 1800.0/9000.0 + (1505.0^2 - 755.0^2)/(2 \cdot 1800.0 \cdot 9000.0))/(1 + (4 \cdot 755.0)/(3 \cdot 9000.0))]$   
 $= -19283.5 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:



$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 45894 \cdot 9000 / 4 \cdot ((1 + 2(1505^2 - 755^2) / (9000^2)) / (1 + (4 \cdot 755) / (3 \cdot 9000)) - 4 \cdot 1800 / 9000) \\
 &= 14151.3 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [ $\sigma_1$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 7.099 \cdot 1505.0 / (2 \cdot 10.0) - 14151.3 / (\pi \cdot 1505.0^2 \cdot 10.0) \\
 &= 51.47 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [ $\sigma_2$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= 7.099 \cdot 1505.0 / (2 \cdot 10.0) + 14151.3 / (\pi \cdot 1505.0^2 \cdot 10.0) \\
 &= 55.37 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [ $\sigma^*3$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 7.099 \cdot 1505.0 / (2 \cdot 10.0) - 19283.5 / (0.1066 \cdot \pi \cdot 1505.0^2 \cdot 10.0) \\
 &= 78.35 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [ $\sigma^*4$ ]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t) \\
 &= 7.099 \cdot 1505.0 / (2 \cdot 10.0) + 19283.5 / (0.1923 \cdot \pi \cdot 1505.0^2 \cdot 10.0) \\
 &= 39.60 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 45894(9000.0 - 2 \cdot 1800.0) / (9000.0 + (4 \cdot 755.0 / 3)) \\
 &= 24766.0 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [ $\tau_2$ ]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 24765.99 / (1505.0 \cdot 10.0) \\
 &= 18.89 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [ $x_1, x_2$ ]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(1505.0 \cdot 10.0)} \\
 &= 95.689 \text{ mm.}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a) \\
 &= \min(280.0 + 1.56 \cdot \sqrt{(1505.0 \cdot 10.0)}, 2 \cdot 1800.0) \\
 &= 471.38 \text{ mm.}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [ $\eta$ ]:

= 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [ $\sigma_6, r$ ]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 45894 \cdot 0.1 / (471.378(10.0 + 1.0 \cdot 10.0)) \\
 &= -3.63 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle,  $L < 8Rm$  (4.15.29) [ $\sigma_7, r^*$ ]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2) \\
 &= -45894 / (4(10.0 + 1.0 \cdot 10.0)471.378) - \\
 &\quad 12 \cdot 0.053 \cdot 45894 \cdot 1505.0 / (9000.0(10.0 + 1.0 \cdot 10.0)^2) \\
 &= -131.27 \text{ N./mm}^2
 \end{aligned}$$

#### Results for Vessel Ribs, Web and Base:

|                                      |        |           |        |
|--------------------------------------|--------|-----------|--------|
| Baseplate Length                     | Bplen  | 2790.0000 | mm.    |
| Baseplate Thickness                  | Bpthk  | 26.0000   | mm.    |
| Baseplate Width                      | Bpwid  | 310.0000  | mm.    |
| Number of Ribs ( inc. outside ribs ) | Nribs  | 6         |        |
| Rib Thickness                        | Ribtk  | 20.0000   | mm.    |
| Web Thickness                        | Webtk  | 20.0000   | mm.    |
| Web Location                         | Webloc | Center    |        |
| Saddle Yield Stress                  | Sy     | 260.0     | N./mm2 |
| Height of Web at Center              | Hw,c   | 607.0     | mm.    |

Friction Coefficient mu 0.400

### Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:  

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1500.0 \cdot 10.0)} + 475.0)10.0^3)/12$$

$$= 5.550 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:  

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 5.55 + 66.606(281.387 - 5.0)^2$$

$$= 50885.516 \text{ cm}^4$$

|           | B<br>mm. | D<br>mm. | Y<br>mm. | A<br>cm <sup>2</sup> | AY<br>mm. <sup>3</sup> | I + AD <sup>(2)</sup><br>cm <sup>4</sup> |
|-----------|----------|----------|----------|----------------------|------------------------|------------------------------------------|
| Shell     | 666.1    | 10.0     | 5.0      | 66.6                 | 33303.0                | 0.509E+05                                |
| Wearplate | 475.0    | 10.0     | 15.0     | 47.5                 | 71250.0                | 0.337E+05                                |
| Web       | 20.0     | 587.0    | 313.5    | 117.4                | 3680490.2              | 0.349E+05                                |
| BasePlate | 310.0    | 26.0     | 620.0    | 80.6                 | 4997200.0              | 0.925E+05                                |
| Totals    | ...      | ...      | ...      | 312.1                | 8782243.0              | 0.212E+06                                |

Distance to Centroid [C1]:  

$$= AY / A$$

$$= 8782243/312.106$$

$$= 281.387 \text{ mm.}$$

Angle [β]:  

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:  

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:  

$$= K1 \cdot Q$$

$$= 0.204 \cdot 45893.516$$

$$= 9340.3252 \text{ Kgf}$$

Tension Stress, St = ( Fh/As ) = 3.7311 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 0.6 \* Yield Str = 155.9972 N./mm<sup>2</sup>

Saddle Splitting Dimension [d]:  

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 2133.0 - 1500.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 892.510 \text{ mm.}$$

Bending Moment, M = Fh \* d = 8336.4990 Kg-m.

Bending Stress, Sb = ( M \* C1 / I ) = 10.8522 N./mm<sup>2</sup>  
 Allowed Stress, Sa = 2/3 \* Yield Str = 173.3302 N./mm<sup>2</sup>

### Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(45894 + 1100)310.0 / (4 \cdot 2790.0 \cdot 173.33))}$$

$$= 14.885 \text{ mm.}$$

### Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [ Web Length ]:  

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1500.0 + 10.0 + 10.0)$$

$$= 2632.717 \text{ mm.}$$

Distance between Ribs [e]:  

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2632.717 / (6 - 1)$$

$$= 526.543 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{pwid} / 2$$

$$= 526.5434 \cdot 310.0 / 2$$

$$= 816.142 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 46993.5 / (2790.0 \cdot 310.0)$$

$$= 5.433 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 816.1 \cdot 5.43$$

$$= 4434.429 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 52.0 + 52.654$$

$$= 104.654 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 4434.4 / 104.6544$$

$$= 4.155 \text{ N/mm}^2$$

#### Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

|         | B    | D     | Y   | A    | AY  | Io        |
|---------|------|-------|-----|------|-----|-----------|
| Rib+Web | 20.0 | 280.0 | ... | 56.0 | ... | 0.366E+04 |

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 280.0 - 20.0$$

$$= 260.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 104.654$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 280.0 / 2$$

$$= 140.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{3658.7 / 104.654}$$

$$= 59.127 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2133.0 - \cos(120.0/2) (1500.0 + 10.0 + 10.0) - 26.0$$

$$= 1347.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1347.0 / 59.127$$

$$= 22.782$$

**Bending Moment [Rm]:**

$$= FI / ( 2 \cdot B_{plen} ) \cdot e \cdot L / 2$$

$$= 622.4 / ( 2 \cdot 2790.0 ) \cdot 526.543 \cdot 1347.0 / 2$$

$$= 39.559 \text{ Kg-m.}$$

**Compressive Allowable,  $KL/r < C_c$  ( 22.7816 < 123.2107 ) per AISC E2-1 [Sca]:**

$$= ( 1 - (Klr)^2 / (2 \cdot Cc^2) ) F_y / ( 5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3) )$$

$$= ( 1 - ( 22.78 )^2 / ( 2 \cdot 123.21^2 ) ) 260 /$$

$$( 5/3 + 3 \cdot (22.78) / (8 \cdot 123.21) - ( 22.78^3 ) / (8 \cdot 123.21^3) )$$

$$= 147.3 \text{ N./mm}^2$$

**AISC Unity Check of Outside Ribs ( must be  $\leq 1$  )**

$$= S_c / S_{ca} + ( R_m \cdot C_1 / I ) / S_{ba}$$

$$= 4.16 / 147.27 + ( 39.56 \cdot 140.0 / 36586676 ) / 173.33$$

$$= 0.037$$

**Check of Inside Ribs:**

**Inertia of Saddle, Inner Ribs - Axial Direction**

|        | B     | D     | Y   | A     | AY  | Io        |
|--------|-------|-------|-----|-------|-----|-----------|
| Rib    | 20.0  | 260.0 | 0.0 | 52.0  | 0.0 | 0.366E+04 |
| Web    | 526.5 | 20.0  | 0.0 | 105.3 | 0.0 | 35.1      |
| Totals | ...   | ...   | ... | 157.3 | ... | 0.369E+04 |

**Distance to Centroid from Datum [ytot]:**

$$= AY / A$$

$$= 0.0 / 157.309$$

$$= 0.000 \text{ mm.}$$

**Distance to Centroid [C1]:**

$$= \text{Saddle Width} / 2$$

$$= 280.0 / 2$$

$$= 140.000 \text{ mm.}$$

**Length of Inner Rib [L]:**

$$= \text{Saddle Height} - \sqrt{ (Ro + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 2133.0 - \sqrt{ (1520.0 + 10.0)^2 - (526.543/2)^2 } - 26.0$$

$$= 609.974 \text{ mm.}$$

**Radius of Gyration [r]:**

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3692.4 / 157.309 }$$

$$= 48.448 \text{ mm.}$$

**Slenderness ratio [KL/r]:**

$$= KL / r$$

$$= 1 \cdot 609.974 / 48.448$$

$$= 12.590$$

**Unit Force [Force,u]:**

$$= FI / ( 2 \cdot \text{Baseplate Length} )$$

$$= 622.444 / ( 2 \cdot 2790.0 )$$

$$= 0.112 \text{ Kg/mm.}$$

**Moment at base of inner Rib [Mbase,c]:**

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.112 \cdot 526.543 \cdot 609.974$$

$$= 35.828 \text{ Kg-m.}$$

**Bending Stress due to Transverse Force and Weight Load [ $\sigma_B$ ,base,c]:**

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 35.828 / 263745.469$$

$$= 1.332 \text{ N./mm}^2$$

**Compressive Allowable,  $KL/r < C_c$  ( 12.5901 < 123.2107 ) per AISC E2-1 [Sca]:**

$$= ( 1 - (Klr)^2 / (2 \cdot Cc^2) ) F_y / ( 5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3) )$$

$$= ( 1 - ( 12.59 )^2 / ( 2 \cdot 123.21^2 ) ) 260 /$$

$$( 5/3 + 3 \cdot (12.59) / (8 \cdot 123.21) - ( 12.59^3 ) / (8 \cdot 123.21^3) )$$

= 151.7 N./mm<sup>2</sup>

**AISC Unity Check of Inside Ribs ( must be ≤ 1 )**

= Sc/Sca + ( Mbase,c • C1/I )/Sba  
 = 5.5/151.71 + ( 35.83 • 140.0/3692.434 )/173.33  
 = 0.044

**Input Data for Base Plate Bolting Calculations:**

|                                            |              |                           |
|--------------------------------------------|--------------|---------------------------|
| Total Number of Bolts per BasePlate        | Nbolts       | 10                        |
| Total Number of Bolts in Tension/Baseplate | Nbt          | 5                         |
| Bolt Material Specification                |              | SA-307 A                  |
| Bolt Allowable Stress                      | Stba         | 138.00 N./mm <sup>2</sup> |
| Bolt Corrosion Allowance                   | Bca          | 1.5 mm.                   |
| Distance from Bolts to Edge                | Edgedis      | 50.8 mm.                  |
| User Defined Area in a Single Bolt         | UserBoltArea | 2.4000 cm <sup>2</sup>    |
| BasePlate Allowable Stress                 | S            | 138.00 N./mm <sup>2</sup> |
| Area Available in a Single Bolt            | BlArea       | 1.6476 cm <sup>2</sup>    |
| Saddle Load QO (Weight)                    | QO           | 45606.9 Kg <sub>f</sub>   |
| Saddle Load QL (Wind/Seismic contribution) | QL           | 245.9 Kg <sub>f</sub>     |
| Maximum Transverse Force                   | Ft           | 570.5 Kg <sub>f</sub>     |
| Maximum Longitudinal Force                 | Fl           | 622.4 Kg <sub>f</sub>     |
| Saddle Bolted to Steel Foundation          |              | No                        |

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

= Fl / ( Bolt Area • Number of Bolts )  
 = 622.4 / ( 1.648 • 10 )  
 = 3.7 N./mm<sup>2</sup>. Must be less than 82.8 N./mm<sup>2</sup>.

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

= Ft / ( Bolt Area • Number of Bolts )  
 = 570.5 / ( 1.648 • 10 )  
 = 3.4 N./mm<sup>2</sup>. Must be less than 82.8 N./mm<sup>2</sup>.

**Bolt Area Calculation per Dennis R. Moss**

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 ( QO > QL --> No Uplift in Longitudinal direction )

Bolt Area due to Shear Load [Bltarears]:

= Fl / ( BoltShearAllowable • Nbolts )  
 = 622.44 / ( 82.8 • 10.0 )  
 = 0.0737 cm<sup>2</sup>

**Bolt Area due to Transverse Load:**

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments  
 = 2133.0 • 570.47 + 0.0  
 = 1216.84 Kg-m.

Eccentricity (e):

= Rmom / QO  
 = 1216.84/45606.93  
 = 26.68 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]  
 = max[0.0, 0.0737, 0.0]  
 = 0.0737 cm<sup>2</sup>

*Review notes about nozzle loadings and supports in the Warnings report.*

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**Input, Nozzle Desc: K6-2 Inch**

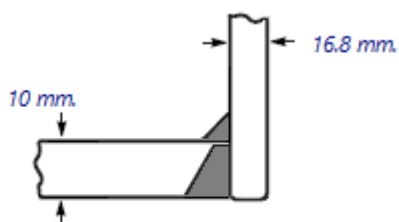
**From: 20**

|                                               |      |         |        |
|-----------------------------------------------|------|---------|--------|
| Pressure for Reinforcement Calculations       | P    | 5.448   | bars   |
| Temperature for Internal Pressure             | Temp | 80      | °C     |
| Parent Material                               |      | SA-516  | 70     |
| Parent Allowable Stress at Temperature        | Sv   | 138.00  | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 138.00  | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00 | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 10.0000 | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 3.0000  | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000  | mm.    |
| Distance from Bottom/Left Tangent             |      | 300.00  | mm.    |
| User Entered Minimum Design Metal Temperature |      | -40.00  | °C     |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |        |
|---------------------------------------------|------|----------------|--------|
| Material                                    |      | SA-105         | +IT    |
| Material UNS Number                         |      | K03504         |        |
| Material Specification/Type                 |      | Forgings       |        |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm2 |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm2 |
| Diameter Basis (for tr calc only)           |      | Inside         |        |
| Layout Angle                                |      | 0.00           | deg    |
| Diameter                                    |      | 2.0000         | in.    |
| Size and Thickness Basis                    |      | Actual         |        |
| Actual Thickness                            | tn   | 16.7640        | mm.    |
| Flange Material                             |      | SA-105         | +IT    |
| Flange Type                                 |      | Long Weld Neck |        |
| Corrosion Allowance                         | can  | 3.0000         | mm.    |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |        |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |        |
| Outside Projection                          | ho   | 330.0000       | mm.    |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.    |
| Groove weld depth between Nozzle and Vessel | Wgnv | 10.0000        | mm.    |
| Inside Projection                           | h    | 0.0000         | mm.    |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.    |
| Flange Class                                |      | 300            |        |
| Flange Grade                                |      | GR 1.1         |        |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: K6-2 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |

**Note:**  
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 0.1124 \text{ mm.}$

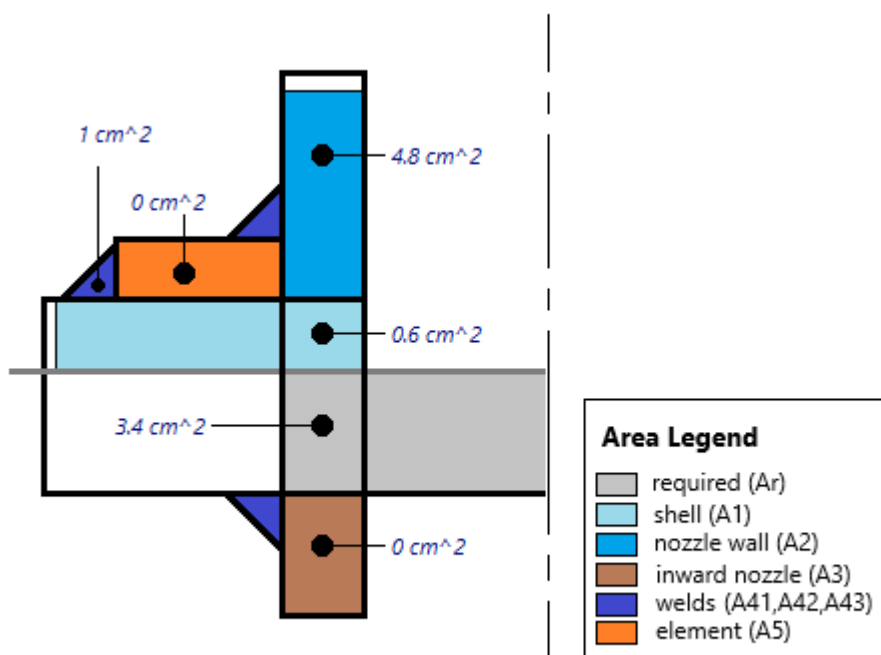
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 113.6000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 56.8000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm²)**

| AREA AVAILABLE, A1 to A5 | MAWP | External | Mapnc |
|--------------------------|------|----------|-------|
|--------------------------|------|----------|-------|



|                       |             |       |    |    |
|-----------------------|-------------|-------|----|----|
| Area Required         | Ar          | 3.378 | NA | NA |
| Area in Shell         | A1          | 0.598 | NA | NA |
| Area in Nozzle Wall   | A2          | 4.778 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000 | NA | NA |
| Area in Welds         | A41+A42+A43 | 1.000 | NA | NA |
| Area in Element       | A5          | 0.000 | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 6.376 | NA | NA |

#### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (56.8 \cdot 5.9475 \cdot 1.0 + 2 \cdot 13.764 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$   
 $= 3.378 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 56.8(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 13.764$   
 $(1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$   
 $= 0.598 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 17.5)(13.76 - 0.11) \cdot 1.$   
 $= 4.778 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$   
 $= 1.000 \text{ cm}^2$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 3.1124 \text{ mm.}$   
 Wall Thickness per UG16(b),  $tr16b = 4.5000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 8.9475 \text{ mm.}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm.}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$   
 Wall Thickness per table UG-45  $tb3 = 7.8000 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 7.8, \max( 8.9475, 4.5 ) ]$   
 $= 7.8000 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 3.1124, 7.8 )$   
 $= 7.8000 \text{ mm.}$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk,  $tg = 10.0$ ,  $tr = 5.948$ ,  $c = 3.0 \text{ mm.}$ ,  $E* = 1.0$   
 Thickness Ratio  $= tr \cdot E*/(tg - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 5.45/51.00 = 0.107

#### Weld Size Calculations, Description: K6-2 Inch

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

#### **Results Per UW-16.1:**

|             | Required Thickness   | Actual Thickness      |
|-------------|----------------------|-----------------------|
| Nozzle Weld | 4.9000 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm. |

#### **Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3.3782 - 0.5978 + 2 \cdot 13.764 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$   
 $= 4320.24 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (4.778 + 0.0 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 8130.62 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (4.778 + 0.0 + 1. + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (4.778 + 0.0 + 1. + 0.0 + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

#### **Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 35.282) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$   
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 9655. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### **Strength of Failure Paths:**

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (9133 + 0 + 9655 + 0) = 18789 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$



= ( 9133 + 9655 + 0 ) = 18789 Kgf

**Summary of Failure Path Calculations:**

|                  |                               |                   |
|------------------|-------------------------------|-------------------|
| Path 1-1 = 24160 | Kgf, must exceed W = 4320 Kgf | or W1 = 8130 Kgf  |
| Path 2-2 = 18788 | Kgf, must exceed W = 4320 Kgf | or W2 = 10842 Kgf |
| Path 3-3 = 18788 | Kgf, must exceed W = 4320 Kgf | or W3 = 10842 Kgf |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5927 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.5927 mm.

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm.*

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**Input, Nozzle Desc: N1-6 Inch**

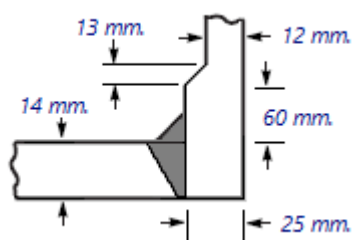
**From: 20**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 |        |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 14.0000   | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.    |
| Distance from Bottom/Left Tangent             |      | 900.00    | mm.    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |        |
|--------------------------------------------------|-----|------------|--------|
| Material                                         |     | SA-105 +IT |        |
| Material UNS Number                              |     | K03504     |        |
| Material Specification/Type                      |     | Forgings   |        |
| Allowable Stress at Temperature                  | Sn  | 138.00     | N./mm2 |
| Allowable Stress At Ambient                      | Sna | 138.00     | N./mm2 |
| Diameter Basis (for tr calc only)                |     | Outside    |        |
| Layout Angle                                     |     | 0.00       | deg    |
| Diameter                                         |     | 6.0000     | in.    |
| Size and Thickness Basis                         |     | Actual     |        |
| Actual Thickness                                 | tn  | 12.0000    | mm.    |
| Flange Material                                  |     | SA-105     |        |
| Hub Height of Integral Nozzle                    | h   | 60.0000    | mm.    |
| Height of Beveled Transition                     | L'  | 13.0000    | mm.    |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 25.0000    | mm.    |
| Corrosion Allowance                              | can | 3.0000     | mm.    |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |        |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |        |
| Outside Projection                               | ho  | 330.0000   | mm.    |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.    |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 10.0000    | mm.    |
| Inside Projection                                | h   | 0.0000     | mm.    |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.    |
| Flange Class                                     |     | 150        |        |
| Flange Grade                                     |     | GR 1.1     |        |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: N1-6 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 6.000 in.  
 Actual Thickness Used in Calculation 0.472 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 5.45 \cdot 76.2 / (138 \cdot 1.0 + 0.4 \cdot 5.45)$   
 $= 0.3003$  mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 22.0$ , less than the hub height, use (e-2)  
 $= 55.0000$  mm.

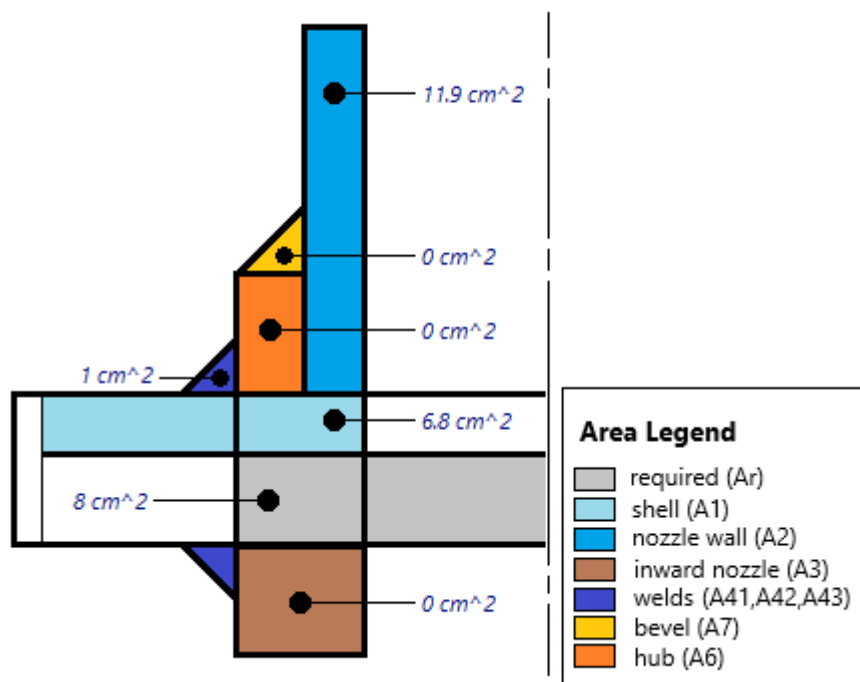
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 268.8000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 134.4000 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 27.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm2)

| AREA AVAILABLE, A1 to A5 |             | MAWP   | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 7.994  | NA       | NA    |
| Area in Shell            | A1          | 6.791  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 11.935 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 0.000  | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 19.725 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (134.4 \cdot 5.9475 \cdot 1.0 + 2 \cdot 22.0 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$$

$$= 7.994 \text{ cm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 134.4(1.0 \cdot 11.0 - 1.0 \cdot 5.948) - 2 \cdot 22.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$$

$$= 6.791 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.5)(22.0 - 0.3) \cdot 1.$$

$$= 11.935 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

*There are no hub area calculations because Figure UG-40 (e-2) was used.*

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $t_a = 3.3003$  mm.  
 Wall Thickness per UG16(b),  $tr_{16b} = 4.5000$  mm.  
 Wall Thickness, shell/head, internal pressure  $tr_{b1} = 8.9475$  mm.  
 Wall Thickness  $tb_1 = \max(tr_{b1}, tr_{16b}) = 8.9475$  mm.  
 Wall Thickness  $tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000$  mm.  
 Wall Thickness per table UG-45  $tb_3 = 9.2200$  mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[ tb_3, \max( tb_1, tb_2 ) ]$   
 $= \min[ 9.22, \max( 8.9475, 4.5 ) ]$   
 $= 8.9475$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max( t_a, t_b )$   
 $= \max( 3.3003, 8.9475 )$   
 $= 8.9475$  mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 37.1, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 307.9 N./mm <sup>2</sup> | Passed |
| Occasional | : | 1.9,  | Allowable | : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear      | : | 23.2, | Allowable | : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, Curve: A**

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 0.3$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= t_r * E^* / (t_g - c) = 0.033$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -3 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D**

Govrn. thk,  $t_g = 14.0$ ,  $t_r = 5.948$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= t_r * E^* / (t_g - c) = 0.541$ , Temp. Reduction = 28 °C

|                                                |        |
|------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve D | -48 °C |
|------------------------------------------------|--------|

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $5.45/19.60 = 0.278$

**Weld Size Calculations, Description: N1-6 Inch**

|                                           |           |             |
|-------------------------------------------|-----------|-------------|
| Intermediate Calc. for nozzle/shell Welds | $T_{min}$ | 11.0000 mm. |
|-------------------------------------------|-----------|-------------|

**Results Per UW-16.1:**

|             |                                |                                  |
|-------------|--------------------------------|----------------------------------|
|             | Required Thickness             | Actual Thickness                 |
| Nozzle Weld | $6.0000 = \text{Min per Code}$ | $7.0700 = 0.7 * W_o \text{ mm.}$ |

**Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (7.9935 - 6.7905 + 2 \cdot 22.0 \cdot 1. \cdot (1.0 \cdot 11.0 - 5.9475)) \cdot 138)$   
 $= 4821.08 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i\text{-Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (11.9346 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 18201.25 \text{ Kgf}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2(\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (11.9346 + 0.0 + 1. + (4.8399)) \cdot 138$   
 $= 25011.87 \text{ Kgf}$

Weld Load [W3]:  
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (11.9346 + 0.0 + 1. + 0.0) \cdot 138$   
 $= 25011.87 \text{ Kgf}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 152.4 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 16506. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 71.7) \cdot (25.0 - 3.0) \cdot 0.7 \cdot 138$   
 $= 48812. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 152.4 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 24928. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (16506 + 48812) = 65318 \text{ Kgf}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (16506 + 0 + 24928 + 0) = 41434 \text{ Kgf}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$   
 $= (16506 + 24928 + 0) = 41434 \text{ Kgf}$

#### Summary of Failure Path Calculations:

|                  |                               |                   |
|------------------|-------------------------------|-------------------|
| Path 1-1 = 65318 | Kgf, must exceed W = 4821 Kgf | or W1 = 18201 Kgf |
| Path 2-2 = 41433 | Kgf, must exceed W = 4821 Kgf | or W2 = 25011 Kgf |
| Path 3-3 = 41433 | Kgf, must exceed W = 4821 Kgf | or W3 = 25011 Kgf |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.9367 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 345.9367 mm.



*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 134.40 mm.*

**Input Echo, WRC297 Item 1, Description: N1-6 Inch :**

|                                      |            |        |
|--------------------------------------|------------|--------|
| Diameter Basis for Cylindrical Shell | ID         |        |
| Shell Internal Corrosion Allowance   | 3.0000     | mm.    |
| Shell External Corrosion Allowance   | 0.0000     | mm.    |
| Shell Diameter                       | 3000.000   | mm.    |
| Shell Thickness                      | 14.0000    | mm.    |
| Shell Stress Concentration Factor    | 1.000      |        |
| Vessel Material                      | SA-516 70  |        |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm2 |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |             |        |
|------------------------------------|-------------|--------|
| Diameter Basis for Nozzle          | OD          |        |
| Nozzle Corrosion Allowance         | 3.0000      | mm.    |
| Nozzle Diameter                    | 178.400     | mm.    |
| Nozzle Thickness                   | 25.0000     | mm.    |
| Nozzle Stress Concentration Factor | 1.000       |        |
| Nozzle Material                    | SA-105      |        |
| Nozzle Cold S.I. Allowable         | SNmc 138.00 | N./mm2 |
| Nozzle Hot S.I. Allowable          | SNmh 138.00 | N./mm2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSained loads.*

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 5.45    | bars  |
| Radial Load              | P  | 1146.82 | Kgf   |
| Circumferential Shear    | Vc | 764.54  | Kgf   |
| Longitudinal Shear       | Vl | 764.54  | Kgf   |
| Circumferential Moment   | Mc | 321.12  | Kg-m. |
| Longitudinal Moment      | Ml | 321.12  | Kg-m. |
| Torsional Moment         | Mt | 458.75  | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (7.136) is < 10  
 check the limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 178.4 + 2 \cdot 1.65 \cdot \sqrt{(1508.5(14.0 - 3.0))}$$

$$= 603.492 \text{ mm.}$$

WRC 297 Curve Access Parameters:

|                         |       |          |     |
|-------------------------|-------|----------|-----|
| Vessel Mean Diameter    | (D) = | 3017.000 | mm. |
| Nozzle Outside Diameter | (d) = | 178.400  | mm. |
| Vessel Thickness used   | (T) = | 11.000   | mm. |
| Nozzle Thickness used   | (t) = | 22.000   | mm. |
| T / t                   | =     | 0.500    |     |
| d / t                   | =     | 10.000   |     |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.979$$

$$Nr/P = 0.106$$



Mr/P = 0.226  
 MØ/P = 0.066  
 NØ/P = 0.135  
 MrD/Mc = 0.307  
 NrDL/Mc = 0.091  
 MØd/Mc = 0.096  
 NØDL/Mc = 0.038  
 MrD/Ml = 0.199  
 NrDL/Ml = 0.088  
 MØD/Ml = 0.064  
 NØDL/Ml = 0.118

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | B1<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -12                  | -12                 | -12                     | -12                    |
| Outplane Bending (P )                                            | -37                  | 37                  | -37                     | 37                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -17                  | -17                 | 17                      | 17                     |
| Outplane Bending (ML)                                            | -56                  | 56                  | 56                      | -56                    |
| Normal Pressure Stress                                           | 74                   | 74                  | 74                      | 74                     |
| Outplane Stress Summary                                          | -49                  | 138                 | 98                      | 60                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | B1<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | -9                   | -9                  | -9                      | -9                     |
| Inplane Bending (P )                                               | -126                 | 126                 | -126                    | 126                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -12                  | -12                 | 12                      | 12                     |
| Inplane Bending (ML)                                               | -173                 | 173                 | 173                     | -173                   |
| Inplane Pressure Stress                                            | 37                   | 37                  | 37                      | 37                     |
| Inplane Stress Summary                                             | -285                 | 314                 | 87                      | -7                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | B1<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 2                    | 2                   | -2                      | -2                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 8                    | 8                   | 8                       | 8                      |
| Shear Stress Summary                                                 | 10                   | 10                  | 5                       | 5                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | B1<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 286                  | 315                 | 100                     | 68                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | D1<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | -12                   | -12                  | -12                    | -12                   |
| Outplane Bending (P )                                                  | -37                   | 37                   | -37                    | 37                    |
| Outplane Membrane (Mc)                                                 | -5                    | -5                   | 5                      | 5                     |
| Outplane Bending (Mc)                                                  | -84                   | 84                   | 84                     | -84                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |



|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Outplane Bending (ML)  | 0  | 0  | 0  | 0  |
| Normal Pressure Stress | 37 | 37 | 37 | 37 |

|                         |      |     |    |     |
|-------------------------|------|-----|----|-----|
| Outplane Stress Summary | -102 | 140 | 77 | -17 |
|-------------------------|------|-----|----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |         |        |         |        |
|----------------------------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE                        | Cu      | Cl     | Du      | Dl     |
| (Stresses parallel to circumferential plane) | Left    | Left   | Right   | Right  |
|                                              | Outside | Inside | Outside | Inside |

|                         |      |     |      |      |
|-------------------------|------|-----|------|------|
| Inplane Membrane (P )   | -9   | -9  | -9   | -9   |
| Inplane Bending (P )    | -126 | 126 | -126 | 126  |
| Inplane Membrane (Mc)   | -13  | -13 | 13   | 13   |
| Inplane Bending (Mc)    | -268 | 268 | 268  | -268 |
| Inplane Membrane (ML)   | 0    | 0   | 0    | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0    | 0    |
| Inplane Pressure Stress | 74   | 74  | 74   | 74   |

|                        |      |     |     |     |
|------------------------|------|-----|-----|-----|
| Inplane Stress Summary | -343 | 446 | 219 | -64 |
|------------------------|------|-----|-----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |         |        |         |        |
|------------------------------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE                          | Cu      | Cl     | Du      | Dl     |
| (Shear stress normal to circumferential plane) | Left    | Left   | Right   | Right  |
|                                                | Outside | Inside | Outside | Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -2 | -2 | 2 | 2 |
| Torsional Shear (Mt) | 8  | 8  | 8 | 8 |

|                      |   |   |    |    |
|----------------------|---|---|----|----|
| Shear Stress Summary | 5 | 5 | 10 | 10 |
|----------------------|---|---|----|----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities)  | Left    | Left   | Right   | Right  |
|                       | Outside | Inside | Outside | Inside |

|                        |     |     |     |    |
|------------------------|-----|-----|-----|----|
| Two * Max Shear Stress | 343 | 446 | 220 | 66 |
|------------------------|-----|-----|-----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |         |        |         |        |
|----------------------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE               | Au      | Al     | Bu      | Bl     |
| (Stresses in the hoop direction) | Top     | Top    | Bottom  | Bottom |
|                                  | Outside | Inside | Outside | Inside |

|                      |     |     |     |     |
|----------------------|-----|-----|-----|-----|
| Hoop Membrane (P )   | -12 | -12 | -12 | -12 |
| Hoop Bending (P )    | 0   | 0   | 0   | 0   |
| Hoop Membrane (Mc)   | 0   | 0   | 0   | 0   |
| Hoop Bending (Mc)    | 0   | 0   | 0   | 0   |
| Hoop Membrane (ML)   | -17 | -17 | 17  | 17  |
| Hoop Bending (ML)    | 0   | 0   | 0   | 0   |
| Hoop Pressure Stress | 1   | 2   | 1   | 2   |

|                     |     |     |   |   |
|---------------------|-----|-----|---|---|
| Hoop Stress Summary | -28 | -27 | 6 | 7 |
|---------------------|-----|-----|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |         |        |         |        |
|-----------------------------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE                      | Au      | Al     | Bu      | Bl     |
| (Stresses Normal to pipe cross-section) | Top     | Top    | Bottom  | Bottom |
|                                         | Outside | Inside | Outside | Inside |

|                       |     |    |     |     |
|-----------------------|-----|----|-----|-----|
| Axial Membrane (P )   | -1  | -1 | -1  | -1  |
| Axial Bending (P )    | -24 | 24 | -24 | 24  |
| Axial Membrane (Mc)   | 0   | 0  | 0   | 0   |
| Axial Bending (Mc)    | 0   | 0  | 0   | 0   |
| Axial Membrane (ML)   | -8  | -8 | 8   | 8   |
| Axial Bending (ML)    | -33 | 33 | 33  | -33 |
| Axial Pressure Stress | 0   | 0  | 0   | 0   |

|                      |     |    |    |    |
|----------------------|-----|----|----|----|
| Axial Stress Summary | -66 | 49 | 17 | -1 |
|----------------------|-----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |    |    |    |    |
|--------------------|----|----|----|----|
| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|



| (Shear stress) | Outside | Inside | Outside | Inside |
|----------------|---------|--------|---------|--------|
|----------------|---------|--------|---------|--------|

|                      |   |   |    |    |
|----------------------|---|---|----|----|
| Shear due to (Vc)    | 1 | 1 | -1 | -1 |
| Shear due to (Vl)    | 0 | 0 | 0  | 0  |
| Shear due to Torsion | 4 | 4 | 4  | 4  |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 5 | 5 | 2 | 2 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
|--------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 67 | 77 | 18 | 10 |
|------------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                      |     |     |     |     |
|----------------------|-----|-----|-----|-----|
| Hoop Membrane (P )   | -12 | -12 | -12 | -12 |
| Hoop Bending (P )    | 0   | 0   | 0   | 0   |
| Hoop Membrane (Mc)   | -5  | -5  | 5   | 5   |
| Hoop Bending (Mc)    | 0   | 0   | 0   | 0   |
| Hoop Membrane (ML)   | 0   | 0   | 0   | 0   |
| Hoop Bending (ML)    | 0   | 0   | 0   | 0   |
| Hoop Pressure Stress | 1   | 2   | 1   | 2   |

|                     |     |     |    |    |
|---------------------|-----|-----|----|----|
| Hoop Stress Summary | -16 | -15 | -5 | -4 |
|---------------------|-----|-----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                       |     |    |     |     |
|-----------------------|-----|----|-----|-----|
| Axial Membrane (P )   | -1  | -1 | -1  | -1  |
| Axial Bending (P )    | -24 | 24 | -24 | 24  |
| Axial Membrane (Mc)   | -8  | -8 | 8   | 8   |
| Axial Bending (Mc)    | -57 | 57 | 57  | -57 |
| Axial Membrane (ML)   | 0   | 0  | 0   | 0   |
| Axial Bending (ML)    | 0   | 0  | 0   | 0   |
| Axial Pressure Stress | 0   | 0  | 0   | 0   |

|                      |     |    |    |     |
|----------------------|-----|----|----|-----|
| Axial Stress Summary | -89 | 72 | 41 | -24 |
|----------------------|-----|----|----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
|-----------------------------------------|---------------|--------------|---------------|--------------|

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Shear due to (Vc)    | 0  | 0  | 0 | 0 |
| Shear due to (Vl)    | -1 | -1 | 1 | 1 |
| Shear due to Torsion | 4  | 4  | 4 | 4 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 5 | 5 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
|-----------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 90 | 88 | 47 | 26 |
|------------------------|----|----|----|----|

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |    |    |     |     |    |    |
|------------------------------------|------------------|-----|----|----|-----|-----|----|----|
|                                    | Au               | Al  | Bu | Bl | Cu  | Cl  | Du | Dl |
| Circ. Pm (SUS)                     | 74               | 74  | 74 | 74 | 74  | 74  | 74 | 74 |
| Circ. Pl (SUS)                     | -29              | -29 | 5  | 5  | -22 | -22 | 4  | 4  |



|                 |       |       |       |      |       |       |       |      |
|-----------------|-------|-------|-------|------|-------|-------|-------|------|
| Circ. Q (SUS)   | -93   | 93    | 19    | -19  | -394  | 394   | 142   | -142 |
| Long. Pm (SUS)  | 37    | 37    | 37    | 37   | 37    | 37    | 37    | 37   |
| Long. Pl (SUS)  | -21   | -21   | 3     | 3    | -17   | -17   | -7    | -7   |
| Long. Q (SUS)   | -299  | 299   | 47    | -47  | -121  | 121   | 47    | -47  |
| Shear Pm (SUS)  | 0     | 0     | 0     | 0    | 0     | 0     | 0     | 0    |
| Shear Pl (SUS)  | 2     | 2     | -2    | -2   | -2    | -2    | 2     | 2    |
| Shear Q (SUS)   | 8     | 8     | 8     | 8    | 8     | 8     | 8     | 8    |
| Pm (SUS)        | 74.0  | 74.0  | 74.0  | 74.0 | 74.0  | 74.0  | 74.0  | 74.0 |
| Pm+Pl (SUS)     | 45.1  | 45.1  | 79.1  | 79.1 | 52.1  | 52.1  | 78.1  | 78.1 |
| Pm+Pl+Q (Total) | 283.4 | 315.6 | 100.6 | 68.1 | 342.1 | 446.1 | 220.7 | 66.0 |

#### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 74.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 79.10     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 446.12    | 414.00         | Failed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

| Type of Stress Int. Location | Stress Values at |      |      |      |      |      |      |      |
|------------------------------|------------------|------|------|------|------|------|------|------|
|                              | Au               | Al   | Bu   | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 1                | 2    | 1    | 2    | 1    | 2    | 1    | 2    |
| Circ. Pl (SUS)               | -29              | -29  | 5    | 5    | -17  | -17  | -7   | -7   |
| Circ. Q (SUS)                | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Long. Pl (SUS)               | -9               | -9   | 7    | 7    | -9   | -9   | 7    | 7    |
| Long. Q (SUS)                | -57              | 57   | 9    | -9   | -81  | 81   | 33   | -33  |
| Shear Pm (SUS)               | 0                | 0    | 0    | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 1                | 1    | -1   | -1   | -1   | -1   | 1    | 1    |
| Shear Q (SUS)                | 4                | 4    | 4    | 4    | 4    | 4    | 4    | 4    |
| Pm (SUS)                     | 1.0              | 2.0  | 1.0  | 2.0  | 1.0  | 2.0  | 1.0  | 2.0  |
| Pm+Pl (SUS)                  | 28.1             | 27.1 | 7.6  | 8.0  | 16.1 | 15.2 | 13.2 | 12.2 |
| Pm+Pl+Q (Total)              | 66.6             | 75.7 | 16.8 | 10.8 | 90.1 | 87.2 | 47.1 | 27.1 |

#### Nozzle Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 2.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 28.05     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 90.12     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: K4-2 Inch**

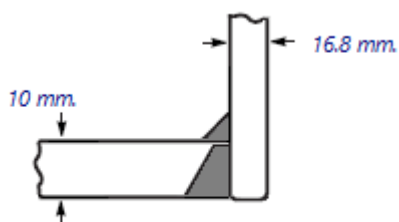
**From: 20**

|                                               |      |         |        |
|-----------------------------------------------|------|---------|--------|
| Pressure for Reinforcement Calculations       | P    | 5.448   | bars   |
| Temperature for Internal Pressure             | Temp | 80      | °C     |
| Parent Material                               |      | SA-516  | 70     |
| Parent Allowable Stress at Temperature        | Sv   | 138.00  | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 138.00  | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00 | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 10.0000 | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 3.0000  | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000  | mm.    |
| Distance from Bottom/Left Tangent             |      | 1400.00 | mm.    |
| User Entered Minimum Design Metal Temperature |      | -40.00  | °C     |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |        |
|---------------------------------------------|------|----------------|--------|
| Material                                    |      | SA-105         | +IT    |
| Material UNS Number                         |      | K03504         |        |
| Material Specification/Type                 |      | Forgings       |        |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm2 |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm2 |
| Diameter Basis (for tr calc only)           |      | Inside         |        |
| Layout Angle                                |      | 0.00           | deg    |
| Diameter                                    |      | 2.0000         | in.    |
| Size and Thickness Basis                    |      | Actual         |        |
| Actual Thickness                            | tn   | 16.7640        | mm.    |
| Flange Material                             |      | SA-105         | +IT    |
| Flange Type                                 |      | Long Weld Neck |        |
| Corrosion Allowance                         | can  | 3.0000         | mm.    |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |        |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |        |
| Outside Projection                          | ho   | 330.0000       | mm.    |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.    |
| Groove weld depth between Nozzle and Vessel | Wgnv | 10.0000        | mm.    |
| Inside Projection                           | h    | 0.0000         | mm.    |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.    |
| Flange Class                                |      | 300            |        |
| Flange Grade                                |      | GR 1.1         |        |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: K4-2 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |

**Note:**  
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 0.1124 \text{ mm.}$

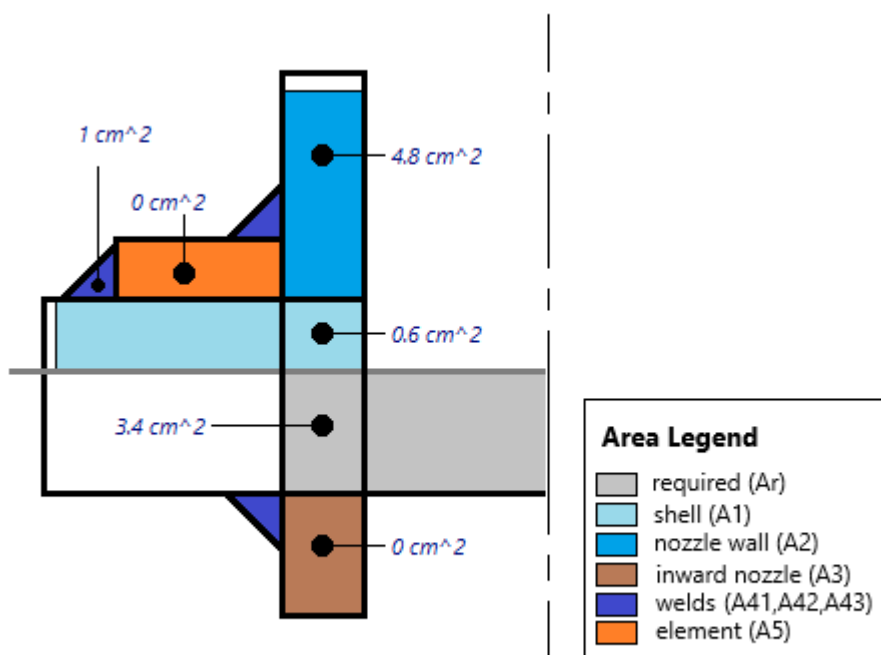
#### UG-40, Limits of Reinforcement: [Internal Pressure]

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 113.6000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 56.8000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

#### Results of Nozzle Reinforcement Area Calculations: (cm²)

| AREA AVAILABLE, A1 to A5 | MAWP | External | Mapnc |
|--------------------------|------|----------|-------|
|--------------------------|------|----------|-------|



|                       |             |       |    |    |
|-----------------------|-------------|-------|----|----|
| Area Required         | Ar          | 3.378 | NA | NA |
| Area in Shell         | A1          | 0.598 | NA | NA |
| Area in Nozzle Wall   | A2          | 4.778 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000 | NA | NA |
| Area in Welds         | A41+A42+A43 | 1.000 | NA | NA |
| Area in Element       | A5          | 0.000 | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 6.376 | NA | NA |

### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

### The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (56.8 \cdot 5.9475 \cdot 1.0 + 2 \cdot 13.764 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$$

$$= 3.378 \text{ cm}^2$$

### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 13.764$$

$$(1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$$

$$= 0.598 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tnp)(tn - trn)fr2$$

$$= (2 \cdot 17.5)(13.76 - 0.11) \cdot 1.$$

$$= 4.778 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures } ta = 3.1124 \text{ mm.}$$

$$\text{Wall Thickness per UG16(b), } tr16b = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, internal pressure } trb1 = 8.9475 \text{ mm.}$$

$$\text{Wall Thickness } tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm.}$$

$$\text{Wall Thickness } tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness per table UG-45 } tb3 = 7.8000 \text{ mm.}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 7.8, \max( 8.9475, 4.5 ) ]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 3.1124, 7.8 )$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govern. thk,  $t_g = 10.0$ ,  $t_r = 5.948$ ,  $c = 3.0 \text{ mm.}$ ,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 5.45/51.00 = 0.107

#### Weld Size Calculations, Description: K4-2 Inch

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

#### **Results Per UW-16.1:**

|             | Required Thickness   | Actual Thickness      |
|-------------|----------------------|-----------------------|
| Nozzle Weld | 4.9000 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm. |

#### **Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3.3782 - 0.5978 + 2 \cdot 13.764 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$   
 $= 4320.24 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (4.778 + 0.0 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 8130.62 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (4.778 + 0.0 + 1. + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (4.778 + 0.0 + 1. + 0.0 + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

#### **Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 35.282) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$   
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 9655. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### **Strength of Failure Paths:**

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (9133 + 0 + 9655 + 0) = 18789 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$



= ( 9133 + 9655 + 0 ) = 18789 Kgf

**Summary of Failure Path Calculations:**

|                  |                               |                   |
|------------------|-------------------------------|-------------------|
| Path 1-1 = 24160 | Kgf, must exceed W = 4320 Kgf | or W1 = 8130 Kgf  |
| Path 2-2 = 18788 | Kgf, must exceed W = 4320 Kgf | or W2 = 10842 Kgf |
| Path 3-3 = 18788 | Kgf, must exceed W = 4320 Kgf | or W3 = 10842 Kgf |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5927 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.5927 mm.

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm.*

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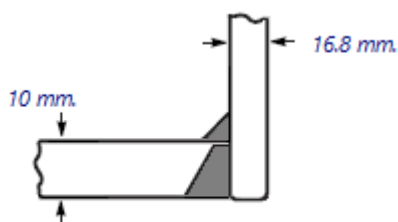
**Input, Nozzle Desc: K5-2 Inch**
**From: 20**

|                                               |      |         |                    |
|-----------------------------------------------|------|---------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.448   | bars               |
| Temperature for Internal Pressure             | Temp | 80      | °C                 |
| Parent Material                               |      | SA-516  | 70                 |
| Parent Allowable Stress at Temperature        | Sv   | 138.00  | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00  | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00 | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 10.0000 | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 3.0000  | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000  | mm.                |
| Distance from Bottom/Left Tangent             |      | 1800.00 | mm.                |
| User Entered Minimum Design Metal Temperature |      | -40.00  | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-105         | +IT                |
| Material UNS Number                         |      | K03504         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | 0.00           | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Actual         |                    |
| Actual Thickness                            | tn   | 16.7640        | mm.                |
| Flange Material                             |      | SA-105         | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 3.0000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 330.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 10.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Overall MAWP.


**Reinforcement CALCULATION, Description: K5-2 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 0.1124 \text{ mm.}$

**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 113.6000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 56.8000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:

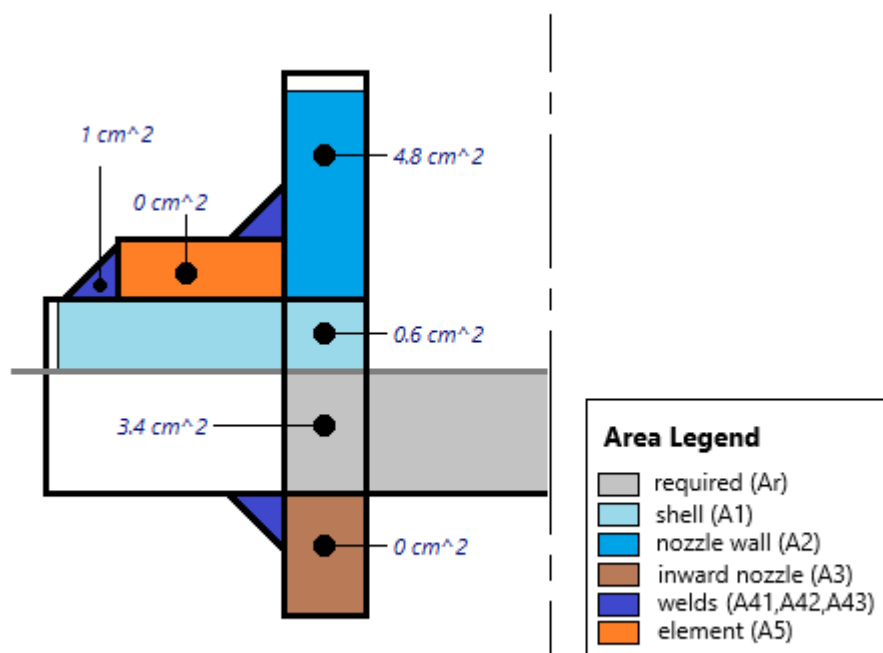
$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm<sup>2</sup>)**

| AREA AVAILABLE, A1 to A5 | MAWP | External | Mapnc |
|--------------------------|------|----------|-------|
|--------------------------|------|----------|-------|



|                       |             |       |    |    |
|-----------------------|-------------|-------|----|----|
| Area Required         | Ar          | 3.378 | NA | NA |
| Area in Shell         | A1          | 0.598 | NA | NA |
| Area in Nozzle Wall   | A2          | 4.778 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000 | NA | NA |
| Area in Welds         | A41+A42+A43 | 1.000 | NA | NA |
| Area in Element       | A5          | 0.000 | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 6.376 | NA | NA |

#### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$  UG-37(c)  
 $= (56.8 \cdot 5.9475 \cdot 1.0 + 2 \cdot 13.764 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$   
 $= 3.378 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 56.8(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 13.764$   
 $(1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$   
 $= 0.598 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 17.5)(13.76 - 0.11) \cdot 1.$   
 $= 4.778 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$   
 $= 1.000 \text{ cm}^2$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 3.1124 \text{ mm}.$   
 Wall Thickness per UG16(b),  $tr16b = 4.5000 \text{ mm}.$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 8.9475 \text{ mm}.$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm}.$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}.$   
 Wall Thickness per table UG-45  $tb3 = 7.8000 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 7.8, \max( 8.9475, 4.5 ) ]$   
 $= 7.8000 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 3.1124, 7.8 )$   
 $= 7.8000 \text{ mm}.$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk,  $tg = 10.0$ ,  $tr = 5.948$ ,  $c = 3.0 \text{ mm}.$ ,  $E* = 1.0$   
 Thickness Ratio  $= tr \cdot E*/(tg - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 5.45/51.00 = 0.107

#### Weld Size Calculations, Description: K5-2 Inch

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

#### **Results Per UW-16.1:**

|             | Required Thickness   | Actual Thickness      |
|-------------|----------------------|-----------------------|
| Nozzle Weld | 4.9000 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm. |

#### **Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3.3782 - 0.5978 + 2 \cdot 13.764 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$   
 $= 4320.24 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (4.778 + 0.0 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 8130.62 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (4.778 + 0.0 + 1. + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (4.778 + 0.0 + 1. + 0.0 + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

#### **Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 35.282) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$   
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 9655. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### **Strength of Failure Paths:**

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (9133 + 0 + 9655 + 0) = 18789 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$



= ( 9133 + 9655 + 0 ) = 18789 Kgf

**Summary of Failure Path Calculations:**

|                                                |                   |
|------------------------------------------------|-------------------|
| Path 1-1 = 24160 Kgf, must exceed W = 4320 Kgf | or W1 = 8130 Kgf  |
| Path 2-2 = 18788 Kgf, must exceed W = 4320 Kgf | or W2 = 10842 Kgf |
| Path 3-3 = 18788 Kgf, must exceed W = 4320 Kgf | or W3 = 10842 Kgf |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5927 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.5927 mm.

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 56.80 mm.*

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**Input, Nozzle Desc: M1-24 Inch**

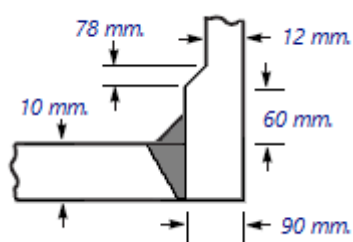
**From: 20**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 10.0000   | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 2400.00   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |                    |
|--------------------------------------------------|-----|------------|--------------------|
| Material                                         |     | SA-105 +IT |                    |
| Material UNS Number                              |     | K03504     |                    |
| Material Specification/Type                      |     | Forgings   |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Inside     |                    |
| Layout Angle                                     |     | 0.00       | deg                |
| Diameter                                         |     | 24.0000    | in.                |
| Size and Thickness Basis                         |     | Actual     |                    |
| Actual Thickness                                 | tn  | 12.0000    | mm.                |
| Flange Material                                  |     | SA-105     |                    |
| Hub Height of Integral Nozzle                    | h   | 60.0000    | mm.                |
| Height of Beveled Transition                     | L'  | 78.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 90.0000    | mm.                |
| Corrosion Allowance                              | can | 3.0000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |                    |
| Outside Projection                               | ho  | 430.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 10.0000    | mm.                |
| Inside Projection                                | h   | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.                |
| This is a Manway, Coupling or Access Opening.    |     |            |                    |
| Flange Class                                     |     | 150        |                    |
| Flange Grade                                     |     | GR 1.1     |                    |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: M1-24 Inch**



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 24.000 in.  
Actual Thickness Used in Calculation 0.472 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
=  $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
= 5.9475 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
=  $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
=  $5.45 \cdot 307.8 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
= 1.2180 mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
= ( Hub Thickness - Neck Thickness ) /  $\tan(30)$   
=  $78.0 / 0.5773$   
= 135.1001 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

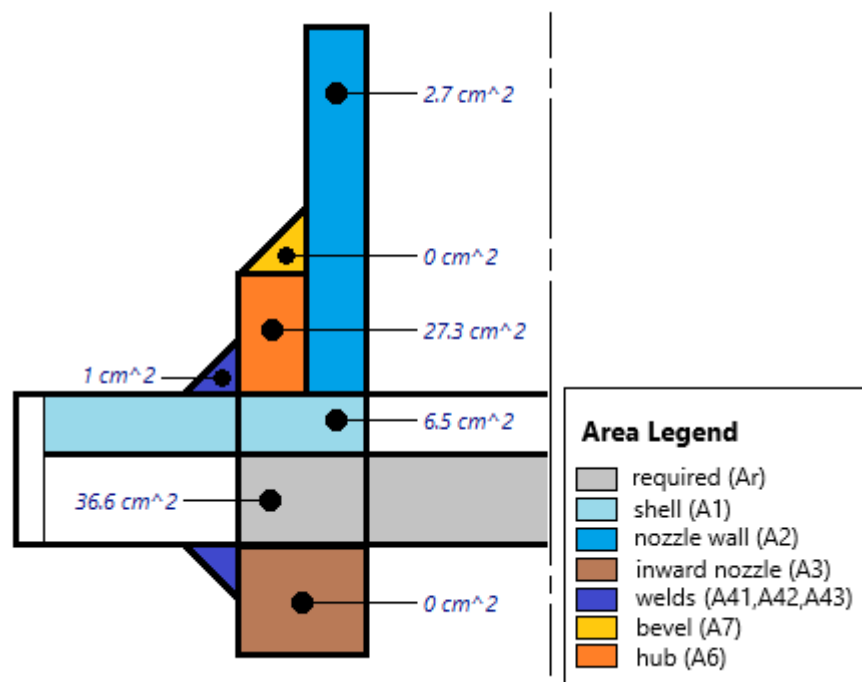
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |           |     |
|-------------------------------------------------|------|-----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | Dl   | 1231.2001 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 615.6000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000   | mm. |

Weld Strength Reduction Factor [fr1]:  
=  $\min(1, S_n / S_v)$   
=  $\min(1, 138.0 / 138.0)$   
= 1.000

Weld Strength Reduction Factor [fr2]:  
=  $\min(1, S_n / S_v)$   
=  $\min(1, 138.0 / 138.0)$   
= 1.000

Weld Strength Reduction Factor [fr3]:  
=  $\min(fr2, fr4)$   
=  $\min(1., 1.0)$   
= 1.000



#### Results of Nozzle Reinforcement Area Calculations: (cm2)

| AREA AVAILABLE, A1 to A5 |             | MAWP   | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 36.613 | NA       | NA    |
| Area in Shell            | A1          | 6.479  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 2.724  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 27.299 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 37.502 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (615.6 \cdot 5.9475 \cdot 1.0 + 2 \cdot 87.0 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.)) \\
 &= 36.613 \text{ cm}^2
 \end{aligned}$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 615.6(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 87.0 \\
 &\quad (1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.) \\
 &= 6.479 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 17.5)(9.0 - 1.22) \cdot 1. \\
 &= 2.724 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

Area Available in the Hub Section [A6]:  

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(17.5, 430.0, 60.0)) \cdot (90.0 - 12.0) \cdot 1. )$$

$$= 27.299 \text{ cm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures  $t_a = 4.2180 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= t_a$   
 $= 4.2180 \text{ mm.}$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, Curve: A

Govrn. thk,  $t_g = 12.0$ ,  $t_r = 1.218$ ,  $c = 3.0 \text{ mm.}$ ,  $E^* = 1.0$   
 Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.135$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -3 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk,  $t_g = 10.0$ ,  $t_r = 5.948$ ,  $c = 3.0 \text{ mm.}$ ,  $E^* = 1.0$   
 Thickness Ratio  $= t_r \cdot E^* / (t_g - c) = 0.85$ , Temp. Reduction = 8 °C

|                                                |        |
|------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve D | -48 °C |
|------------------------------------------------|--------|

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $5.45/19.60 = 0.278$

##### Weld Size Calculations, Description: M1-24 Inch

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 7.0000 \text{ mm.}$

##### Results Per UW-16.1:

|             | Required Thickness           | Actual Thickness                     |
|-------------|------------------------------|--------------------------------------|
| Nozzle Weld | $4.9000 = 0.7 \cdot t_{min}$ | $7.0700 = 0.7 \cdot W_o \text{ mm.}$ |

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A_1 + 2 \cdot t_n \cdot fr_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (36.6131 - 6.4791 + 2 \cdot 87.0 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$$

$$= 44980.95 \text{ Kg}$$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A_2 + A_4 - (W_i - C_{an} / .707)^2 \cdot fr_2) \cdot S_v$$

$$= (30.0231 + 1. - 0.0 \cdot 1. ) \cdot 138$$

$$= 43655.07 \text{ Kg}$$

Weld Load [W2]:

$$= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot fr_1)) \cdot S_v$$

$$= (30.0231 + 0.0 + 1. + (12.1797) ) \cdot 138$$

= 60794.19 Kgf

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S$$

$$= (30.0231 + 0.0 + 1. + 0.0 + (12.1797)) \cdot 138$$

$$= 60794.19 \text{ Kgf}$$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 789.6 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 85519. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 351.3) \cdot (90.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 945772. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 789.6 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 90408. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SONW + SNW) = (85519 + 945772) = 1031292 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (85519 + 0 + 90408 + 0) = 175927 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Sonw + Tngw + Sinw)$$

$$= (85519 + 90408 + 0) = 175927 \text{ Kgf}$$

#### Summary of Failure Path Calculations:

|                                                   |                   |
|---------------------------------------------------|-------------------|
| Path 1-1 = 1031291 Kgf, must exceed W = 44980 Kgf | or W1 = 43655 Kgf |
| Path 2-2 = 175927 Kgf, must exceed W = 44980 Kgf  | or W2 = 60794 Kgf |
| Path 3-3 = 175927 Kgf, must exceed W = 44980 Kgf  | or W3 = 60794 Kgf |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 52.8881 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 492.8880 mm.

*This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 615.60 mm.*

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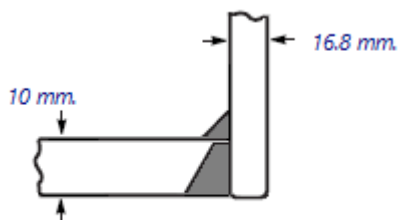
**Input, Nozzle Desc: D1-2 Inch**
**From: 20**

|                                               |      |         |                    |
|-----------------------------------------------|------|---------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.448   | bars               |
| Temperature for Internal Pressure             | Temp | 80      | °C                 |
| Parent Material                               |      | SA-516  | 70                 |
| Parent Allowable Stress at Temperature        | Sv   | 138.00  | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00  | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00 | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 10.0000 | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 3.0000  | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000  | mm.                |
| Distance from Bottom/Left Tangent             |      | 2500.00 | mm.                |
| User Entered Minimum Design Metal Temperature |      | -40.00  | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-105         | +IT                |
| Material UNS Number                         |      | K03504         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | 180.00         | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Actual         |                    |
| Actual Thickness                            | tn   | 16.7640        | mm.                |
| Flange Material                             |      | SA-105         | +IT                |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 3.0000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 330.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 10.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Overall MAWP.


**Reinforcement CALCULATION, Description: D1-2 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |

**Note:**  
 Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 0.1124 \text{ mm.}$

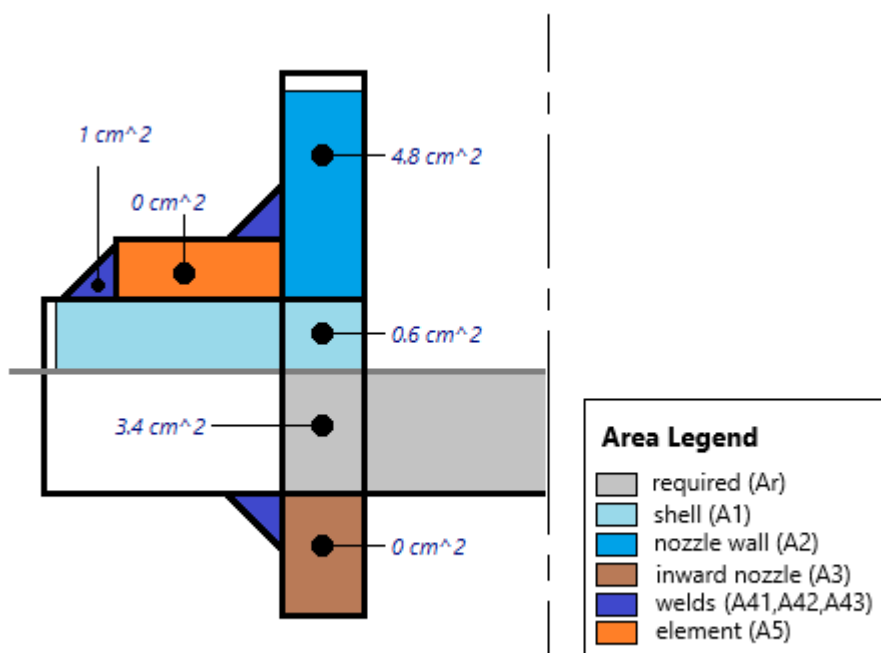
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 113.6000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 56.8000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm<sup>2</sup>)**

| AREA AVAILABLE, A1 to A5 | MAWP | External | Mapnc |
|--------------------------|------|----------|-------|
|--------------------------|------|----------|-------|



|                       |             |       |    |    |
|-----------------------|-------------|-------|----|----|
| Area Required         | Ar          | 3.378 | NA | NA |
| Area in Shell         | A1          | 0.598 | NA | NA |
| Area in Nozzle Wall   | A2          | 4.778 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000 | NA | NA |
| Area in Welds         | A41+A42+A43 | 1.000 | NA | NA |
| Area in Element       | A5          | 0.000 | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 6.376 | NA | NA |

### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

### The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (56.8 \cdot 5.9475 \cdot 1.0 + 2 \cdot 13.764 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$$

$$= 3.378 \text{ cm}^2$$

### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 13.764$$

$$(1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$$

$$= 0.598 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 17.5)(13.76 - 0.11) \cdot 1.$$

$$= 4.778 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures } ta = 3.1124 \text{ mm.}$$

$$\text{Wall Thickness per UG16(b), } tr16b = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, internal pressure } trb1 = 8.9475 \text{ mm.}$$

$$\text{Wall Thickness } tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm.}$$

$$\text{Wall Thickness } tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness per table UG-45 } tb3 = 7.8000 \text{ mm.}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 7.8, \max( 8.9475, 4.5 ) ]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 3.1124, 7.8 )$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk, tg = 10.0, tr = 5.948, c = 3.0 mm., E\* = 1.0  
Thickness Ratio = tr \* E\*/(tg - c) = 0.85, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 5.45/51.00 = 0.107

#### Weld Size Calculations, Description: D1-2 Inch

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

#### **Results Per UW-16.1:**

|             | Required Thickness   | Actual Thickness      |
|-------------|----------------------|-----------------------|
| Nozzle Weld | 4.9000 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm. |

#### **Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3.3782 - 0.5978 + 2 \cdot 13.764 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$   
 $= 4320.24 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (4.778 + 0.0 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 8130.62 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (4.778 + 0.0 + 1. + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (4.778 + 0.0 + 1. + 0.0 + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

#### **Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 35.282) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$   
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 9655. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### **Strength of Failure Paths:**

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (9133 + 0 + 9655 + 0) = 18789 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$



$$= ( 9133 + 9655 + 0 ) = 18789 \text{ Kgf}$$

**Summary of Failure Path Calculations:**

|                                                |                   |
|------------------------------------------------|-------------------|
| Path 1-1 = 24160 Kgf, must exceed W = 4320 Kgf | or W1 = 8130 Kgf  |
| Path 2-2 = 18788 Kgf, must exceed W = 4320 Kgf | or W2 = 10842 Kgf |
| Path 3-3 = 18788 Kgf, must exceed W = 4320 Kgf | or W3 = 10842 Kgf |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5927 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.5927 mm.

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## ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 10.000 in.  
 Actual Thickness Used in Calculation 0.472 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 5.45 \cdot 127.0 / (138 \cdot 1.0 + 0.4 \cdot 5.45)$   
 $= 0.5006$  mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 33.0 / 0.5773$   
 $= 57.1577$  mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

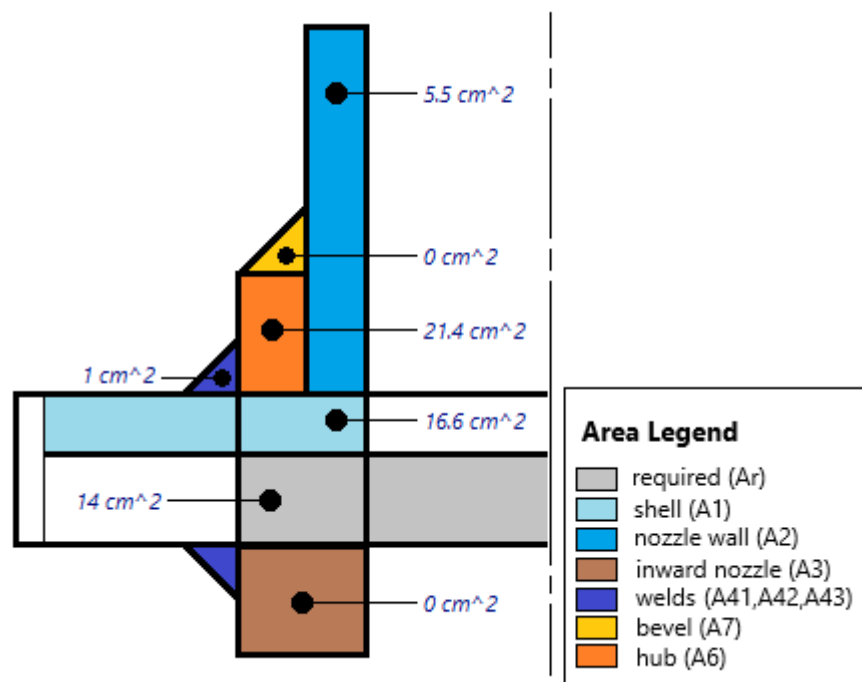
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 472.0000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 236.0000 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 32.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 138.0 / 138.0 )$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 138.0 / 138.0 )$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min( fr2, fr4 )$   
 $= \min( 1., 1.0 )$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm2)

| AREA AVAILABLE, A1 to A5 |             | MAWP   | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 14.036 | NA       | NA    |
| Area in Shell            | A1          | 16.644 | NA       | NA    |
| Area in Nozzle Wall      | A2          | 5.525  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 21.450 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 44.618 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (236.0 \cdot 5.9475 \cdot 1.0 + 2 \cdot 42.0 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$   
 $= 14.036 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 236.0(1.0 \cdot 13.0 - 1.0 \cdot 5.948) - 2 \cdot 42.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$   
 $= 16.644 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 32.5)(9.0 - 0.5) \cdot 1.$   
 $= 5.525 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$   
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:  

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(32.5, 330.0, 75.0)) \cdot (45.0 - 12.0) \cdot 1. )$$

$$= 21.450 \text{ cm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 3.5006 \text{ mm.}$   
 Wall Thickness per UG16(b),  $tr16b = 4.5000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 8.9475 \text{ mm.}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm.}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$   
 Wall Thickness per table UG-45  $tb3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 11.103, \max( 8.9475, 4.5 ) ]$   
 $= 8.9475 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 3.5006, 8.9475 )$   
 $= 8.9475 \text{ mm.}$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 36.3, | Allowable | : | 138.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 308.7 N./mm <sup>2</sup> | Passed |
| Occasional | : | 3.4,  | Allowable | : | 183.5 N./mm <sup>2</sup> | Passed |
| Shear      | : | 21.2, | Allowable | : | 96.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, Curve: A

Govrn. thk,  $tg = 12.0$ ,  $tr = 0.501$ ,  $c = 3.0 \text{ mm.}$ ,  $E* = 1.0$   
 Thickness Ratio =  $tr \cdot E*/(tg - c) = 0.056$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -3 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk,  $tg = 16.0$ ,  $tr = 5.948$ ,  $c = 3.0 \text{ mm.}$ ,  $E* = 1.0$   
 Thickness Ratio =  $tr \cdot E*/(tg - c) = 0.458$ , Temp. Reduction = 39 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve D   | -46 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $5.45/19.60 = 0.278$

#### Weld Size Calculations, Description: N2-10 Inch

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 13.0000 \text{ mm.}$

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

#### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (14.0362 - 16.6438 + 2 \cdot 42.0 \cdot 1. \cdot (1.0 \cdot 13.0 - 5.9475)) \cdot 138)$   
 $= 4666.82 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (26.9741 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 39364.57 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (26.9741 + 0.0 + 1. + (10.9198)) \cdot 138$   
 $= 54730.67 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (26.9741 + 0.0 + 1. + 0.0 + (10.9198)) \cdot 138$   
 $= 54730.67 \text{ Kg}$

#### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 254.0 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 27510. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 122.5) \cdot (45.0 - 3.0) \cdot 0.7 \cdot 138$   
 $= 159212. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{g_{nvi}} \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 254.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 41547. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (27510 + 159212) = 186722 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (27510 + 0 + 41547 + 0) = 69057 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$   
 $= (27510 + 41547 + 0) = 69057 \text{ Kg}$

#### Summary of Failure Path Calculations:

|                   |                              |                  |
|-------------------|------------------------------|------------------|
| Path 1-1 = 186721 | Kgf, must exceed W = 4666 Kg | or W1 = 39364 Kg |
| Path 2-2 = 69056  | Kgf, must exceed W = 4666 Kg | or W2 = 54730 Kg |
| Path 3-3 = 69056  | Kgf, must exceed W = 4666 Kg | or W3 = 54730 Kg |

#### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 5.3860 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.3860 mm.

*This nozzle or its diameter limit intersects weld seam # [5] on this element. The distance between the nozzle and seam is, 131.16 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 236.00 mm.*

**Input Echo, WRC297 Item 1, Description: N2-10 Inch :**

|                                      |            |                    |
|--------------------------------------|------------|--------------------|
| Diameter Basis for Cylindrical Shell | ID         |                    |
| Shell Internal Corrosion Allowance   | 3.0000     | mm.                |
| Shell External Corrosion Allowance   | 0.0000     | mm.                |
| Shell Diameter                       | 3000.000   | mm.                |
| Shell Thickness                      | 16.0000    | mm.                |
| Shell Stress Concentration Factor    | 1.000      |                    |
| Vessel Material                      | SA-516 70  |                    |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm <sup>2</sup> |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |             |                    |
|------------------------------------|-------------|--------------------|
| Diameter Basis for Nozzle          | OD          |                    |
| Nozzle Corrosion Allowance         | 3.0000      | mm.                |
| Nozzle Diameter                    | 320.000     | mm.                |
| Nozzle Thickness                   | 45.0000     | mm.                |
| Nozzle Stress Concentration Factor | 1.000       |                    |
| Nozzle Material                    | SA-105      |                    |
| Nozzle Cold S.I. Allowable         | SNmc 138.00 | N./mm <sup>2</sup> |
| Nozzle Hot S.I. Allowable          | SNmh 138.00 | N./mm <sup>2</sup> |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUStained loads.*

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 5.45    | bars  |
| Radial Load              | P  | 1911.36 | Kgf   |
| Circumferential Shear    | Vc | 1274.24 | Kgf   |
| Longitudinal Shear       | Vl | 1274.24 | Kgf   |
| Circumferential Moment   | Mc | 892.01  | Kg-m. |
| Longitudinal Moment      | Ml | 892.01  | Kg-m. |
| Torsional Moment         | Mt | 1274.31 | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (7.111) is < 10  
 check the Limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 320.0 + 2 \cdot 1.65 \cdot \sqrt{(1509.5(16.0 - 3.0))}$$

$$= 782.277 \text{ mm.}$$

WRC 297 Curve Access Parameters:

|                         |       |          |     |
|-------------------------|-------|----------|-----|
| Vessel Mean Diameter    | (D) = | 3019.000 | mm. |
| Nozzle Outside Diameter | (d) = | 320.000  | mm. |
| Vessel Thickness used   | (T) = | 13.000   | mm. |
| Nozzle Thickness used   | (t) = | 42.000   | mm. |
| T / t                   | =     | 0.310    |     |
| d / t                   | =     | 10.000   |     |



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.615$$

|         |   |       |
|---------|---|-------|
| Nr/P    | = | 0.079 |
| Mr/P    | = | 0.167 |
| M0/P    | = | 0.049 |
| N0/P    | = | 0.097 |
| MrD/Mc  | = | 0.296 |
| NrDL/Mc | = | 0.090 |
| M0d/Mc  | = | 0.091 |
| N0DL/Mc | = | 0.066 |
| MrD/M1  | = | 0.134 |
| NrDL/M1 | = | 0.087 |
| M0D/M1  | = | 0.042 |
| N0DL/M1 | = | 0.117 |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
|-----------------------------------------|-------------|------------|----------------|---------------|
| (Stresses Normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Membrane (P )                  | -10         | -10        | -10            | -10           |
| Outplane Bending (P )                   | -32         | 32         | -32            | 32            |
| Outplane Membrane (Mc)                  | 0           | 0          | 0              | 0             |
| Outplane Bending (Mc)                   | 0           | 0          | 0              | 0             |
| Outplane Membrane (ML)                  | -18         | -18        | 18             | 18            |
| Outplane Bending (ML)                   | -40         | 40         | 40             | -40           |
| Normal Pressure Stress                  | 62          | 63         | 62             | 63            |
| Outplane Stress Summary                 | -40         | 106        | 79             | 62            |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | Bl            |
|-------------------------------------------|-------------|------------|----------------|---------------|
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Inplane Membrane (P )                     | -8          | -8         | -8             | -8            |
| Inplane Bending (P )                      | -111        | 111        | -111           | 111           |
| Inplane Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Inplane Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Inplane Membrane (ML)                     | -14         | -14        | 14             | 14            |
| Inplane Bending (ML)                      | -130        | 130        | 130            | -130          |
| Inplane Pressure Stress                   | 31          | 31         | 31             | 31            |
| Inplane Stress Summary                    | -233        | 249        | 55             | 17            |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | Bl            |
|---------------------------------------------|-------------|------------|----------------|---------------|
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Shear (Vc)                         | 1           | 1          | -1             | -1            |
| Outplane Shear (Vl)                         | 0           | 0          | 0              | 0             |
| Outplane Shear (Mt)                         | 5           | 5          | 5              | 5             |
| Shear Stress Summary                        | 7           | 7          | 4              | 4             |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE     | Au          | Al         | Bu             | Bl            |
|------------------------|-------------|------------|----------------|---------------|
| (Stress Intensities)   | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Two * Max Shear Stress | 233         | 250        | 80             | 63            |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | Dl           |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Membrane (P )                     | -10          | -10         | -10           | -10          |



|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Outplane Bending (P )  | -32 | 32  | -32 | 32  |
| Outplane Membrane (Mc) | -10 | -10 | 10  | 10  |
| Outplane Bending (Mc)  | -88 | 88  | 88  | -88 |
| Outplane Membrane (ML) | 0   | 0   | 0   | 0   |
| Outplane Bending (ML)  | 0   | 0   | 0   | 0   |
| Normal Pressure Stress | 31  | 31  | 31  | 31  |

|                         |      |     |    |     |
|-------------------------|------|-----|----|-----|
| Outplane Stress Summary | -111 | 131 | 87 | -25 |
|-------------------------|------|-----|----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |              |             |               |              |
|----------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                        | Cu           | Cl          | Du            | Dl           |
| (Stresses parallel to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                         |      |     |      |      |
|-------------------------|------|-----|------|------|
| Inplane Membrane (P )   | -8   | -8  | -8   | -8   |
| Inplane Bending (P )    | -111 | 111 | -111 | 111  |
| Inplane Membrane (Mc)   | -14  | -14 | 14   | 14   |
| Inplane Bending (Mc)    | -287 | 287 | 287  | -287 |
| Inplane Membrane (ML)   | 0    | 0   | 0    | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0    | 0    |
| Inplane Pressure Stress | 62   | 63  | 62   | 63   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -358 | 438 | 244 | -106 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |              |             |               |              |
|------------------------------------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE                          | Cu           | Cl          | Du            | Dl           |
| (Shear stress normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 5  | 5  | 5 | 5 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 7 | 7 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |              |             |               |              |
|-----------------------|--------------|-------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE | Cu           | Cl          | Du            | Dl           |
| (Stress Intensities)  | Left Outside | Left Inside | Right Outside | Right Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 358 | 438 | 244 | 107 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |             |            |                |               |
|----------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE               | Au          | Al         | Bu             | Bl            |
| (Stresses in the hoop direction) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                      |     |     |     |     |
|----------------------|-----|-----|-----|-----|
| Hoop Membrane (P )   | -10 | -10 | -10 | -10 |
| Hoop Bending (P )    | 0   | 0   | 0   | 0   |
| Hoop Membrane (Mc)   | 0   | 0   | 0   | 0   |
| Hoop Bending (Mc)    | 0   | 0   | 0   | 0   |
| Hoop Membrane (ML)   | -18 | -18 | 18  | 18  |
| Hoop Bending (ML)    | 0   | 0   | 0   | 0   |
| Hoop Pressure Stress | 1   | 2   | 1   | 2   |

|                     |     |     |   |    |
|---------------------|-----|-----|---|----|
| Hoop Stress Summary | -28 | -27 | 9 | 10 |
|---------------------|-----|-----|---|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |             |            |                |               |
|-----------------------------------------|-------------|------------|----------------|---------------|
| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | Bl            |
| (Stresses Normal to pipe cross-section) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |

|                       |    |    |    |    |
|-----------------------|----|----|----|----|
| Axial Membrane (P )   | 0  | 0  | 0  | 0  |
| Axial Bending (P )    | -8 | 8  | -8 | 8  |
| Axial Membrane (Mc)   | 0  | 0  | 0  | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0  | 0  |
| Axial Membrane (ML)   | -3 | -3 | 3  | 3  |
| Axial Bending (ML)    | -8 | 8  | 8  | -8 |
| Axial Pressure Stress | 0  | 0  | 0  | 0  |



Axial Stress Summary -20 12 4 3

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE   |               |              |               |              |
|----------------------|---------------|--------------|---------------|--------------|
| (Shear stress)       | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Shear due to (Vc)    | 0             | 0            | 0             | 0            |
| Shear due to (Vl)    | 0             | 0            | 0             | 0            |
| Shear due to Torsion | 1             | 1            | 1             | 1            |
| Shear Stress Summary | 2             | 2            | 1             | 1            |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| LONGITUDINAL PLANE     |               |              |               |              |
|------------------------|---------------|--------------|---------------|--------------|
| (Stress Intensities    | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Two * Max Shear Stress | 28            | 40           | 10            | 10           |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE |                       |                      |                        |                       |
|-----------------------|-----------------------|----------------------|------------------------|-----------------------|
| (Stresses in the      | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Hoop Membrane (P )    | -10                   | -10                  | -10                    | -10                   |
| Hoop Bending (P )     | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)    | -10                   | -10                  | 10                     | 10                    |
| Hoop Bending (Mc)     | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)    | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)     | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress  | 1                     | 2                    | 1                      | 2                     |
| Hoop Stress Summary   | -19                   | -19                  | 1                      | 2                     |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE |                       |                      |                        |                       |
|-----------------------|-----------------------|----------------------|------------------------|-----------------------|
| (Stresses Normal to   | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Axial Membrane (P )   | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )    | -8                    | 8                    | -8                     | 8                     |
| Axial Membrane (Mc)   | -3                    | -3                   | 3                      | 3                     |
| Axial Bending (Mc)    | -23                   | 23                   | 23                     | -23                   |
| Axial Membrane (ML)   | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)    | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress | 0                     | 0                    | 0                      | 0                     |
| Axial Stress Summary  | -35                   | 27                   | 19                     | -11                   |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE |               |              |               |              |
|-----------------------|---------------|--------------|---------------|--------------|
| (Shear stress)        | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
| Shear due to (Vc)     | 0             | 0            | 0             | 0            |
| Shear due to (Vl)     | 0             | 0            | 0             | 0            |
| Shear due to Torsion  | 1             | 1            | 1             | 1            |
| Shear Stress Summary  | 1             | 1            | 2             | 2            |

#### Nozzle Stresses (N./mm<sup>2</sup>):

| CIRCUMFERENTIAL PLANE  |               |              |               |              |
|------------------------|---------------|--------------|---------------|--------------|
| (Stress Intensities    | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
| Two * Max Shear Stress | 35            | 47           | 19            | 14           |

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

#### Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):

| Type of<br>Stress Int. | Stress Values at |
|------------------------|------------------|
|------------------------|------------------|



| Location        | Au    | Al    | Bu   | Bl   | Cu    | Cl    | Du    | Dl    |
|-----------------|-------|-------|------|------|-------|-------|-------|-------|
| Circ. Pm (SUS)  | 62    | 63    | 62   | 63   | 62    | 63    | 62    | 63    |
| Circ. Pl (SUS)  | -28   | -28   | 8    | 8    | -22   | -22   | 6     | 6     |
| Circ. Q (SUS)   | -72   | 72    | 8    | -8   | -398  | 398   | 176   | -176  |
| Long. Pm (SUS)  | 31    | 31    | 31   | 31   | 31    | 31    | 31    | 31    |
| Long. Pl (SUS)  | -22   | -22   | 6    | 6    | -20   | -20   | 0     | 0     |
| Long. Q (SUS)   | -241  | 241   | 19   | -19  | -120  | 120   | 56    | -56   |
| Shear Pm (SUS)  | 0     | 0     | 0    | 0    | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1     | 1     | -1   | -1   | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 5     | 5     | 5    | 5    | 5     | 5     | 5     | 5     |
| Pm (SUS)        | 62.0  | 63.0  | 62.0 | 63.0 | 62.0  | 63.0  | 62.0  | 63.0  |
| Pm+Pl (SUS)     | 34.0  | 35.0  | 70.0 | 71.0 | 40.0  | 41.0  | 68.0  | 69.0  |
| Pm+Pl+Q (Total) | 232.2 | 250.3 | 78.7 | 63.4 | 358.1 | 439.1 | 244.2 | 107.4 |

**Vessel Stress Summation Comparison (N./mm2):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 63.00     | 138.00         | Passed |
| Pm+Pl (SUS)         | 71.03     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 439.05    | 414.00         | Failed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):**

| Type of Stress Int. Location | Stress Values at |      |     |      |      |      |      |      |
|------------------------------|------------------|------|-----|------|------|------|------|------|
|                              | Au               | Al   | Bu  | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 1                | 2    | 1   | 2    | 1    | 2    | 1    | 2    |
| Circ. Pl (SUS)               | -28              | -28  | 8   | 8    | -20  | -20  | 0    | 0    |
| Circ. Q (SUS)                | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Long. Pl (SUS)               | -3               | -3   | 3   | 3    | -3   | -3   | 3    | 3    |
| Long. Q (SUS)                | -16              | 16   | 0   | 0    | -31  | 31   | 15   | -15  |
| Shear Pm (SUS)               | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 1                | 1    | 1   | 1    | 1    | 1    | 1    | 1    |
| Pm (SUS)                     | 1.0              | 2.0  | 1.0 | 2.0  | 1.0  | 2.0  | 1.0  | 2.0  |
| Pm+Pl (SUS)                  | 27.0             | 26.0 | 9.0 | 10.0 | 19.0 | 18.0 | 3.0  | 3.0  |
| Pm+Pl+Q (Total)              | 27.1             | 39.1 | 9.2 | 10.1 | 34.1 | 46.0 | 18.1 | 14.1 |

**Nozzle Stress Summation Comparison (N./mm2):**

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 2.00      | 138.00         | Passed |
| Pm+Pl (SUS)         | 27.00     | 207.00         | Passed |
| Pm+Pl+Q (TOTAL)     | 46.04     | 413.99         | Passed |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: K3-2 Inch**

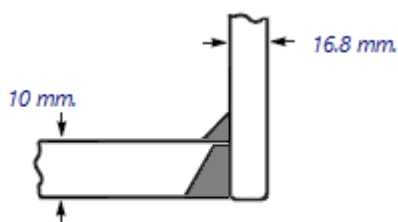
**From: 30**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 |        |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 10.0000   | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.    |
| Distance from Bottom/Left Tangent             |      | 4200.00   | mm.    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |        |
|---------------------------------------------|------|----------------|--------|
| Material                                    |      | SA-105 +IT     |        |
| Material UNS Number                         |      | K03504         |        |
| Material Specification/Type                 |      | Forgings       |        |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm2 |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm2 |
| Diameter Basis (for tr calc only)           |      | Inside         |        |
| Layout Angle                                |      | 0.00           | deg    |
| Diameter                                    |      | 2.0000         | in.    |
| Size and Thickness Basis                    |      | Actual         |        |
| Actual Thickness                            | tn   | 16.7640        | mm.    |
| Flange Material                             |      | SA-105 +IT     |        |
| Flange Type                                 |      | Long Weld Neck |        |
| Corrosion Allowance                         | can  | 3.0000         | mm.    |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |        |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |        |
| Outside Projection                          | ho   | 330.0000       | mm.    |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.    |
| Groove weld depth between Nozzle and Vessel | Wgnv | 10.0000        | mm.    |
| Inside Projection                           | h    | 0.0000         | mm.    |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.    |
| Flange Class                                |      | 300            |        |
| Flange Grade                                |      | GR 1.1         |        |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: K3-2 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 0.1124 \text{ mm.}$

**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 113.6000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 56.8000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:

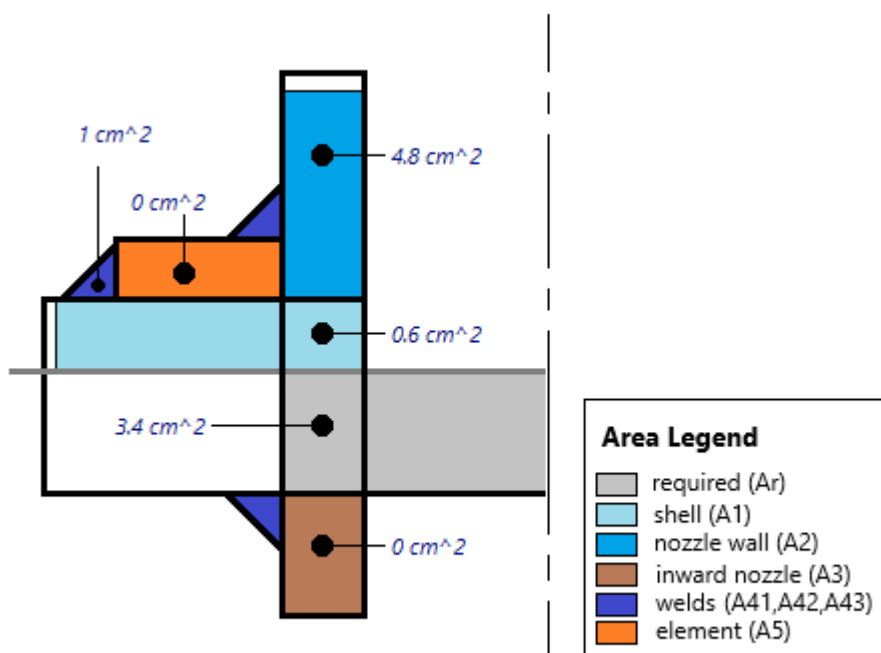
$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm<sup>2</sup>)**

| AREA AVAILABLE, A1 to A5 | MAWP | External | Mapnc |
|--------------------------|------|----------|-------|
|--------------------------|------|----------|-------|



|                       |             |       |    |    |
|-----------------------|-------------|-------|----|----|
| Area Required         | Ar          | 3.378 | NA | NA |
| Area in Shell         | A1          | 0.598 | NA | NA |
| Area in Nozzle Wall   | A2          | 4.778 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000 | NA | NA |
| Area in Welds         | A41+A42+A43 | 1.000 | NA | NA |
| Area in Element       | A5          | 0.000 | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 6.376 | NA | NA |

### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

### The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (56.8 \cdot 5.9475 \cdot 1.0 + 2 \cdot 13.764 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$$

$$= 3.378 \text{ cm}^2$$

### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 56.8(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 13.764$$

$$(1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$$

$$= 0.598 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 17.5)(13.76 - 0.11) \cdot 1.$$

$$= 4.778 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$$

$$= 1.000 \text{ cm}^2$$

### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

$$\text{Wall Thickness for Internal/External pressures } ta = 3.1124 \text{ mm.}$$

$$\text{Wall Thickness per UG16(b), } tr16b = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness, shell/head, internal pressure } trb1 = 8.9475 \text{ mm.}$$

$$\text{Wall Thickness } tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm.}$$

$$\text{Wall Thickness } tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$$

$$\text{Wall Thickness per table UG-45 } tb3 = 7.8000 \text{ mm.}$$

Determine Nozzle Thickness candidate [tb]:

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 7.8, \max( 8.9475, 4.5 ) ]$$

$$= 7.8000 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max( ta, tb )$$

$$= \max( 3.1124, 7.8 )$$

$$= 7.8000 \text{ mm.}$$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

#### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk,  $t_g = 10.0$ ,  $t_r = 5.948$ ,  $c = 3.0 \text{ mm.}$ ,  $E^* = 1.0$   
 Thickness Ratio =  $t_r \cdot E^* / (t_g - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 5.45/51.00 = 0.107

#### Weld Size Calculations, Description: K3-2 Inch

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

#### **Results Per UW-16.1:**

|             | Required Thickness   | Actual Thickness      |
|-------------|----------------------|-----------------------|
| Nozzle Weld | 4.9000 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm. |

#### **Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$   
 $= \max(0, (3.3782 - 0.5978 + 2 \cdot 13.764 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$   
 $= 4320.24 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (4.778 + 0.0 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 8130.62 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$   
 $= (4.778 + 0.0 + 1. + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$   
 $= (4.778 + 0.0 + 1. + 0.0 + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

#### **Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 35.282) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$   
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 9655. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### **Strength of Failure Paths:**

Failure Path [Path 1-1]:  
 $= (SONW + SNW) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (Sonw + Tpgw + Tngw + Sinw)$   
 $= (9133 + 0 + 9655 + 0) = 18789 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (Sonw + Tngw + Sinw)$



= ( 9133 + 9655 + 0 ) = 18789 Kgf

**Summary of Failure Path Calculations:**

|                  |                               |                   |
|------------------|-------------------------------|-------------------|
| Path 1-1 = 24160 | Kgf, must exceed W = 4320 Kgf | or W1 = 8130 Kgf  |
| Path 2-2 = 18788 | Kgf, must exceed W = 4320 Kgf | or W2 = 10842 Kgf |
| Path 3-3 = 18788 | Kgf, must exceed W = 4320 Kgf | or W3 = 10842 Kgf |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5927 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.5927 mm.

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**Input, Nozzle Desc: N4-4 Inch**

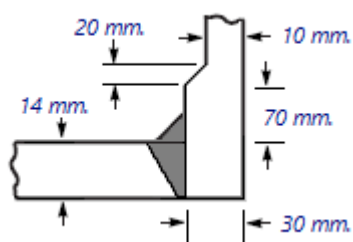
**From: 30**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 14.0000   | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 4550.00   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |                    |
|--------------------------------------------------|-----|------------|--------------------|
| Material                                         |     | SA-106 B   |                    |
| Material UNS Number                              |     | K03006     |                    |
| Material Specification/Type                      |     | Smls. pipe |                    |
| Allowable Stress at Temperature                  | Sn  | 118.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 118.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Outside    |                    |
| Layout Angle                                     |     | 180.00     | deg                |
| Diameter                                         |     | 4.0000     | in.                |
| Size and Thickness Basis                         |     | Actual     |                    |
| Actual Thickness                                 | tn  | 10.0000    | mm.                |
| Flange Material                                  |     | SA-105     |                    |
| Hub Height of Integral Nozzle                    | h   | 70.0000    | mm.                |
| Height of Beveled Transition                     | L'  | 20.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 30.0000    | mm.                |
| Corrosion Allowance                              | can | 3.0000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |                    |
| Outside Projection                               | ho  | 330.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 10.0000    | mm.                |
| Inside Projection                                | h   | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.                |
| Flange Class                                     |     | 150        |                    |
| Flange Grade                                     |     | GR 1.1     |                    |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: N4-4 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 4.000 in.  
 Actual Thickness Used in Calculation 0.394 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 5.45 \cdot 50.8 / (118 \cdot 1.0 + 0.4 \cdot 5.45)$   
 $= 0.2341$  mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 27.0$ , less than the hub height, use (e-2)  
 $= 67.5000$  mm.

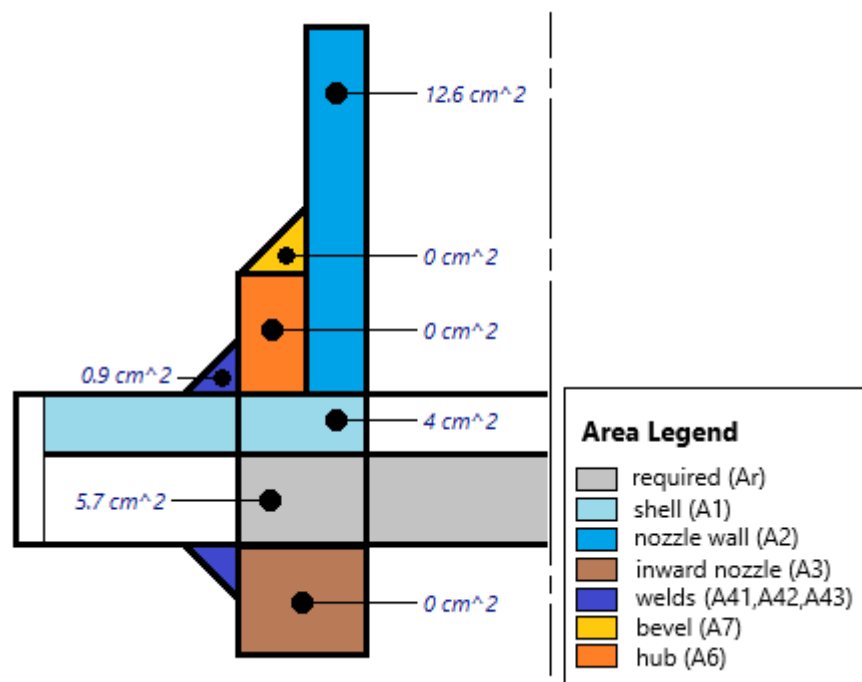
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 175.2000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 87.6000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 27.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 118.0 / 138.0)$   
 $= 0.855$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(0.855, 1.0)$   
 $= 0.855$



#### Results of Nozzle Reinforcement Area Calculations: (cm2)

| AREA AVAILABLE, A1 to A5 |             | MAWP   | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 5.675  | NA       | NA    |
| Area in Shell            | A1          | 4.031  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 12.588 | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 0.855  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 0.000  | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 17.473 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (87.6 \cdot 5.9475 \cdot 1.0 + 2 \cdot 27.0 \cdot 5.9475 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 5.675 \text{ cm}^2$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 87.6(1.0 \cdot 11.0 - 1.0 \cdot 5.948) - 2 \cdot 27.0$$

$$(1.0 \cdot 11.0 - 1.0 \cdot 5.9475) \cdot (1 - 0.855)$$

$$= 4.031 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:  

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.5)(27.0 - 0.23)0.8551$$

$$= 12.588 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  

$$= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$$

$$= 10.0^2 \cdot 0.8551 + (0.0)^2 \cdot 0.8551$$

$$= 0.855 \text{ cm}^2$$

*There are no hub area calculations because Figure UG-40 (e-2) was used.*

**UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]**

Wall Thickness for Internal/External pressures  $t_a = 3.2341$  mm.  
 Wall Thickness per UG16(b),  $tr_{16b} = 4.5000$  mm.  
 Wall Thickness, shell/head, internal pressure  $trb_1 = 8.9475$  mm.  
 Wall Thickness  $tb_1 = \max(trb_1, tr_{16b}) = 8.9475$  mm.  
 Wall Thickness  $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$  mm.  
 Wall Thickness per table UG-45  $tb_3 = 8.0200$  mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[ tb_3, \max( tb_1, tb_2 ) ]$   
 $= \min[ 8.02, \max( 8.9475, 4.5 ) ]$   
 $= 8.0200$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max( t_a, t_b )$   
 $= \max( 3.2341, 8.02 )$   
 $= 8.0200$  mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

**Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):**

|            |   |       |           |   |                          |        |
|------------|---|-------|-----------|---|--------------------------|--------|
| Sustained  | : | 48.2, | Allowable | : | 118.0 N./mm <sup>2</sup> | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 246.8 N./mm <sup>2</sup> | Passed |
| Occasional | : | 1.6,  | Allowable | : | 156.9 N./mm <sup>2</sup> | Passed |
| Shear      | : | 31.0, | Allowable | : | 82.6 N./mm <sup>2</sup>  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

**Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:**

**Nozzle Neck to Flange Weld, min( Curve:C, Curve:A)**

Govrn. thk,  $t_g = 10.0$ ,  $tr = 0.234$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.033$ , Temp. Reduction = 78 °C

|                                                  |         |
|--------------------------------------------------|---------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -8 °C   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 °C |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 °C  |

**Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min( Curve:C, Curve:D)**

Govrn. thk,  $t_g = 14.0$ ,  $tr = 5.948$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.541$ , Temp. Reduction = 28 °C

|                                                  |        |
|--------------------------------------------------|--------|
| Min Metal Temp. w/o impact per UCS-66, Curve C   | -36 °C |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 °C |

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

**Flange MDMT including Temperature reduction per UCS-66.1:**

|                                                        |         |
|--------------------------------------------------------|---------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 °C  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 °C |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating =  $5.45/19.60 = 0.278$

**Weld Size Calculations, Description: N4-4 Inch**

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 11.0000$  mm.

**Results Per UW-16.1:**

|             | Required Thickness             | Actual Thickness                 |
|-------------|--------------------------------|----------------------------------|
| Nozzle Weld | $6.0000 = \text{Min per Code}$ | $7.0700 = 0.7 * W_o \text{ mm.}$ |



### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$   
 $= \max(0, (5.6755 - 4.0306 + 2 \cdot 27.0 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 5.9475)) \cdot 138)$   
 $= 5597.54 \text{ Kg}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$   
 $= (12.5877 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$   
 $= 18916.47 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$   
 $= (12.5877 + 0.0 + 0.8551 + (5.0791)) \cdot 138$   
 $= 26063.74 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+(2 \cdot (\text{Hub Thickness}) \cdot t \cdot fr1)) \cdot Sv$   
 $= (12.5877 + 0.0 + 0.8551 + 0.0) \cdot 138$   
 $= 26063.74 \text{ Kg}$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$   
 $= (3.1416/2.0) \cdot 101.6 \cdot 10.0 \cdot 0.49 \cdot 118$   
 $= 9409. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$   
 $= (3.1416 \cdot 47.3) \cdot (30.0 - 3.0) \cdot 0.7 \cdot 118$   
 $= 33793. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$   
 $= (3.1416/2.0) \cdot 101.6 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 16619. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (9409 + 33793) = 43202 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (9409 + 0 + 16619 + 0) = 26028 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$   
 $= (9409 + 16619 + 0) = 26028 \text{ Kg}$

### Summary of Failure Path Calculations:

|                  |                              |                  |
|------------------|------------------------------|------------------|
| Path 1-1 = 43202 | Kgf, must exceed W = 5597 Kg | or W1 = 18916 Kg |
| Path 2-2 = 26028 | Kgf, must exceed W = 5597 Kg | or W2 = 26063 Kg |
| Path 3-3 = 26028 | Kgf, must exceed W = 5597 Kg | or W3 = 26063 Kg |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.8604 mm.



The Cut Length for this Nozzle is, Drop + Ho + H + T : 344.8604 mm.

**Input Echo, WRC297 Item 1, Description: N4-4 Inch :**

|                                      |           |               |
|--------------------------------------|-----------|---------------|
| Diameter Basis for Cylindrical Shell | ID        |               |
| Shell Internal Corrosion Allowance   | 3.0000    | mm.           |
| Shell External Corrosion Allowance   | 0.0000    | mm.           |
| Shell Diameter                       | 3000.000  | mm.           |
| Shell Thickness                      | 14.0000   | mm.           |
| Shell Stress Concentration Factor    | 1.000     |               |
| Vessel Material                      | SA-516 70 |               |
| Vessel Cold S.I. Allowable           | Smc       | 138.00 N./mm2 |
| Vessel Hot S.I. Allowable            | Smh       | 138.00 N./mm2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |          |               |
|------------------------------------|----------|---------------|
| Diameter Basis for Nozzle          | OD       |               |
| Nozzle Corrosion Allowance         | 3.0000   | mm.           |
| Nozzle Diameter                    | 141.600  | mm.           |
| Nozzle Thickness                   | 30.0000  | mm.           |
| Nozzle Stress Concentration Factor | 1.000    |               |
| Nozzle Material                    | SA-106 B |               |
| Nozzle Cold S.I. Allowable         | SNmc     | 118.00 N./mm2 |
| Nozzle Hot S.I. Allowable          | SNmh     | 118.00 N./mm2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSained loads.*

|                          |    |        |       |
|--------------------------|----|--------|-------|
| Design Internal Pressure | Dp | 5.45   | bars  |
| Radial Load              | P  | 764.54 | Kgf   |
| Circumferential Shear    | Vc | 509.70 | Kgf   |
| Longitudinal Shear       | Vl | 509.70 | Kgf   |
| Circumferential Moment   | Mc | 142.72 | Kg-m. |
| Longitudinal Moment      | Ml | 142.72 | Kg-m. |
| Torsional Moment         | Mt | 203.89 | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (4.720) is < 10  
 check the limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$   
 $= 141.6 + 2 \cdot 1.65 \cdot \sqrt{(1508.5(14.0 - 3.0))}$   
 $= 566.692 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |              |
|-------------------------|-------|--------------|
| Vessel Mean Diameter    | (D) = | 3017.000 mm. |
| Nozzle Outside Diameter | (d) = | 141.600 mm.  |
| Vessel Thickness used   | (T) = | 11.000 mm.   |
| Nozzle Thickness used   | (t) = | 27.000 mm.   |
| T / t                   | =     | 0.407        |
| d / T                   | =     | 10.000       |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.777$$

|      |   |       |
|------|---|-------|
| Nr/P | = | 0.118 |
| Mr/P | = | 0.261 |
| M0/P | = | 0.076 |
| N0/P | = | 0.143 |



MrD/Mc = 0.311  
 NrDL/Mc = 0.078  
 MØd/Mc = 0.097  
 NØDL/Mc = 0.027  
 MrD/Ml = 0.227  
 NrDL/Ml = 0.089  
 MØD/Ml = 0.073  
 NØDL/Ml = 0.097

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE                      | Au          | Al         | Bu             | B1            |
|-----------------------------------------|-------------|------------|----------------|---------------|
| (Stresses Normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Membrane (P )                  | -8          | -8         | -8             | -8            |
| Outplane Bending (P )                   | -28         | 28         | -28            | 28            |
| Outplane Membrane (Mc)                  | 0           | 0          | 0              | 0             |
| Outplane Bending (Mc)                   | 0           | 0          | 0              | 0             |
| Outplane Membrane (ML)                  | -7          | -7         | 7              | 7             |
| Outplane Bending (ML)                   | -35         | 35         | 35             | -35           |
| Normal Pressure Stress                  | 74          | 74         | 74             | 74            |
| Outplane Stress Summary                 | -6          | 122        | 80             | 66            |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE                        | Au          | Al         | Bu             | B1            |
|-------------------------------------------|-------------|------------|----------------|---------------|
| (Stresses parallel to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Inplane Membrane (P )                     | -7          | -7         | -7             | -7            |
| Inplane Bending (P )                      | -96         | 96         | -96            | 96            |
| Inplane Membrane (Mc)                     | 0           | 0          | 0              | 0             |
| Inplane Bending (Mc)                      | 0           | 0          | 0              | 0             |
| Inplane Membrane (ML)                     | -7          | -7         | 7              | 7             |
| Inplane Bending (ML)                      | -111        | 111        | 111            | -111          |
| Inplane Pressure Stress                   | 37          | 37         | 37             | 37            |
| Inplane Stress Summary                    | -185        | 230        | 51             | 22            |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE                          | Au          | Al         | Bu             | B1            |
|---------------------------------------------|-------------|------------|----------------|---------------|
| (Shear stress normal to longitudinal plane) | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Outplane Shear (Vc)                         | 2           | 2          | -2             | -2            |
| Outplane Shear (Vl)                         | 0           | 0          | 0              | 0             |
| Outplane Shear (Mt)                         | 5           | 5          | 5              | 5             |
| Shear Stress Summary                        | 7           | 7          | 3              | 3             |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE     | Au          | Al         | Bu             | B1            |
|------------------------|-------------|------------|----------------|---------------|
| (Stress Intensities)   | Top Outside | Top Inside | Bottom Outside | Bottom Inside |
| Two * Max Shear Stress | 185         | 231        | 81             | 66            |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE                      | Cu           | Cl          | Du            | D1           |
|--------------------------------------------|--------------|-------------|---------------|--------------|
| (Stresses Normal to circumferential plane) | Left Outside | Left Inside | Right Outside | Right Inside |
| Outplane Membrane (P )                     | -8           | -8          | -8            | -8           |
| Outplane Bending (P )                      | -28          | 28          | -28           | 28           |
| Outplane Membrane (Mc)                     | -2           | -2          | 2             | 2            |
| Outplane Bending (Mc)                      | -47          | 47          | 47            | -47          |
| Outplane Membrane (ML)                     | 0            | 0           | 0             | 0            |
| Outplane Bending (ML)                      | 0            | 0           | 0             | 0            |
| Normal Pressure Stress                     | 37           | 37          | 37            | 37           |



|                         |     |     |    |    |
|-------------------------|-----|-----|----|----|
| Outplane Stress Summary | -50 | 102 | 49 | 11 |
|-------------------------|-----|-----|----|----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                        |         |        |         |        |
|------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE  | Cu      | Cl     | Du      | Dl     |
| (Stresses parallel to  | Left    | Left   | Right   | Right  |
| circumferential plane) | Outside | Inside | Outside | Inside |

|                         |      |     |     |      |
|-------------------------|------|-----|-----|------|
| Inplane Membrane (P )   | -7   | -7  | -7  | -7   |
| Inplane Bending (P )    | -96  | 96  | -96 | 96   |
| Inplane Membrane (Mc)   | -6   | -6  | 6   | 6    |
| Inplane Bending (Mc)    | -152 | 152 | 152 | -152 |
| Inplane Membrane (ML)   | 0    | 0   | 0   | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0   | 0    |
| Inplane Pressure Stress | 74   | 74  | 74  | 74   |

|                        |      |     |     |    |
|------------------------|------|-----|-----|----|
| Inplane Stress Summary | -189 | 310 | 128 | 18 |
|------------------------|------|-----|-----|----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                         |         |        |         |        |
|-------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE   | Cu      | Cl     | Du      | Dl     |
| (Shear stress normal to | Left    | Left   | Right   | Right  |
| circumferential plane)  | Outside | Inside | Outside | Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -2 | -2 | 2 | 2 |
| Torsional Shear (Mt) | 5  | 5  | 5 | 5 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 3 | 3 | 7 | 7 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities   | Left    | Left   | Right   | Right  |
|                       | Outside | Inside | Outside | Inside |

|                        |     |     |     |    |
|------------------------|-----|-----|-----|----|
| Two * Max Shear Stress | 189 | 310 | 129 | 23 |
|------------------------|-----|-----|-----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |         |        |         |        |
|--------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE | Au      | Al     | Bu      | Bl     |
| (Stresses in the   | Top     | Top    | Bottom  | Bottom |
| hoop direction)    | Outside | Inside | Outside | Inside |

|                      |    |    |    |    |
|----------------------|----|----|----|----|
| Hoop Membrane (P )   | -8 | -8 | -8 | -8 |
| Hoop Bending (P )    | 0  | 0  | 0  | 0  |
| Hoop Membrane (Mc)   | 0  | 0  | 0  | 0  |
| Hoop Bending (Mc)    | 0  | 0  | 0  | 0  |
| Hoop Membrane (ML)   | -7 | -7 | 7  | 7  |
| Hoop Bending (ML)    | 0  | 0  | 0  | 0  |
| Hoop Pressure Stress | 0  | 1  | 0  | 1  |

|                     |     |     |   |   |
|---------------------|-----|-----|---|---|
| Hoop Stress Summary | -15 | -15 | 0 | 0 |
|---------------------|-----|-----|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                     |         |        |         |        |
|---------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE  | Au      | Al     | Bu      | Bl     |
| (Stresses Normal to | Top     | Top    | Bottom  | Bottom |
| pipe cross-section) | Outside | Inside | Outside | Inside |

|                       |     |    |     |     |
|-----------------------|-----|----|-----|-----|
| Axial Membrane (P )   | 0   | 0  | 0   | 0   |
| Axial Bending (P )    | -12 | 12 | -12 | 12  |
| Axial Membrane (Mc)   | 0   | 0  | 0   | 0   |
| Axial Bending (Mc)    | 0   | 0  | 0   | 0   |
| Axial Membrane (ML)   | -5  | -5 | 5   | 5   |
| Axial Bending (ML)    | -14 | 14 | 14  | -14 |
| Axial Pressure Stress | 0   | 0  | 0   | 0   |

|                      |     |    |   |   |
|----------------------|-----|----|---|---|
| Axial Stress Summary | -33 | 20 | 7 | 3 |
|----------------------|-----|----|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |         |        |         |        |
|--------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE | Au      | Al     | Bu      | Bl     |
| (Shear stress)     | Outside | Inside | Outside | Inside |

|                   |   |   |   |   |
|-------------------|---|---|---|---|
| Shear due to (Vc) | 0 | 0 | 0 | 0 |
|-------------------|---|---|---|---|



|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear due to (V1)    | 0 | 0 | 0 | 0 |
| Shear due to Torsion | 2 | 2 | 2 | 2 |
| Shear Stress Summary | 3 | 3 | 1 | 1 |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                           |               |              |               |              |
|-------------------------------------------|---------------|--------------|---------------|--------------|
| LONGITUDINAL PLANE<br>(Stress Intensities | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
| Two * Max Shear Stress                    | 34            | 36           | 8             | 4            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                                              |                       |                      |                        |                       |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Hoop Membrane (P )                                           | -8                    | -8                   | -8                     | -8                    |
| Hoop Bending (P )                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (Mc)                                           | -2                    | -2                   | 2                      | 2                     |
| Hoop Bending (Mc)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Membrane (ML)                                           | 0                     | 0                    | 0                      | 0                     |
| Hoop Bending (ML)                                            | 0                     | 0                    | 0                      | 0                     |
| Hoop Pressure Stress                                         | 0                     | 1                    | 0                      | 1                     |
| Hoop Stress Summary                                          | -10                   | -9                   | -5                     | -5                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                                                     |                       |                      |                        |                       |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
| Axial Membrane (P )                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (P )                                                  | -12                   | 12                   | -12                    | 12                    |
| Axial Membrane (Mc)                                                 | -5                    | -5                   | 5                      | 5                     |
| Axial Bending (Mc)                                                  | -22                   | 22                   | 22                     | -22                   |
| Axial Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |
| Axial Bending (ML)                                                  | 0                     | 0                    | 0                      | 0                     |
| Axial Pressure Stress                                               | 0                     | 0                    | 0                      | 0                     |
| Axial Stress Summary                                                | -40                   | 28                   | 15                     | -4                    |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |               |              |               |              |
|-----------------------------------------|---------------|--------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
| Shear due to (Vc)                       | 0             | 0            | 0             | 0            |
| Shear due to (V1)                       | 0             | 0            | 0             | 0            |
| Shear due to Torsion                    | 2             | 2            | 2             | 2            |
| Shear Stress Summary                    | 1             | 1            | 3             | 3            |

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                              |               |              |               |              |
|----------------------------------------------|---------------|--------------|---------------|--------------|
| CIRCUMFERENTIAL PLANE<br>(Stress Intensities | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
| Two * Max Shear Stress                       | 40            | 38           | 21            | 7            |

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |    |    |      |     |    |     |
|------------------------------------|------------------|-----|----|----|------|-----|----|-----|
|                                    | Au               | Al  | Bu | Bl | Cu   | Cl  | Du | Dl  |
| Circ. Pm (SUS)                     | 74               | 74  | 74 | 74 | 74   | 74  | 74 | 74  |
| Circ. Pl (SUS)                     | -15              | -15 | -1 | -1 | -13  | -13 | -1 | -1  |
| Circ. Q (SUS)                      | -63              | 63  | 7  | -7 | -248 | 248 | 56 | -56 |
| Long. Pm (SUS)                     | 37               | 37  | 37 | 37 | 37   | 37  | 37 | 37  |



|                 |       |       |      |      |       |       |       |      |
|-----------------|-------|-------|------|------|-------|-------|-------|------|
| Long. P1 (SUS)  | -14   | -14   | 0    | 0    | -10   | -10   | -6    | -6   |
| Long. Q (SUS)   | -207  | 207   | 15   | -15  | -75   | 75    | 19    | -19  |
| Shear Pm (SUS)  | 0     | 0     | 0    | 0    | 0     | 0     | 0     | 0    |
| Shear P1 (SUS)  | 2     | 2     | -2   | -2   | -2    | -2    | 2     | 2    |
| Shear Q (SUS)   | 5     | 5     | 5    | 5    | 5     | 5     | 5     | 5    |
| Pm (SUS)        | 74.0  | 74.0  | 74.0 | 74.0 | 74.0  | 74.0  | 74.0  | 74.0 |
| Pm+P1 (SUS)     | 59.1  | 59.1  | 73.1 | 73.1 | 61.1  | 61.1  | 73.1  | 73.1 |
| Pm+P1+Q (Total) | 184.3 | 230.5 | 80.3 | 66.2 | 187.1 | 309.0 | 129.6 | 21.9 |

#### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 74.00     | 138.00         | Passed |
| Pm+P1 (SUS)         | 73.11     | 207.00         | Passed |
| Pm+P1+Q (TOTAL)     | 309.04    | 414.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

| Type of Stress Int. Location | Stress Values at |      |     |     |      |      |      |      |
|------------------------------|------------------|------|-----|-----|------|------|------|------|
|                              | Au               | Al   | Bu  | Bl  | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 0                | 1    | 0   | 1   | 0    | 1    | 0    | 1    |
| Circ. P1 (SUS)               | -15              | -15  | -1  | -1  | -10  | -10  | -6   | -6   |
| Circ. Q (SUS)                | 0                | 0    | 0   | 0   | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 0                | 0    | 0   | 0   | 0    | 0    | 0    | 0    |
| Long. P1 (SUS)               | -5               | -5   | 5   | 5   | -5   | -5   | 5    | 5    |
| Long. Q (SUS)                | -26              | 26   | 2   | -2  | -34  | 34   | 10   | -10  |
| Shear Pm (SUS)               | 0                | 0    | 0   | 0   | 0    | 0    | 0    | 0    |
| Shear P1 (SUS)               | 0                | 0    | 0   | 0   | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 2                | 2    | 2   | 2   | 2    | 2    | 2    | 2    |
| Pm (SUS)                     | 0.0              | 1.0  | 0.0 | 1.0 | 0.0  | 1.0  | 0.0  | 1.0  |
| Pm+P1 (SUS)                  | 15.0             | 14.0 | 6.0 | 5.0 | 10.0 | 9.0  | 11.0 | 10.0 |
| Pm+P1+Q (Total)              | 31.2             | 35.2 | 8.9 | 5.0 | 39.1 | 38.2 | 21.4 | 7.0  |

#### Nozzle Stress Summation Comparison (N./mm2):

| Type of Stress Int. | Max. S.I. | S.I. Allowable | Result |
|---------------------|-----------|----------------|--------|
| Pm (SUS)            | 1.00      | 118.00         | Passed |
| Pm+P1 (SUS)         | 15.00     | 177.00         | Passed |
| Pm+P1+Q (TOTAL)     | 39.14     | 354.00         | Passed |

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: K1-6 Inch**

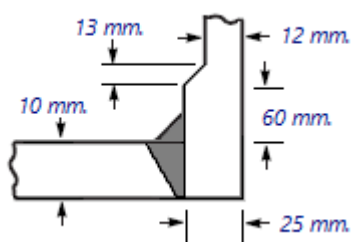
**From: 40**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 10.0000   | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 6700.00   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |                    |
|--------------------------------------------------|-----|------------|--------------------|
| Material                                         |     | SA-105 +IT |                    |
| Material UNS Number                              |     | K03504     |                    |
| Material Specification/Type                      |     | Forgings   |                    |
| Allowable Stress at Temperature                  | Sn  | 138.00     | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                      | Sna | 138.00     | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)                |     | Outside    |                    |
| Layout Angle                                     |     | 0.00       | deg                |
| Diameter                                         |     | 6.0000     | in.                |
| Size and Thickness Basis                         |     | Actual     |                    |
| Actual Thickness                                 | tn  | 12.0000    | mm.                |
| Flange Material                                  |     | SA-105     |                    |
| Hub Height of Integral Nozzle                    | h   | 60.0000    | mm.                |
| Height of Beveled Transition                     | L'  | 13.0000    | mm.                |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 25.0000    | mm.                |
| Corrosion Allowance                              | can | 3.0000     | mm.                |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |                    |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |                    |
| Outside Projection                               | ho  | 330.0000   | mm.                |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.                |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 10.0000    | mm.                |
| Inside Projection                                | h   | 0.0000     | mm.                |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.                |
| Flange Class                                     |     | 150        |                    |
| Flange Grade                                     |     | GR 1.1     |                    |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: K1-6 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 6.000 in.  
 Actual Thickness Used in Calculation 0.472 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 5.45 \cdot 76.2 / (138 \cdot 1.0 + 0.4 \cdot 5.45)$   
 $= 0.3003$  mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):  
 $= 2.5 \cdot \text{Corroded Hub Thickness}$   
 $= 2.5 \cdot 22.0$ , less than the hub height, use (e-2)  
 $= 55.0000$  mm.

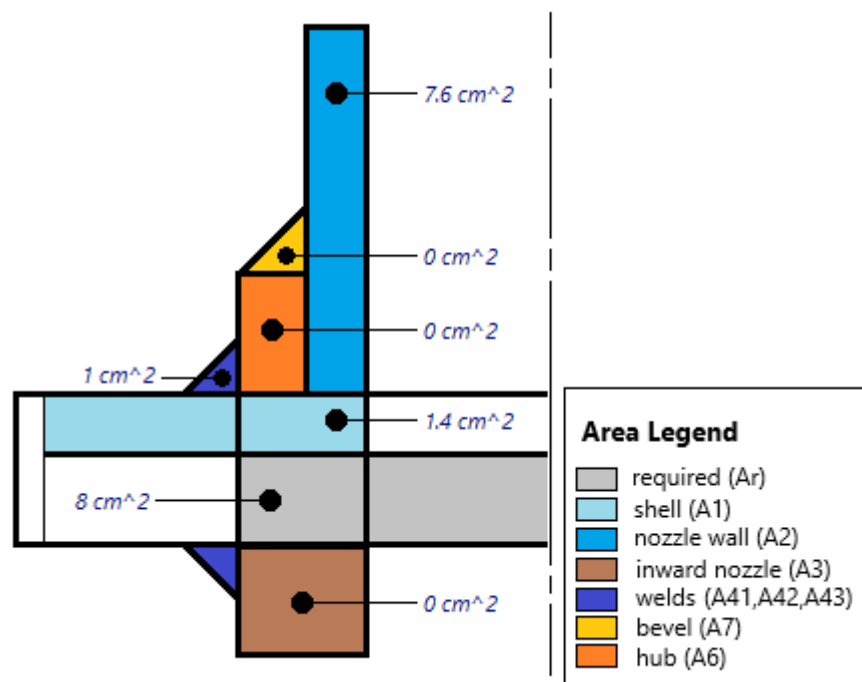
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 268.8000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 134.4000 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm2)

| AREA AVAILABLE, A1 to A5 |             | MAWP   | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 7.994  | NA       | NA    |
| Area in Shell            | A1          | 1.415  | NA       | NA    |
| Area in Nozzle Wall      | A2          | 7.595  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 0.000  | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 10.009 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (134.4 \cdot 5.9475 \cdot 1.0 + 2 \cdot 22.0 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.)) \\
 &= 7.994 \text{ cm}^2
 \end{aligned}$$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 134.4(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 22.0 \\
 &\quad (1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.) \\
 &= 1.415 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot tlnp)(tn - trn)fr2 \\
 &= (2 \cdot 17.5)(22.0 - 0.3) \cdot 1. \\
 &= 7.595 \text{ cm}^2
 \end{aligned}$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2 \\
 &= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. \\
 &= 1.000 \text{ cm}^2
 \end{aligned}$$

*There are no hub area calculations because Figure UG-40 (e-2) was used.*

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $t_a = 3.3003$  mm.  
 Wall Thickness per UG16(b),  $tr_{16b} = 4.5000$  mm.  
 Wall Thickness, shell/head, internal pressure  $trb_1 = 8.9475$  mm.  
 Wall Thickness  $tb_1 = \max(trb_1, tr_{16b}) = 8.9475$  mm.  
 Wall Thickness  $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$  mm.  
 Wall Thickness per table UG-45  $tb_3 = 9.2200$  mm.

Determine Nozzle Thickness candidate [tb]:

$= \min[ tb_3, \max( tb_1, tb_2 ) ]$   
 $= \min[ 9.22, \max( 8.9475, 4.5 ) ]$   
 $= 8.9475$  mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max( t_a, t_b )$   
 $= \max( 3.3003, 8.9475 )$   
 $= 8.9475$  mm.

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, Curve: A

Govrn. thk,  $t_g = 12.0$ ,  $tr = 0.3$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.033$ , Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A  $-3$  °C  
 Min Metal Temp. at Required thickness (UCS 66.1)  $-104$  °C  
 Min Metal Temp. w/o impact per UG-20(f)  $-29$  °C

##### Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: D

Govrn. thk,  $t_g = 10.0$ ,  $tr = 5.948$ ,  $c = 3.0$  mm.,  $E^* = 1.0$   
 Thickness Ratio  $= tr * E^* / (t_g - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D  $-48$  °C

Gov. MDMT of the nozzle to shell joint welded assembly :  $-48$  °C

##### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)  $-18$  °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)  $-104$  °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $5.45/19.60 = 0.278$

#### Weld Size Calculations, Description: K1-6 Inch

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 7.0000$  mm.

##### Results Per UW-16.1:

Required Thickness Actual Thickness  
 Nozzle Weld  $4.9000 = 0.7 * t_{min}$   $7.0700 = 0.7 * W_o$  mm.

##### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max( 0, (A-A_1+2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) S_v )$   
 $= \max( 0, (7.9935 - 1.4145 + 2 \cdot 22.0 \cdot 1.0 \cdot (1.0 \cdot 7.0 - 5.9475)) 138 )$   
 $= 9909.45$  Kg

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  

$$= (A2 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (7.5947 + 1. - 0.0 \cdot 1.) \cdot 138$$

$$= 12094.31 \text{ Kgf}$$

Weld Load [W2]:  

$$= (A2 + A3 + A4 + (2(Hub \text{ Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (7.5947 + 0.0 + 1. + (3.0799)) \cdot 138$$

$$= 16428.34 \text{ Kgf}$$

Weld Load [W3]:  

$$= (A2 + A3 + A4 + (2(Hub \text{ Thickness}) \cdot t \cdot fr1)) \cdot Sv$$

$$= (7.5947 + 0.0 + 1. + 0.0) \cdot 138$$

$$= 16428.34 \text{ Kgf}$$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 152.4 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 16506. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:  

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 71.7) \cdot (25.0 - 3.0) \cdot 0.7 \cdot 138$$

$$= 48812. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:  

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 152.4 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 17450. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  

$$= (SONW + SNW) = (16506 + 48812) = 65318 \text{ Kgf}$$

Failure Path [Path 2-2]:  

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (16506 + 0 + 17450 + 0) = 33956 \text{ Kgf}$$

Failure Path [Path 3-3]:  

$$= (Sonw + Tngw + Sinw)$$

$$= (16506 + 17450 + 0) = 33956 \text{ Kgf}$$

### Summary of Failure Path Calculations:

|                  |                               |                   |
|------------------|-------------------------------|-------------------|
| Path 1-1 = 65318 | Kgf, must exceed W = 9909 Kgf | or W1 = 12094 Kgf |
| Path 2-2 = 33955 | Kgf, must exceed W = 9909 Kgf | or W2 = 16428 Kgf |
| Path 3-3 = 33955 | Kgf, must exceed W = 9909 Kgf | or W3 = 16428 Kgf |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.9367 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 341.9367 mm.

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**Input, Nozzle Desc: N3-10 Inch**

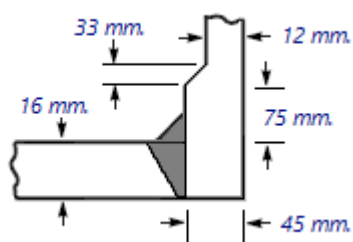
**From: 40**

|                                               |      |           |        |
|-----------------------------------------------|------|-----------|--------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars   |
| Temperature for Internal Pressure             | Temp | 80        | °C     |
| Parent Material                               |      | SA-516 70 |        |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm2 |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm2 |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.    |
| Shell Finished (Minimum) Thickness            | t    | 16.0000   | mm.    |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.    |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.    |
| Distance from Bottom/Left Tangent             |      | 7550.00   | mm.    |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C     |

**Type of Element Connected to the Parent : Nozzle**

|                                                  |     |            |        |
|--------------------------------------------------|-----|------------|--------|
| Material                                         |     | SA-105 +IT |        |
| Material UNS Number                              |     | K03504     |        |
| Material Specification/Type                      |     | Forgings   |        |
| Allowable Stress at Temperature                  | Sn  | 137.98     | N./mm2 |
| Allowable Stress At Ambient                      | Sna | 137.98     | N./mm2 |
| Diameter Basis (for tr calc only)                |     | Outside    |        |
| Layout Angle                                     |     | 0.00       | deg    |
| Diameter                                         |     | 10.0000    | in.    |
| Size and Thickness Basis                         |     | Actual     |        |
| Actual Thickness                                 | tn  | 12.0000    | mm.    |
| Flange Material                                  |     | SA-105     |        |
| Hub Height of Integral Nozzle                    | h   | 75.0000    | mm.    |
| Height of Beveled Transition                     | L'  | 33.0000    | mm.    |
| Hub Thickness of Integral Nozzle ( tn or x+tp )  |     | 45.0000    | mm.    |
| Corrosion Allowance                              | can | 3.0000     | mm.    |
| Joint Efficiency of Shell Seam at Nozzle         | E1  | 1.00       |        |
| Joint Efficiency of Nozzle Neck                  | En  | 1.00       |        |
| Outside Projection                               | ho  | 330.0000   | mm.    |
| Weld leg size between Nozzle and Pad/Shell       | Wo  | 10.0000    | mm.    |
| Groove weld depth between Nozzle and Vessel Wgnv |     | 10.0000    | mm.    |
| Inside Projection                                | h   | 0.0000     | mm.    |
| Weld leg size, Inside Element to Shell           | Wi  | 0.0000     | mm.    |
| Flange Class                                     |     | 150        |        |
| Flange Grade                                     |     | GR 1.1     |        |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: N3-10 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 10.000 in.  
 Actual Thickness Used in Calculation 0.472 in.

**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475$  mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$  per Appendix 1-1 (a)(1)  
 $= 5.45 \cdot 127.0 / (138 \cdot 1.0 + 0.4 \cdot 5.45)$   
 $= 0.5006$  mm.

**Intermediate Hub Nozzle Calculations:**

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:  
 $= ( \text{Hub Thickness} - \text{Neck Thickness} ) / \tan(30)$   
 $= 33.0 / 0.5773$   
 $= 57.1577$  mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

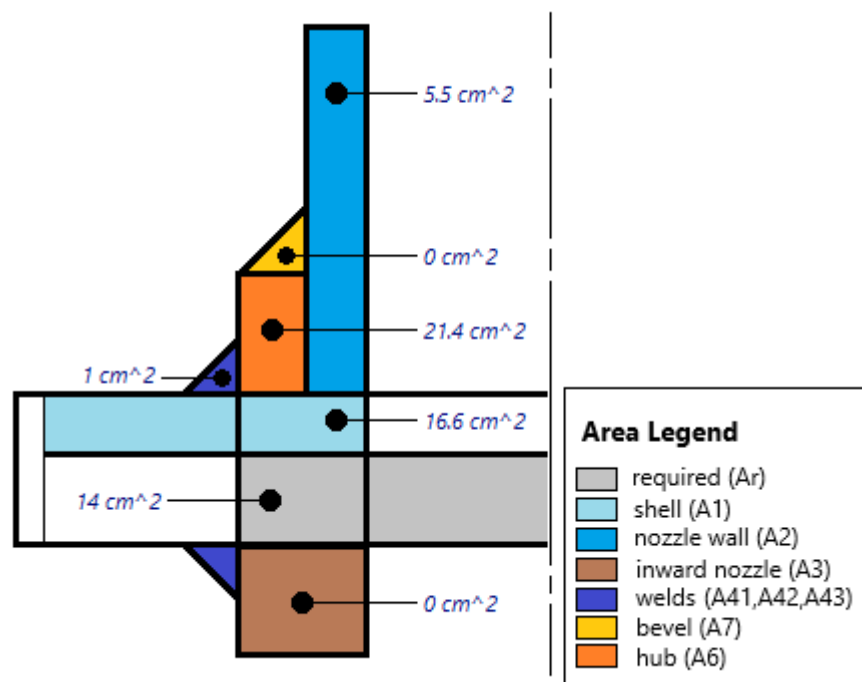
**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 472.0000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 236.0000 | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 32.5000  | mm. |

Weld Strength Reduction Factor [fr1]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 138.0 / 138.0 )$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:  
 $= \min( 1, S_n / S_v )$   
 $= \min( 1, 138.0 / 138.0 )$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:  
 $= \min( fr2, fr4 )$   
 $= \min( 1., 1.0 )$   
 $= 1.000$



#### Results of Nozzle Reinforcement Area Calculations: (cm2)

| AREA AVAILABLE, A1 to A5 |             | MAWP   | External | Mapnc |
|--------------------------|-------------|--------|----------|-------|
| Area Required            | Ar          | 14.037 | NA       | NA    |
| Area in Shell            | A1          | 16.643 | NA       | NA    |
| Area in Nozzle Wall      | A2          | 5.524  | NA       | NA    |
| Area in Inward Nozzle    | A3          | 0.000  | NA       | NA    |
| Area in Welds            | A41+A42+A43 | 1.000  | NA       | NA    |
| Area in Element          | A5          | 0.000  | NA       | NA    |
| Area in Hub              | A6          | 21.448 | NA       | NA    |
| TOTAL AREA AVAILABLE     | Atot        | 44.615 | NA       | NA    |

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (236.0 \cdot 5.9475 \cdot 1.0 + 2 \cdot 42.0 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$   
 $= 14.037 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 236.0(1.0 \cdot 13.0 - 1.0 \cdot 5.948) - 2 \cdot 42.0$   
 $(1.0 \cdot 13.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$   
 $= 16.643 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 32.5)(9.0 - 0.5) \cdot 0.9999$   
 $= 5.524 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 0.9999 + (0.0)^2 \cdot 0.9999$   
 $= 1.000 \text{ cm}^2$

Area Available in the Hub Section [A6]:  

$$= (2 \cdot \min(T_{lnp}, h_o, Hubht)) \cdot (Hubtk - t_n) \cdot fr2$$

$$= (2 \cdot \min(32.5, 330.0, 75.0)) \cdot (45.0 - 12.0) \cdot 1. )$$

$$= 21.448 \text{ cm}^2$$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 3.5006 \text{ mm.}$   
 Wall Thickness per UG16(b),  $tr16b = 4.5000 \text{ mm.}$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 8.9475 \text{ mm.}$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm.}$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm.}$   
 Wall Thickness per table UG-45  $tb3 = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:  

$$= \min[ tb3, \max( tb1, tb2 ) ]$$

$$= \min[ 11.103, \max( 8.9475, 4.5 ) ]$$

$$= 8.9475 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  

$$= \max( ta, tb )$$

$$= \max( 3.5006, 8.9475 )$$

$$= 8.9475 \text{ mm.}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

#### Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

|            |   |       |           |   |              |        |
|------------|---|-------|-----------|---|--------------|--------|
| Sustained  | : | 36.3, | Allowable | : | 138.0 N./mm2 | Passed |
| Expansion  | : | 0.0,  | Allowable | : | 308.6 N./mm2 | Passed |
| Occasional | : | 3.4,  | Allowable | : | 183.5 N./mm2 | Passed |
| Shear      | : | 21.2, | Allowable | : | 96.6 N./mm2  | Passed |

*The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.*

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle Neck to Flange Weld, Curve: A

Govrn. thk,  $tg = 12.0$ ,  $tr = 0.501$ ,  $c = 3.0 \text{ mm.}$ ,  $E* = 1.0$   
 Thickness Ratio =  $tr \cdot E*/(tg - c) = 0.056$ , Temp. Reduction =  $78 \text{ }^\circ\text{C}$

|                                                  |                       |
|--------------------------------------------------|-----------------------|
| Min Metal Temp. w/o impact per UCS-66, Curve A   | -3 $^\circ\text{C}$   |
| Min Metal Temp. at Required thickness (UCS 66.1) | -104 $^\circ\text{C}$ |
| Min Metal Temp. w/o impact per UG-20(f)          | -29 $^\circ\text{C}$  |

##### Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk,  $tg = 16.0$ ,  $tr = 5.948$ ,  $c = 3.0 \text{ mm.}$ ,  $E* = 1.0$   
 Thickness Ratio =  $tr \cdot E*/(tg - c) = 0.458$ , Temp. Reduction =  $39 \text{ }^\circ\text{C}$

|                                                  |                      |
|--------------------------------------------------|----------------------|
| Min Metal Temp. w/o impact per UCS-66, Curve D   | -46 $^\circ\text{C}$ |
| Min Metal Temp. at Required thickness (UCS 66.1) | -48 $^\circ\text{C}$ |

Gov. MDMT of the nozzle to shell joint welded assembly : -48  $^\circ\text{C}$

##### Flange MDMT including Temperature reduction per UCS-66.1:

|                                                        |                       |
|--------------------------------------------------------|-----------------------|
| Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) | -18 $^\circ\text{C}$  |
| Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)   | -104 $^\circ\text{C}$ |

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating =  $5.45/19.60 = 0.278$

#### Weld Size Calculations, Description: N3-10 Inch

Intermediate Calc. for nozzle/shell Welds  $T_{min} = 13.0000 \text{ mm.}$

#### Results Per UW-16.1:

|             | Required Thickness    | Actual Thickness      |
|-------------|-----------------------|-----------------------|
| Nozzle Weld | 6.0000 = Min per Code | 7.0700 = 0.7 * Wo mm. |

### Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (14.0367 - 16.6432 + 2 \cdot 42.0 \cdot 0.9999 \cdot (1.0 \cdot 13.0 - 5.9475)) \cdot 138)$   
 $= 4667.47 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (26.9715 + 0.9999 - 0.0 \cdot 1.) \cdot 138$   
 $= 39360.89 \text{ Kgf}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (26.9715 + 0.0 + 0.9999 + (10.9188)) \cdot 138$   
 $= 54725.58 \text{ Kgf}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (26.9715 + 0.0 + 0.9999 + 0.0 + (10.9188)) \cdot 138$   
 $= 54725.58 \text{ Kgf}$

### Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 254.0 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 27507. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 122.5) \cdot (45.0 - 3.0) \cdot 0.7 \cdot 138$   
 $= 159197. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_{g_{nv}} \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 254.0 \cdot (10.0 - 0.0) \cdot 0.74 \cdot 138$   
 $= 41547. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

### Strength of Failure Paths:

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (27507 + 159197) = 186704 \text{ Kgf}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (27507 + 0 + 41547 + 0) = 69054 \text{ Kgf}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$   
 $= (27507 + 41547 + 0) = 69054 \text{ Kgf}$

### Summary of Failure Path Calculations:

|                                                 |                   |
|-------------------------------------------------|-------------------|
| Path 1-1 = 186704 Kgf, must exceed W = 4667 Kgf | or W1 = 39360 Kgf |
| Path 2-2 = 69054 Kgf, must exceed W = 4667 Kgf  | or W2 = 54725 Kgf |
| Path 3-3 = 69054 Kgf, must exceed W = 4667 Kgf  | or W3 = 54725 Kgf |

### Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 5 bars



Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 5.3860 mm.  
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 351.3860 mm.

**Input Echo, WRC297 Item 1, Description: N3-10 Inch :**

|                                      |            |        |
|--------------------------------------|------------|--------|
| Diameter Basis for Cylindrical Shell | ID         |        |
| Shell Internal Corrosion Allowance   | 3.0000     | mm.    |
| Shell External Corrosion Allowance   | 0.0000     | mm.    |
| Shell Diameter                       | 3000.000   | mm.    |
| Shell Thickness                      | 16.0000    | mm.    |
| Shell Stress Concentration Factor    | 1.000      |        |
| Vessel Material                      | SA-516 70  |        |
| Vessel Cold S.I. Allowable           | Smc 138.00 | N./mm2 |
| Vessel Hot S.I. Allowable            | Smh 138.00 | N./mm2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

|                                    |             |        |
|------------------------------------|-------------|--------|
| Diameter Basis for Nozzle          | OD          |        |
| Nozzle Corrosion Allowance         | 3.0000      | mm.    |
| Nozzle Diameter                    | 320.000     | mm.    |
| Nozzle Thickness                   | 45.0000     | mm.    |
| Nozzle Stress Concentration Factor | 1.000       |        |
| Nozzle Material                    | SA-105      |        |
| Nozzle Cold S.I. Allowable         | SNmc 137.98 | N./mm2 |
| Nozzle Hot S.I. Allowable          | SNmh 137.98 | N./mm2 |

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

**Note:**

*External Forces and Moments in WRC 107/537 Convention:  
 These loads are assumed to be SUSstained loads.*

|                          |    |         |       |
|--------------------------|----|---------|-------|
| Design Internal Pressure | Dp | 5.45    | bars  |
| Radial Load              | P  | 1911.36 | Kgf   |
| Circumferential Shear    | Vc | 1274.24 | Kgf   |
| Longitudinal Shear       | Vl | 1274.24 | Kgf   |
| Circumferential Moment   | Mc | 892.01  | Kg-m. |
| Longitudinal Moment      | Ml | 892.01  | Kg-m. |
| Torsional Moment         | Mt | 1274.31 | Kg-m. |

|                                                 |    |
|-------------------------------------------------|----|
| Include Axial Pressure Thrust                   | No |
| Include Pressure Stress Indices per Div. 2      | No |
| Local Loads applied at end of Nozzle/Attachment | No |

*Warning - The ratio, Dn/Tn (7.111) is < 10  
 check the limitations of WRC bulletin 297.*

**Stress Computations at the Edge of the Nozzle:**

Stress Attenuation Diameter (for Insert Plates) per WRC 297:  
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$   
 $= 320.0 + 2 \cdot 1.65 \cdot \sqrt{(1509.5(16.0 - 3.0))}$   
 $= 782.277 \text{ mm.}$

WRC 297 Curve Access Parameters:

|                         |       |          |     |
|-------------------------|-------|----------|-----|
| Vessel Mean Diameter    | (D) = | 3019.000 | mm. |
| Nozzle Outside Diameter | (d) = | 320.000  | mm. |
| Vessel Thickness used   | (T) = | 13.000   | mm. |
| Nozzle Thickness used   | (t) = | 42.000   | mm. |
| T / t                   | =     | 0.310    |     |
| d / t                   | =     | 10.000   |     |

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.615$$

$$Nr/P = 0.079$$



Mr/P = 0.167  
 MØ/P = 0.049  
 NØ/P = 0.097  
 MrD/Mc = 0.296  
 NrDL/Mc = 0.090  
 MØd/Mc = 0.091  
 NØDL/Mc = 0.066  
 MrD/Ml = 0.134  
 NrDL/Ml = 0.087  
 MØD/Ml = 0.042  
 NØDL/Ml = 0.117

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses Normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Membrane (P )                                           | -10                  | -10                 | -10                     | -10                    |
| Outplane Bending (P )                                            | -32                  | 32                  | -32                     | 32                     |
| Outplane Membrane (Mc)                                           | 0                    | 0                   | 0                       | 0                      |
| Outplane Bending (Mc)                                            | 0                    | 0                   | 0                       | 0                      |
| Outplane Membrane (ML)                                           | -18                  | -18                 | 18                      | 18                     |
| Outplane Bending (ML)                                            | -40                  | 40                  | 40                      | -40                    |
| Normal Pressure Stress                                           | 62                   | 63                  | 62                      | 63                     |
| Outplane Stress Summary                                          | -40                  | 106                 | 79                      | 62                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stresses parallel to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Inplane Membrane (P )                                              | -8                   | -8                  | -8                      | -8                     |
| Inplane Bending (P )                                               | -111                 | 111                 | -111                    | 111                    |
| Inplane Membrane (Mc)                                              | 0                    | 0                   | 0                       | 0                      |
| Inplane Bending (Mc)                                               | 0                    | 0                   | 0                       | 0                      |
| Inplane Membrane (ML)                                              | -14                  | -14                 | 14                      | 14                     |
| Inplane Bending (ML)                                               | -130                 | 130                 | 130                     | -130                   |
| Inplane Pressure Stress                                            | 31                   | 31                  | 31                      | 31                     |
| Inplane Stress Summary                                             | -233                 | 249                 | 55                      | 17                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Shear stress normal to<br>longitudinal plane) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|----------------------------------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Outplane Shear (Vc)                                                  | 1                    | 1                   | -1                      | -1                     |
| Outplane Shear (Vl)                                                  | 0                    | 0                   | 0                       | 0                      |
| Outplane Shear (Mt)                                                  | 5                    | 5                   | 5                       | 5                      |
| Shear Stress Summary                                                 | 7                    | 7                   | 4                       | 4                      |

**Vessel Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Top<br>Outside | Al<br>Top<br>Inside | Bu<br>Bottom<br>Outside | Bl<br>Bottom<br>Inside |
|--------------------------------------------|----------------------|---------------------|-------------------------|------------------------|
| Two * Max Shear Stress                     | 233                  | 250                 | 80                      | 63                     |

**Vessel Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>circumferential plane) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|------------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
| Outplane Membrane (P )                                                 | -10                   | -10                  | -10                    | -10                   |
| Outplane Bending (P )                                                  | -32                   | 32                   | -32                    | 32                    |
| Outplane Membrane (Mc)                                                 | -10                   | -10                  | 10                     | 10                    |
| Outplane Bending (Mc)                                                  | -88                   | 88                   | 88                     | -88                   |
| Outplane Membrane (ML)                                                 | 0                     | 0                    | 0                      | 0                     |



|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Outplane Bending (ML)  | 0  | 0  | 0  | 0  |
| Normal Pressure Stress | 31 | 31 | 31 | 31 |

|                         |      |     |    |     |
|-------------------------|------|-----|----|-----|
| Outplane Stress Summary | -111 | 131 | 87 | -25 |
|-------------------------|------|-----|----|-----|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                              |         |        |         |        |
|----------------------------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE                        | Cu      | Cl     | Du      | Dl     |
| (Stresses parallel to circumferential plane) | Left    | Left   | Right   | Right  |
|                                              | Outside | Inside | Outside | Inside |

|                         |      |     |      |      |
|-------------------------|------|-----|------|------|
| Inplane Membrane (P )   | -8   | -8  | -8   | -8   |
| Inplane Bending (P )    | -111 | 111 | -111 | 111  |
| Inplane Membrane (Mc)   | -14  | -14 | 14   | 14   |
| Inplane Bending (Mc)    | -287 | 287 | 287  | -287 |
| Inplane Membrane (ML)   | 0    | 0   | 0    | 0    |
| Inplane Bending (ML)    | 0    | 0   | 0    | 0    |
| Inplane Pressure Stress | 62   | 63  | 62   | 63   |

|                        |      |     |     |      |
|------------------------|------|-----|-----|------|
| Inplane Stress Summary | -358 | 438 | 244 | -106 |
|------------------------|------|-----|-----|------|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                                                |         |        |         |        |
|------------------------------------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE                          | Cu      | Cl     | Du      | Dl     |
| (Shear stress normal to circumferential plane) | Left    | Left   | Right   | Right  |
|                                                | Outside | Inside | Outside | Inside |

|                      |    |    |   |   |
|----------------------|----|----|---|---|
| Outplane Shear (Vc)  | 0  | 0  | 0 | 0 |
| Outplane Shear (Vl)  | -1 | -1 | 1 | 1 |
| Torsional Shear (Mt) | 5  | 5  | 5 | 5 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 4 | 4 | 7 | 7 |
|----------------------|---|---|---|---|

**Vessel Stresses (N./mm<sup>2</sup>):**

|                       |         |        |         |        |
|-----------------------|---------|--------|---------|--------|
| CIRCUMFERENTIAL PLANE | Cu      | Cl     | Du      | Dl     |
| (Stress Intensities)  | Left    | Left   | Right   | Right  |
|                       | Outside | Inside | Outside | Inside |

|                        |     |     |     |     |
|------------------------|-----|-----|-----|-----|
| Two * Max Shear Stress | 358 | 438 | 244 | 107 |
|------------------------|-----|-----|-----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                  |         |        |         |        |
|----------------------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE               | Au      | Al     | Bu      | Bl     |
| (Stresses in the hoop direction) | Top     | Top    | Bottom  | Bottom |
|                                  | Outside | Inside | Outside | Inside |

|                      |     |     |     |     |
|----------------------|-----|-----|-----|-----|
| Hoop Membrane (P )   | -10 | -10 | -10 | -10 |
| Hoop Bending (P )    | 0   | 0   | 0   | 0   |
| Hoop Membrane (Mc)   | 0   | 0   | 0   | 0   |
| Hoop Bending (Mc)    | 0   | 0   | 0   | 0   |
| Hoop Membrane (ML)   | -18 | -18 | 18  | 18  |
| Hoop Bending (ML)    | 0   | 0   | 0   | 0   |
| Hoop Pressure Stress | 1   | 2   | 1   | 2   |

|                     |     |     |   |    |
|---------------------|-----|-----|---|----|
| Hoop Stress Summary | -28 | -27 | 9 | 10 |
|---------------------|-----|-----|---|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                                         |         |        |         |        |
|-----------------------------------------|---------|--------|---------|--------|
| LONGITUDINAL PLANE                      | Au      | Al     | Bu      | Bl     |
| (Stresses Normal to pipe cross-section) | Top     | Top    | Bottom  | Bottom |
|                                         | Outside | Inside | Outside | Inside |

|                       |    |    |    |    |
|-----------------------|----|----|----|----|
| Axial Membrane (P )   | 0  | 0  | 0  | 0  |
| Axial Bending (P )    | -8 | 8  | -8 | 8  |
| Axial Membrane (Mc)   | 0  | 0  | 0  | 0  |
| Axial Bending (Mc)    | 0  | 0  | 0  | 0  |
| Axial Membrane (ML)   | -3 | -3 | 3  | 3  |
| Axial Bending (ML)    | -8 | 8  | 8  | -8 |
| Axial Pressure Stress | 0  | 0  | 0  | 0  |

|                      |     |    |   |   |
|----------------------|-----|----|---|---|
| Axial Stress Summary | -20 | 12 | 4 | 3 |
|----------------------|-----|----|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

|                    |    |    |    |    |
|--------------------|----|----|----|----|
| LONGITUDINAL PLANE | Au | Al | Bu | Bl |
|--------------------|----|----|----|----|



| (Shear stress) | Outside | Inside | Outside | Inside |
|----------------|---------|--------|---------|--------|
|----------------|---------|--------|---------|--------|

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear due to (Vc)    | 0 | 0 | 0 | 0 |
| Shear due to (Vl)    | 0 | 0 | 0 | 0 |
| Shear due to Torsion | 1 | 1 | 1 | 1 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 2 | 2 | 1 | 1 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| LONGITUDINAL PLANE<br>(Stress Intensities) | Au<br>Outside | Al<br>Inside | Bu<br>Outside | Bl<br>Inside |
|--------------------------------------------|---------------|--------------|---------------|--------------|
|--------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 28 | 40 | 10 | 10 |
|------------------------|----|----|----|----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses in the<br>hoop direction) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|--------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                      |     |     |     |     |
|----------------------|-----|-----|-----|-----|
| Hoop Membrane (P )   | -10 | -10 | -10 | -10 |
| Hoop Bending (P )    | 0   | 0   | 0   | 0   |
| Hoop Membrane (Mc)   | -10 | -10 | 10  | 10  |
| Hoop Bending (Mc)    | 0   | 0   | 0   | 0   |
| Hoop Membrane (ML)   | 0   | 0   | 0   | 0   |
| Hoop Bending (ML)    | 0   | 0   | 0   | 0   |
| Hoop Pressure Stress | 1   | 2   | 1   | 2   |

|                     |     |     |   |   |
|---------------------|-----|-----|---|---|
| Hoop Stress Summary | -19 | -19 | 1 | 2 |
|---------------------|-----|-----|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stresses Normal to<br>pipe cross-section) | Cu<br>Left<br>Outside | Cl<br>Left<br>Inside | Du<br>Right<br>Outside | Dl<br>Right<br>Inside |
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|
|---------------------------------------------------------------------|-----------------------|----------------------|------------------------|-----------------------|

|                       |     |    |    |     |
|-----------------------|-----|----|----|-----|
| Axial Membrane (P )   | 0   | 0  | 0  | 0   |
| Axial Bending (P )    | -8  | 8  | -8 | 8   |
| Axial Membrane (Mc)   | -3  | -3 | 3  | 3   |
| Axial Bending (Mc)    | -23 | 23 | 23 | -23 |
| Axial Membrane (ML)   | 0   | 0  | 0  | 0   |
| Axial Bending (ML)    | 0   | 0  | 0  | 0   |
| Axial Pressure Stress | 0   | 0  | 0  | 0   |

|                      |     |    |    |     |
|----------------------|-----|----|----|-----|
| Axial Stress Summary | -35 | 27 | 19 | -11 |
|----------------------|-----|----|----|-----|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Shear stress) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------|---------------|--------------|---------------|--------------|
|-----------------------------------------|---------------|--------------|---------------|--------------|

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear due to (Vc)    | 0 | 0 | 0 | 0 |
| Shear due to (Vl)    | 0 | 0 | 0 | 0 |
| Shear due to Torsion | 1 | 1 | 1 | 1 |

|                      |   |   |   |   |
|----------------------|---|---|---|---|
| Shear Stress Summary | 1 | 1 | 2 | 2 |
|----------------------|---|---|---|---|

**Nozzle Stresses (N./mm<sup>2</sup>):**

| CIRCUMFERENTIAL PLANE<br>(Stress Intensities) | Cu<br>Outside | Cl<br>Inside | Du<br>Outside | Dl<br>Inside |
|-----------------------------------------------|---------------|--------------|---------------|--------------|
|-----------------------------------------------|---------------|--------------|---------------|--------------|

|                        |    |    |    |    |
|------------------------|----|----|----|----|
| Two * Max Shear Stress | 35 | 47 | 19 | 14 |
|------------------------|----|----|----|----|

**WRC 297 Stress Summations per ASME Sec. VIII Div. 2:****Vessel Stress Summation at Vessel-Nozzle Junction (N./mm<sup>2</sup>):**

| Type of<br>Stress Int.<br>Location | Stress Values at |     |    |    |     |     |    |    |
|------------------------------------|------------------|-----|----|----|-----|-----|----|----|
|                                    | Au               | Al  | Bu | Bl | Cu  | Cl  | Du | Dl |
| Circ. Pm (SUS)                     | 62               | 63  | 62 | 63 | 62  | 63  | 62 | 63 |
| Circ. Pl (SUS)                     | -28              | -28 | 8  | 8  | -22 | -22 | 6  | 6  |



|                 |       |       |      |      |       |       |       |       |
|-----------------|-------|-------|------|------|-------|-------|-------|-------|
| Circ. Q (SUS)   | -72   | 72    | 8    | -8   | -398  | 398   | 176   | -176  |
| Long. Pm (SUS)  | 31    | 31    | 31   | 31   | 31    | 31    | 31    | 31    |
| Long. Pl (SUS)  | -22   | -22   | 6    | 6    | -20   | -20   | 0     | 0     |
| Long. Q (SUS)   | -241  | 241   | 19   | -19  | -120  | 120   | 56    | -56   |
| Shear Pm (SUS)  | 0     | 0     | 0    | 0    | 0     | 0     | 0     | 0     |
| Shear Pl (SUS)  | 1     | 1     | -1   | -1   | -1    | -1    | 1     | 1     |
| Shear Q (SUS)   | 5     | 5     | 5    | 5    | 5     | 5     | 5     | 5     |
| Pm (SUS)        | 62.0  | 63.0  | 62.0 | 63.0 | 62.0  | 63.0  | 62.0  | 63.0  |
| Pm+Pl (SUS)     | 34.0  | 35.0  | 70.0 | 71.0 | 40.0  | 41.0  | 68.0  | 69.0  |
| Pm+Pl+Q (Total) | 232.2 | 250.3 | 78.7 | 63.4 | 358.1 | 439.1 | 244.2 | 107.4 |

#### Vessel Stress Summation Comparison (N./mm2):

| Type of Stress Int.    | Max. S.I.     | S.I. Allowable | Result        |
|------------------------|---------------|----------------|---------------|
| Pm (SUS)               | 63.00         | 138.00         | Passed        |
| Pm+Pl (SUS)            | 71.03         | 207.00         | Passed        |
| <b>Pm+Pl+Q (TOTAL)</b> | <b>439.05</b> | <b>414.00</b>  | <b>Failed</b> |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.

#### WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

##### Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

| Type of Stress Int. Location | Stress Values at |      |     |      |      |      |      |      |
|------------------------------|------------------|------|-----|------|------|------|------|------|
|                              | Au               | Al   | Bu  | Bl   | Cu   | Cl   | Du   | Dl   |
| Circ. Pm (SUS)               | 1                | 2    | 1   | 2    | 1    | 2    | 1    | 2    |
| Circ. Pl (SUS)               | -28              | -28  | 8   | 8    | -20  | -20  | 0    | 0    |
| Circ. Q (SUS)                | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Long. Pm (SUS)               | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Long. Pl (SUS)               | -3               | -3   | 3   | 3    | -3   | -3   | 3    | 3    |
| Long. Q (SUS)                | -16              | 16   | 0   | 0    | -31  | 31   | 15   | -15  |
| Shear Pm (SUS)               | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Shear Pl (SUS)               | 0                | 0    | 0   | 0    | 0    | 0    | 0    | 0    |
| Shear Q (SUS)                | 1                | 1    | 1   | 1    | 1    | 1    | 1    | 1    |
| Pm (SUS)                     | 1.0              | 2.0  | 1.0 | 2.0  | 1.0  | 2.0  | 1.0  | 2.0  |
| Pm+Pl (SUS)                  | 27.0             | 26.0 | 9.0 | 10.0 | 19.0 | 18.0 | 3.0  | 3.0  |
| Pm+Pl+Q (Total)              | 27.1             | 39.1 | 9.2 | 10.1 | 34.1 | 46.0 | 18.1 | 14.1 |

#### Nozzle Stress Summation Comparison (N./mm2):

| Type of Stress Int.    | Max. S.I.    | S.I. Allowable | Result        |
|------------------------|--------------|----------------|---------------|
| Pm (SUS)               | 2.00         | 137.98         | Passed        |
| Pm+Pl (SUS)            | 27.00        | 206.98         | Passed        |
| <b>Pm+Pl+Q (TOTAL)</b> | <b>46.04</b> | <b>413.95</b>  | <b>Passed</b> |

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 \* Smh.



**Input, Nozzle Desc: V1-2 Inch**

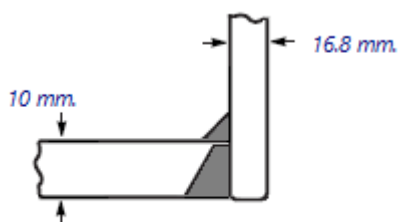
**From: 40**

|                                               |      |           |                    |
|-----------------------------------------------|------|-----------|--------------------|
| Pressure for Reinforcement Calculations       | P    | 5.448     | bars               |
| Temperature for Internal Pressure             | Temp | 80        | °C                 |
| Parent Material                               |      | SA-516 70 |                    |
| Parent Allowable Stress at Temperature        | Sv   | 138.00    | N./mm <sup>2</sup> |
| Parent Allowable Stress At Ambient            | Sva  | 138.00    | N./mm <sup>2</sup> |
| Inside Diameter of Cylindrical Shell          | D    | 3000.00   | mm.                |
| Shell Finished (Minimum) Thickness            | t    | 10.0000   | mm.                |
| Shell Internal Corrosion Allowance            | cas  | 3.0000    | mm.                |
| Shell External Corrosion Allowance            | co   | 0.0000    | mm.                |
| Distance from Bottom/Left Tangent             |      | 8450.00   | mm.                |
| User Entered Minimum Design Metal Temperature |      | -40.00    | °C                 |

**Type of Element Connected to the Parent : Nozzle**

|                                             |      |                |                    |
|---------------------------------------------|------|----------------|--------------------|
| Material                                    |      | SA-105 +IT     |                    |
| Material UNS Number                         |      | K03504         |                    |
| Material Specification/Type                 |      | Forgings       |                    |
| Allowable Stress at Temperature             | Sn   | 138.00         | N./mm <sup>2</sup> |
| Allowable Stress At Ambient                 | Sna  | 138.00         | N./mm <sup>2</sup> |
| Diameter Basis (for tr calc only)           |      | Inside         |                    |
| Layout Angle                                |      | 0.00           | deg                |
| Diameter                                    |      | 2.0000         | in.                |
| Size and Thickness Basis                    |      | Actual         |                    |
| Actual Thickness                            | tn   | 16.7640        | mm.                |
| Flange Material                             |      | SA-105 +IT     |                    |
| Flange Type                                 |      | Long Weld Neck |                    |
| Corrosion Allowance                         | can  | 3.0000         | mm.                |
| Joint Efficiency of Shell Seam at Nozzle    | E1   | 1.00           |                    |
| Joint Efficiency of Nozzle Neck             | En   | 1.00           |                    |
| Outside Projection                          | ho   | 330.0000       | mm.                |
| Weld leg size between Nozzle and Pad/Shell  | Wo   | 10.0000        | mm.                |
| Groove weld depth between Nozzle and Vessel | Wgnv | 10.0000        | mm.                |
| Inside Projection                           | h    | 0.0000         | mm.                |
| Weld leg size, Inside Element to Shell      | Wi   | 0.0000         | mm.                |
| Flange Class                                |      | 300            |                    |
| Flange Grade                                |      | GR 1.1         |                    |

The Pressure Design option was Overall MAWP.



**Reinforcement CALCULATION, Description: V1-2 Inch**

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

|                                            |       |     |
|--------------------------------------------|-------|-----|
| Actual Inside Diameter Used in Calculation | 2.000 | in. |
| Actual Thickness Used in Calculation       | 0.660 | in. |



**Note:**

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]  
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 1503.0 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 5.9475 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]  
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$  per UG-27 (c)(1)  
 $= 5.45 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 5.45)$   
 $= 0.1124 \text{ mm.}$

**UG-40, Limits of Reinforcement: [Internal Pressure]**

|                                                 |      |          |     |
|-------------------------------------------------|------|----------|-----|
| Parallel to Vessel Wall (Diameter Limit)        | D1   | 113.6000 | mm. |
| Parallel to Vessel Wall, opening length         | d    | 56.8000  | mm. |
| Normal to Vessel Wall (Thickness Limit), no pad | Tlnp | 17.5000  | mm. |

Weld Strength Reduction Factor [fr1]:

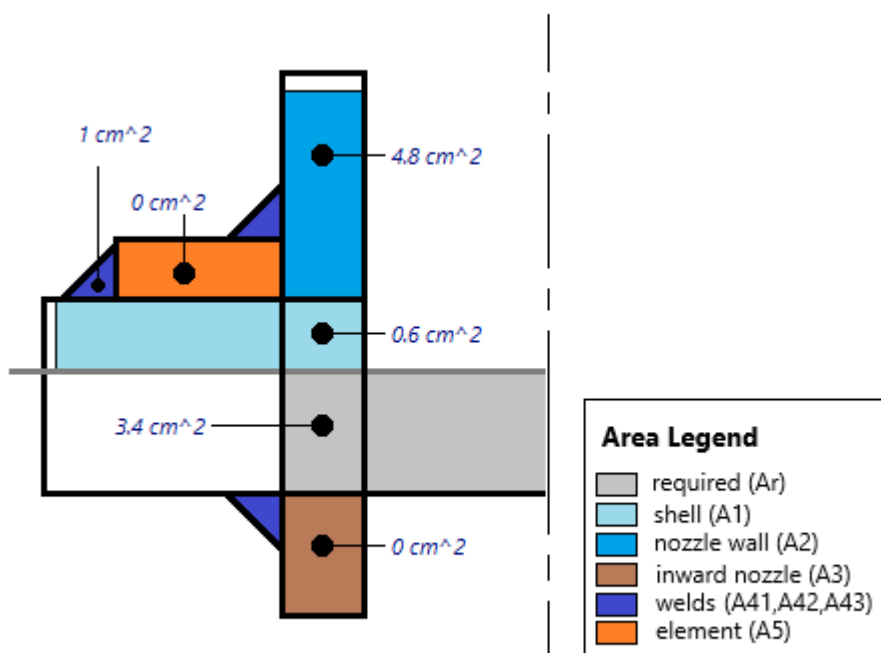
$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr2]:

$= \min(1, S_n / S_v)$   
 $= \min(1, 138.0 / 138.0)$   
 $= 1.000$

Weld Strength Reduction Factor [fr3]:

$= \min(fr2, fr4)$   
 $= \min(1., 1.0)$   
 $= 1.000$



**Figure UG 37.1 Nozzle Areas**

**Results of Nozzle Reinforcement Area Calculations: (cm<sup>2</sup>)**

| AREA AVAILABLE, A1 to A5 | MAWP | External | Mapnc |
|--------------------------|------|----------|-------|
|--------------------------|------|----------|-------|



|                       |             |       |    |    |
|-----------------------|-------------|-------|----|----|
| Area Required         | Ar          | 3.378 | NA | NA |
| Area in Shell         | A1          | 0.598 | NA | NA |
| Area in Nozzle Wall   | A2          | 4.778 | NA | NA |
| Area in Inward Nozzle | A3          | 0.000 | NA | NA |
| Area in Welds         | A41+A42+A43 | 1.000 | NA | NA |
| Area in Element       | A5          | 0.000 | NA | NA |
| TOTAL AREA AVAILABLE  | Atot        | 6.376 | NA | NA |

#### The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

#### The area available without a pad is Sufficient.

Area Required [A]:  
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$   
 $= (56.8 \cdot 5.9475 \cdot 1.0 + 2 \cdot 13.764 \cdot 5.9475 \cdot 1.0 \cdot (1 - 1.))$   
 $= 3.378 \text{ cm}^2$

#### Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:  
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$   
 $= 56.8(1.0 \cdot 7.0 - 1.0 \cdot 5.948) - 2 \cdot 13.764$   
 $(1.0 \cdot 7.0 - 1.0 \cdot 5.9475) \cdot (1 - 1.)$   
 $= 0.598 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:  
 $= (2 \cdot tlnp)(tn - trn)fr2$   
 $= (2 \cdot 17.5)(13.76 - 0.11) \cdot 1.$   
 $= 4.778 \text{ cm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:  
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$   
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1.$   
 $= 1.000 \text{ cm}^2$

#### UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures  $ta = 3.1124 \text{ mm}.$   
 Wall Thickness per UG16(b),  $tr16b = 4.5000 \text{ mm}.$   
 Wall Thickness, shell/head, internal pressure  $trb1 = 8.9475 \text{ mm}.$   
 Wall Thickness  $tb1 = \max(trb1, tr16b) = 8.9475 \text{ mm}.$   
 Wall Thickness  $tb2 = \max(trb2, tr16b) = 4.5000 \text{ mm}.$   
 Wall Thickness per table UG-45  $tb3 = 7.8000 \text{ mm}.$

Determine Nozzle Thickness candidate [tb]:  
 $= \min[ tb3, \max( tb1, tb2 ) ]$   
 $= \min[ 7.8, \max( 8.9475, 4.5 ) ]$   
 $= 7.8000 \text{ mm}.$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:  
 $= \max( ta, tb )$   
 $= \max( 3.1124, 7.8 )$   
 $= 7.8000 \text{ mm}.$

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

#### Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

##### Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: D

Govrn. thk,  $tg = 10.0$ ,  $tr = 5.948$ ,  $c = 3.0 \text{ mm}.$ ,  $E* = 1.0$   
 Thickness Ratio  $= tr \cdot E*/(tg - c) = 0.85$ , Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -48 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

#### Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C  
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :  
 Design Pressure/Ambient Rating = 5.45/51.00 = 0.107

#### Weld Size Calculations, Description: V1-2 Inch

Intermediate Calc. for nozzle/shell Welds Tmin 7.0000 mm.

#### **Results Per UW-16.1:**

|             | Required Thickness   | Actual Thickness      |
|-------------|----------------------|-----------------------|
| Nozzle Weld | 4.9000 = 0.7 * tmin. | 7.0700 = 0.7 * Wo mm. |

#### **Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)**

Weld Load [W]:  
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$   
 $= \max(0, (3.3782 - 0.5978 + 2 \cdot 13.764 \cdot 1. \cdot (1.0 \cdot 7.0 - 5.9475)) \cdot 138)$   
 $= 4320.24 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:  
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$   
 $= (4.778 + 0.0 + 1. - 0.0 \cdot 1. ) \cdot 138$   
 $= 8130.62 \text{ Kg}$

Weld Load [W2]:  
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$   
 $= (4.778 + 0.0 + 1. + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

Weld Load [W3]:  
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$   
 $= (4.778 + 0.0 + 1. + 0.0 + (1.9269)) \cdot 138$   
 $= 10842.15 \text{ Kg}$

#### **Strength of Connection Elements for Failure Path Analysis**

Shear, Outward Nozzle Weld [Sonw]:  
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$   
 $= (3.1416/2.0) \cdot 84.328 \cdot 10.0 \cdot 0.49 \cdot 138$   
 $= 9133. \text{ Kg}$

Shear, Nozzle Wall [Snw]:  
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$   
 $= (3.1416 \cdot 35.282) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$   
 $= 15028. \text{ Kg}$

Tension, Shell Groove Weld [Tngw]:  
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$   
 $= (3.1416/2.0) \cdot 84.328 \cdot (10.0 - 3.0) \cdot 0.74 \cdot 138$   
 $= 9655. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

#### **Strength of Failure Paths:**

Failure Path [Path 1-1]:  
 $= (\text{SONW} + \text{SNW}) = (9133 + 15028) = 24161 \text{ Kg}$

Failure Path [Path 2-2]:  
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$   
 $= (9133 + 0 + 9655 + 0) = 18789 \text{ Kg}$

Failure Path [Path 3-3]:  
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$



$$= ( 9133 + 9655 + 0 ) = 18789 \text{ Kgf}$$

**Summary of Failure Path Calculations:**

|                                                |                   |
|------------------------------------------------|-------------------|
| Path 1-1 = 24160 Kgf, must exceed W = 4320 Kgf | or W1 = 8130 Kgf  |
| Path 2-2 = 18788 Kgf, must exceed W = 4320 Kgf | or W2 = 10842 Kgf |
| Path 3-3 = 18788 Kgf, must exceed W = 4320 Kgf | or W3 = 10842 Kgf |

**Maximum Allowable Pressure for this Nozzle at this Location:**

Converged Maximum Allowable Pressure in the Operating case: 5 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.5927 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 340.5927 mm.

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**Noted issues for the current model:**

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**\*\* Warning: One or more nozzles do not meet proper Spacing Requirements!**

**Please review all reports carefully!**

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### PV Elite<sup>®</sup> Pressure Vessel Design Notes

| Project Information  |  |
|----------------------|--|
| Project Number       |  |
| Client               |  |
| Date                 |  |
| Designer             |  |
| Design Code          |  |
| Additional Standards |  |

| Design Conditions         |  |  |
|---------------------------|--|--|
| Internal Design Pressure: |  |  |
| External Design Pressure: |  |  |
| Design Int. Temperature:  |  |  |
| Design Ext. Temperature:  |  |  |
| Corrosion Allowance:      |  |  |

| Vessel Geometry           |  |  |
|---------------------------|--|--|
| Vessel Type               |  |  |
| Diameter                  |  |  |
| Tangent-to-Tangent Length |  |  |
| Head Type                 |  |  |
| Number of Segments        |  |  |

| Material Specifications |  |
|-------------------------|--|
| Shell:                  |  |
| Head:                   |  |
| Nozzle:                 |  |

| Nozzle/Opening Details | ID/OD | Size | Rating | Matl. | Reinf. |
|------------------------|-------|------|--------|-------|--------|
|                        |       |      |        |       |        |

| Miscellaneous      |  |  |
|--------------------|--|--|
| Support Type       |  |  |
| Number of Supports |  |  |
| Support Material   |  |  |
| Seismic Code       |  |  |
| Wind Code          |  |  |
| Insulation         |  |  |
| Pressure Test      |  |  |
| Radiography        |  |  |
| Fatigue Analysis   |  |  |

### Analysis Notes: